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Context of Evaluating Endodontic Outcomes

The development of medical and dental practices has been guided by prevailing philosophies and consensus of expert opinions. The foundations of modern endodontics were shaken when Billings⁹⁷ brought the apparent relationship between oral sepsis and bacterial endocarditis to the attention of dentistry and medicine. He had strengthened the concept further from Miller,¹³⁴ who had coined the term *focus of infection* to highlight a possible link between “mouth germs” and systemic disease. The disastrous consequences of the focal infection era were sealed when Hunter⁹⁶ delivered his famous address at McGill University. Fears of fatal oral sepsis from deficient root canal treatments led to widespread extraction of pulpless teeth. Endodontics virtually disappeared from many dental schools, whereas in some areas, treatment was restricted to anterior teeth. The focal infection theory reigned for about 50 years⁸³ until around 1940.

The discipline of endodontics was rescued by individual, diligent practitioners in Europe and the United States who meticulously recorded their treatments, as well as their

outcomes, to demonstrate the effectiveness of the procedures in controlling root canal infection. It was through these individual endeavors of merit that the reputation of endodontic procedures was restored and the discipline granted its specialist status in 1952 in the United States.⁵⁹

Toward the 1990s, the improvements in general health and longer survival of Western populations as well as their teeth, coupled with the rising costs of health care for longer-living individuals, prompted a host of reevaluations of the altered economic burdens on society. Among them, the *cost-effectiveness of treatment procedures* for management of diseases loomed large. Thus began the era of evidence-based medicine and dentistry, with its emphasis on costs, benefits, and outcomes of treatment. Attempts to pool outcome data for greater power brought with it the realization that studies originating from different centers varied in diverse ways, leading to very heterogeneous data that challenged formal attempts at drawing definitive conclusions. Characterization of data types and its quality prompted calls for standardization of approaches in measuring outcomes to enable more meaningful pooling.

Because of the better established science of evidence assessment, a modern revisitation of the “focal infection

era”⁹⁴ did not lead to the same threat to the endodontic discipline that was evident at the beginning of the 20th century. In fact, it was suggested as an opportunity to secure research funding for assessing and managing the importance of dental care on systemic health.

Interestingly, the modern threat to endodontics was posed by the cost-economic pressures exerted on treatment planning decisions centering on the question of whether to “save the tooth” or extract it with replacement by an implant-supported crown.²³⁸ Once again, the science of evidence-based practice has helped to avert irrational treatment recommendations leaning toward extraction of savable teeth.^{54,100,103}

TYPES OF DISEASE AND THEIR TREATMENT

The statement, “Endodontists provide endodontic treatment to manage endodontic disease” is a gross oversimplification that masks important subtleties in recognition of the nature of the diseases and how best to address them. Endodontists manage inflammation of the specialized connective tissues within and surrounding the teeth; more specifically, they deal with inflammation that generally commences in the pulp tissue and progresses to the periradicular tissues via portals of communication that convey the neurovascular supply. Incipient or established pulpitis may be managed by vital pulp therapy when judged reversible, largely because a sufficient body of pulp tissue remains healthy and unaffected. More advanced pulpitis that approaches the foramina into the periodontal ligament may require root canal treatment. When pulpal inflammation, necrosis, or infection encroaches on the apical periodontal ligament, root canal treatment is required. Apical periodontitis that persists despite technically adequate root canal treatment may require root canal retreatment, periradicular surgery, or extraction to manage the source of the persistent disease. *Endodontic treatment* is therefore a collective and nonspecific term for a range of procedures directed at managing the spread of pulpal inflammation or infection. Endodontic treatment encompasses the following procedures:

1. Vital pulp therapy (indirect pulp therapy, direct pulp capping, pulpotomy, regenerative pulp therapy)
2. Nonsurgical root canal treatment
3. Nonsurgical root canal retreatment
4. Surgical retreatment

The ideal outcome for endodontic treatment consists of the controlled reduction of inflammation, accompanied by healing through regeneration, although sometimes repair may follow instead. Unfortunately, none of the involved tissues are within the direct sight of the clinician, hidden as they are by their housings of hard tooth structure or alveolar bone and their gingival/mucosal coverings. Consequently, surrogate measures must be employed to evaluate the presence or absence of the disease process and its resolution. The evaluation process is further complicated by the lack of direct correlation between measures of the disease process and its clinical manifestation.

What Are Surrogate Outcome Measures?

Signs of acute inflammation are classically described in the “triple response” exhibited by mechanically injured skin,

which includes altered color (redness), texture/contour (swelling), and sensation (pain), which are directly accessible and viewable. These changes have a direct correlation with histopathologic and molecular alterations. Chronic inflammation does not necessarily exhibit the same highly visible signs and symptoms of its histopathologic character. When hidden from view, as are pulpal and periradicular tissues, the task of recognizing the presence or absence of chronic inflammation is even more challenging. The clinician is therefore left with indirect or associated (proxy) changes by which to judge the presence or absence of disease; these are called “surrogate” measures. Some of these may have to be inferred through associations, some may be directly visualized, and others are only indirectly “viewable” through various imaging techniques (such as radiography). The process therefore calls upon the clinician to assimilate various sources of information to form a judgment about the presence or absence of disease.

TYPES OF OUTCOME MEASURES

In its broadest sense, an outcome measure for a treatment intervention may constitute any consistently anticipated and measurable consequence of the treatment. The prepared shape of the root canal system, bacterial load reduction, and technical quality of the root filling may all be regarded as outcome measures by this definition. But the ultimate *clinical* measure of a treatment outcome is assessing the prevention and resolution of disease.

The outcome of endodontic treatment may be assessed in four dimensions as in other medical disciplines.⁹ The first dimension is physical/physiologic and related to presence or absence of pulpal/periapical health/disease, pain, and function. The second dimension assesses longevity or tooth survival. The third dimension relates to economics and assesses direct and indirect costs. Finally, the fourth dimension examines psychologic aspects involving perceptions of oral health-related quality of life (OHRQoL) and aesthetics.

What Is the Purpose of Evaluating Outcomes?

Apart from the importance of developing a solid foundation of evidence-based practice, it is important to evaluate treatment outcomes for a number of reasons.

EFFECTIVENESS OF PROCEDURES

First, treatment procedures must be effective. Otherwise, there is no reason to recommend them to patients as a treatment option. The patient must be properly informed as to the risks, benefits, and potential outcomes of the offered treatment. The availability of pooled outcome data and consensus guidelines offers both patients and endodontists reassurance and confidence in the validity and predictability of the offered procedure. However, the pooled average outcome data may not pertain if the clinician does not have the requisite experience, skill, and personal outcomes matching at least these average figures of performance. Personal outcome data offer the patient more precise measures for comparison and expectation. It also aids practitioners in refining

their technique and knowledge to further improve their outcomes and, ultimately, to help improve the overall pooled data.³⁴ The treating clinician must assess his or her own personal outcome data and personal expectation of success. If the clinician is uncertain of a result or feels that the outcome could be enhanced by treatment from another clinician, then referral to someone else more qualified is paramount.

FACTORS AFFECTING OUTCOMES

Pooling data (that is preferably homogeneous) offers the potential to evaluate and prioritize the factors that exert a dominant influence on outcomes. In this way, protocols for treatment may be improved in a progressive fashion, bringing about the perfect meld of technical, clinical, and biologic insight necessary for the highest performance and most predictable outcome. Evaluation of factors affecting treatment outcomes arguably provides the strongest influence for making changes that will improve the clinical outcomes. Weighing the relative importance of individual factors should help to identify the key biologic factors at play and how best to manage them from a clinical or technical perspective.

VALUE FOR PROGNOSTICATION

Prognostication, which could be defined as the prediction, projection, prophesizing, or foretelling the likely outcome of treatment, is not often well defined in endodontics. The overall prognosis of a tooth depends on the interaction among three individual and often independent variables, including endodontic, periodontal, and restorative prognoses. Each has a set of subsidiary factors that must be considered to derive an overall prognosis. Finally, the tooth in question must be considered from a strategic perspective, relative to its position in the dental arch and the contribution it makes in the dynamic occlusion.

A deeper analysis of the factors affecting outcomes as prognostic indicators is merited in order to clarify the degree of complexity inherent to the problem. A comparison may be made with the randomized controlled trial of a drug as a treatment intervention. The drug therapy is delivered as a clearly prescribed and standardized dose regimen delivered by specified timings to effect an anticipated blood or target tissue concentration for a specified duration. Data recording is confined to compliance with the prescription and possibly to the checking of actual blood levels achieved as well as the final outcome effect.

In stark contrast, a surgical intervention, regardless of any standardization of the described procedural protocol, is subject to enormous variation in its delivery dependent on interpretation and execution of the protocol by each operator. To add to the levels of complexity and variation, surgical protocols are often multistep, sequential procedures, each prospective step dependent on the previous for its efficacy. Even characterizing and accurately recording any variations in treatment protocols becomes a challenge because many aspects and steps must be documented. This leads to the next challenge, that of demonstrating compliance with the accuracy of such complex data gathering. Furthermore, it becomes important from an analytic perspective to not only consider the effect of individual steps

(factors) alone but also any interactive effect between the multiple steps inherent in the procedure. Even when such comprehensive prospective outcome data are available, utilizing these data to individually calculate the prognosis in a particular scenario may be challenging. Clinically, there may be two distinct ways to estimate prognosis: (1) to apply heuristic principles and intuitively gauge the effect of dominant factors or, at the other end of the spectrum, (2) to mathematically input comprehensive data into an algorithmic model to calculate the estimated outcome. An even more sophisticated approach may be to use mathematical modeling to calculate iterations on variations within the system. At present, the evidence base is insufficient to allow such a sophisticated approach.

It therefore follows that evaluating outcomes and the factors that affect them are fundamental to the creation of a suitable data foundation for future application to endodontic prognostication. The endodontic treatment option is in competition with other alternative therapies and therefore for the discipline to survive and thrive, a suitable blend of efficacy, efficiency, utility, predictability, and cost-effectiveness data must be generated for its principal therapies. Achievement of such a goal will require a diligent and conscientious collection of data from many sources. Analyses of such powerful datasets will ultimately yield proper insight into outcomes of the procedures we use. Coupled with biologic insight of the problem, it will become possible to derive new therapeutic solutions necessary to deliver them.

Outcome Measures for Endodontic Treatment

Before determining the success or failure of an endodontic procedure, the parameters as to what is considered a success or a failure have to be discussed. These are termed the *outcome measures* of the procedures.

OUTCOME MEASURES FOR VITAL PULP THERAPY PROCEDURES

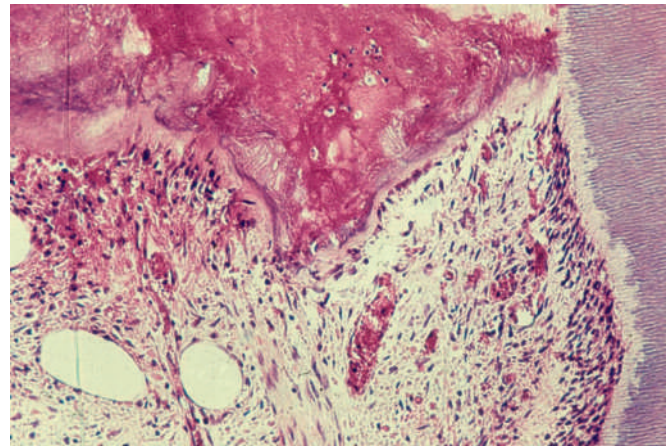
The techniques used to maintain the vitality and health of pulps in teeth with extensive caries or those with traumatic/mechanical exposures include (1) indirect pulp capping with one-step or stepwise caries excavation, (2) direct pulp capping of exposed pulps, and (3) partial/full pulpotomy procedures for more extensively involved pulps. The surrogate outcome measures adopted in studies include (1) clinical success (pulp sensitivity to cold test and absence of pain, soft-tissue swelling, sinus tract, periradicular radiolucency, or pathologic root resorption), (2) patient satisfaction, (3) adverse events (pain, swelling, tooth fracture), and (4) tooth extraction.^{89,136} Although not providing a specific follow-up strategy, the quality guidelines of the European Society of Endodontology (ESE)⁶¹ suggests “initial review at no longer than 6 months and thereafter at further regular intervals.” They also provide the criteria for judging the favorable outcome of vital pulp therapy (Table 17.1). There are substantial differences in the criteria adopted in published clinical studies on outcomes of vital pulp therapy^{89,136} with few studies adopting all of the criteria. The frequency of radiologic review adopted in published studies also varies substantially, with some recommending the first review at

Table 17.1 Criteria for Assessing the Outcome of Pulp Therapy

Quality Guidelines for Endodontic Treatment: Consensus Report of the European Society of Endodontology (2006)	Guidelines on Pulp Therapy for Primary and Immature Permanent Teeth (American Academy of Pediatric Dentistry, 2014)
1. Normal response to pulp sensitivity tests (when feasible) 2. Absence of pain and other symptoms 3. Radiologic evidence of dentinal bridge formation 4. Radiologic evidence of continued root formation in immature teeth 5. Absence of clinical and radiographic signs of internal root resorption and apical periodontitis	1. Tooth vitality maintained 2. Absence of posttreatment signs or symptoms such as sensitivity, pain, or swelling 3. Occurrence of pulp healing and reparative dentin formation 4. Absence of radiographic evidence of internal or external root resorption, periapical radiolucency, abnormal calcification, or other pathologic changes 5. Teeth with immature roots should show continued root development and apexogenesis

1 month followed by subsequent 3 monthly reviews.^{172,242} The relative benefit of acquired information versus unnecessary radiation to patients has been questioned.⁶⁶ An initial assessment at 6 to 12 weeks, followed by a review 6 and 12 months after treatment, seems to have been accepted and is recommended. Some studies reported up to 10 years follow-up on case series demonstrating lower success rates, inferring the possibility of slow spread of pulpal inflammation and late failures. The conclusion is that longer follow-up periods are merited.

The process at each review consists of obtaining a history of symptoms, coupled with an examination to determine the presence or absence of tenderness to palpation of adjacent soft tissues, tenderness to pressure and percussion of the tooth, signs of radiographic pulpal, periapical changes, and responses to pulp tests. The accuracy of pulp tests may be limited in pulpotomized teeth because of the distance of the remaining pulp tissue from the tooth's surface. In the case of pulp capping and pulpotomy, additional tests include radiographic verification of the presence of the calcific barrier (Fig. 17.1) and its integrity by removal

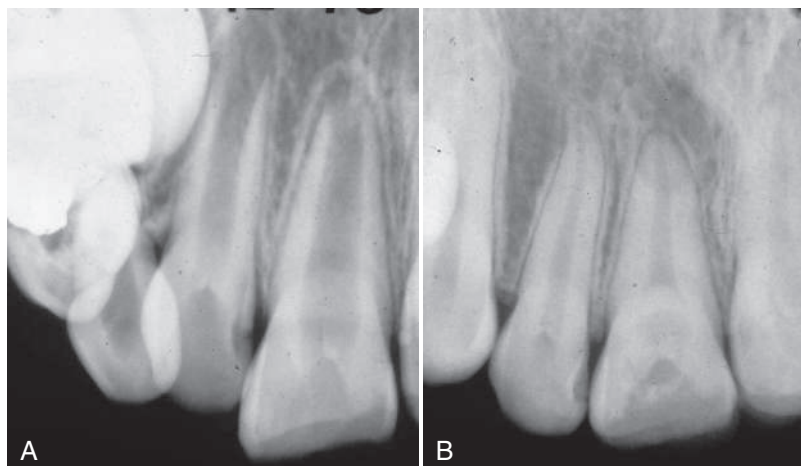
**Fig. 17.1** Histologic view of intact calcific bridge formation following pulpotomy.

of the dressing and direct probing. Although an initial examination at 6 weeks has been suggested, this can be modified for the radiographic assessment. If there is no evidence of complete bridge formation, the treatment is considered failed and root canal treatment should be considered. In addition, in the case of incompletely formed roots there should be radiographic evidence of continued root development (Fig. 17.2).

OUTCOME MEASURES FOR NONSURGICAL ROOT CANAL TREATMENT AND RETREATMENT

Root canal treatment may be employed either to prevent or resolve periapical disease. Given that periapical lesions develop as a result of the *interaction* between bacteria (and their by-products) and the host defenses, it is clear that prevention or resolution of the disease process depends on preventing or terminating this interaction.

Prevention of apical periodontitis applies to the clinical situation where it is judged that the pulp is irreversibly inflamed, necrotic, or infected to the extent that vital pulp therapy would not resolve the problem, which therefore requires pulpectomy. The implication is that the procedure

**Fig. 17.2** Immature maxillary right lateral and central incisors (A) showing continued root development following pulpotomies (B).

demands asepsis as a prime requirement. The precise technical details of the protocol may not be so important, as long as they are focused on effective and aseptic removal of the pulp tissue. This is validated by the high chance of retaining periapical health (judged by conventional radiography), regardless of the clinical protocol used.¹⁵³

Once the periapical lesion has become established, the challenge is a different one because the purpose now is to remove the bacterial biofilm and to effect switching off of the periapical host response. The challenge seems to be greater still if the periapical lesion is larger as it is associated with a more diverse infection. A number of approaches (protocols) have been used to achieve this general aim. The periapical healing process that occurs after root canal treatment is less clear. Nevertheless, ideal healing would eventually result in regeneration and the formation of cementum over the apical termini, isolating the root canal system from the periapex (Fig. 17.3), but this is not an inevitable result. Incomplete removal of the infection will reduce but not eliminate the periradicular inflammatory reaction, and in fact this is generally what happens.¹⁴⁷ This implies that residual infection in the apical anatomy is typical following completion of root canal treatment and that an ongoing interaction continues between the residual infection, root filling material, and host defenses, which plays a definitive role in determining the final healing outcome.

The culture test has been used during root canal treatment as an interim measure of the efficacy of the chemo-mechanical procedure; however, it has fallen out of favor in contemporary practice for a variety of reasons.¹³⁹ The outcome measures that quantitate healing subsequent to root

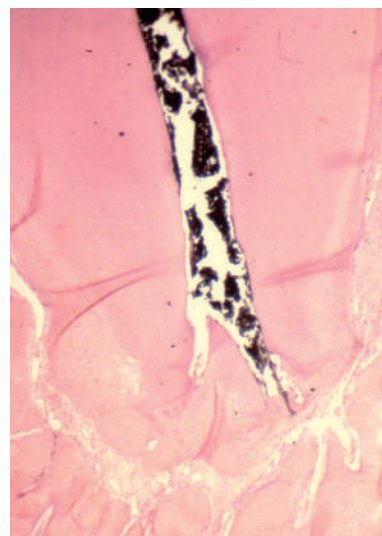


Fig. 17.3 Low-power view of healing by cementum formation when Sealapex is used (black particles are residual Sealapex and root filling material). (Courtesy Prof. M. Tagger.)

canal treatment are the absence of clinical signs and symptoms of persistent periapical disease.^{17,18} The definitive outcome measure (in conjunction with the absence of signs and symptoms), however, is periapical healing, because the treatment is aimed at resolution of periapical disease (Fig. 17.4).¹⁵² Clinical judgment of the outcome of treatment is based on the absence of signs of infection and inflammation, such as pain, tenderness to pressure/percussion of the tooth, tenderness to palpation of the related soft

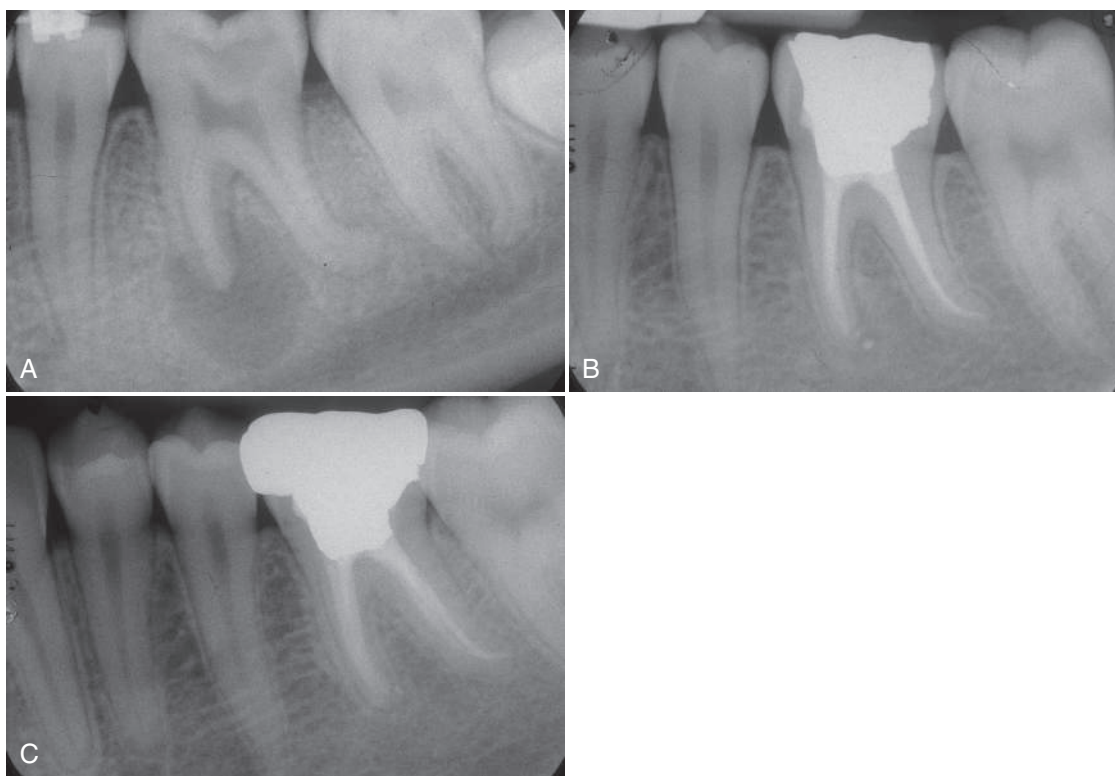


Fig. 17.4 Resolution of periapical disease after root canal treatment of the mandibular left molar with preoperative periapical radiolucency associated with mesial and distal roots (A), delayed healing caused by extruded filling material from the distal canal (B) and nearing complete healing as extruded material is resorbed (C).

tissues, absence of swelling and sinus tract, and radiographic demonstration of reduction in the size of the periapical lesion (if sufficient time has lapsed), with a completely normal development of the periodontal ligament space. Although the majority of periapical lesions heal within 1 year, healing may continue for up to 4 years or longer.²²⁹

Absence of signs and symptoms of periapical disease with radiographic evidence of a persistent periapical radiolucency may indicate either fibrous repair (Fig. 17.5) or persistent chronic inflammation or infection. Only time and acute exacerbation would identify the latter, whereas the former should remain asymptomatic.

Longevity measures include survival of the root canal fillings or treatment^{118,228,236} and tooth retention or survival.^{111,149,190} The term *functional retention* was coined by Friedman in 2004⁶⁹ to mean retention of the tooth in the absence of signs and symptoms regardless of the radiographic presence of a lesion. The term *functionality* should further and more specifically cover the functional utility of the tooth—that is, some patients may complain that despite the absence of specific signs and symptoms of infection or inflammation, they find it impossible to use the tooth because it “feels” weak.

Modern definitions of health embrace a broader spectrum of measures including psychological well-being.¹¹⁶ Instruments to assess these aspects have been developed in general medicine and adapted for application to dentistry, such as the Oral Health Related Quality of Life instrument.^{55,87,115} There is no definitive instrument for measuring these aspects in endodontics as yet. Studies published so far have mainly adapted versions of the Oral Health Impact Profile (OHIP).

The periapical status of root-treated teeth has traditionally been assessed using two-dimensional conventional radiographic imaging. The use and limitations of two-dimensional radiographic images for assessing treatment outcome has been reviewed thoroughly and is reported to suffer from low sensitivity, particularly in the posterior parts of the mouth.⁹⁸ The development of digital imaging technology⁸⁰ brought the possibility of image manipulation, including digital subtraction,^{50,158,243,259} densitometric analysis,¹⁵⁶ correction of gray values,³² and the manipulation of brightness and contrast.⁷⁵ However, none

of these approaches has addressed the main deficiencies of accurately elucidating and quantitating the actual existence or progression of periapical bone loss.

Cone-beam computed tomography (CBCT), a new three-dimensional imaging technique requiring only 8% of the effective dose of conventional computed tomography,¹¹⁹ has been proposed as a means of overcoming the problem of superimposition of tissue layers and structures. The diagnostic values of periapical radiography and CBCT were compared on dogs' teeth using histology as the comparative method of determining apical periodontitis 180 days after root canal treatment.⁴⁴ CBCT was found to be significantly more accurate in detecting minor bone lesions compared with two-dimensional radiology.^{44,174} Other evidence to support the superior sensitivity of CBCT is available from studies using a variety of simulated defects created in bone in pig²²⁶ or human^{165,221} jaws. Although the validity of clinical outcome data derived using conventional radiographic techniques has been questioned,²⁵⁶ the routine use of CBCT is not recommended^{62,91,197} owing to its higher radiation dosage.^{5,91} The use of CBCT, which may be more sensitive for detection of periapical healing,^{114,167} may give lower healing rates and longer durations to complete healing.

Many studies consider the treatment successful only when both radiographic and clinical criteria are satisfied.⁶⁹ It has been documented that a small proportion of cases present with persistent symptoms despite complete radiographic resolution of the periapical lesion.¹⁷⁶ Comparison of success rates estimated with or without clinical examination revealed no⁹³ or only a very small difference (1%).¹⁵⁰

The widely accepted definition for endodontic success and failure by Strindberg²²⁹ embraces both radiographic and clinical findings (Table 17.2). Friedman and Mor⁶⁹ preferred the terms *healed*, *healing*, and *diseased* instead of *success* and *failure* because of the potential of the latter to confuse patients. The “healed” category corresponds to “success” as defined by Strindberg,²²⁹ whereas “healing” corresponds to “success” as defined by Bender and colleagues (see Table 17.2).^{17,18}

The long duration of the periapical healing process coupled with the reduced recall rates at longer follow-up periods has provided opportunity for setting the thresholds of success at either complete healing or partial healing (reduced lesion size). The success criteria used for complete resolution have been described either as “strict”¹⁵² or “stringent,”⁶⁹ whereas the criteria established for reduction in periapical radiolucency size has been described as either “loose”¹⁵² or “lenient.”⁶⁹ The frequency of adoption of these two thresholds has been similar in previous studies; the expected success rates using “strict” criteria would be expected to be lower than those based on “loose” criteria. The literature finds differences that vary from 4% to 48%.¹⁵³

A periapical index (PAI) consisting of five points on the scale was devised for measuring the periapical status and used in some studies.^{157,159} However, the studies only reported the increase or decrease in mean scores for the clinical factors under investigation coupled with the proportion of successful cases. This approach precluded direct comparison of their data with studies reporting conventional dichotomized outcomes. Other studies using the index dichotomized the scores into “healthy” (PAI 1 or 2) or “diseased” (PAI 3 to 5) categories,¹⁶⁰ thus allowing the data

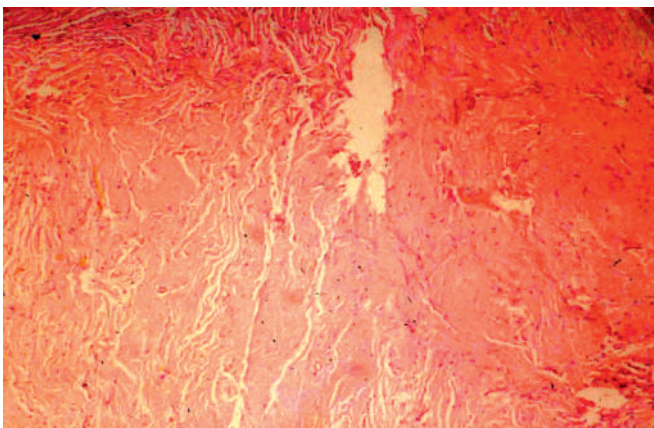


Fig. 17.5 Histologic view of fibrous periapical healing.

Table 17.2 Criteria for Determination of Periapical Status

Strindberg (1956)	Bender et al. (1966, a and b)	Friedman and Mor (2004)
Success	Success	Healed
<i>Clinical:</i> No symptoms <i>Radiographic:</i> The contours, width, and structure of the periodontal margin were normal OR The periodontal contours were widened mainly around the excess filling	<i>Clinical:</i> Absence of pain/swelling; disappearance of fistula; no loss of function; no evidence of tissue destruction <i>Radiographic:</i> An eliminated or arrested area of rarefaction after a posttreatment interval of 6 months to 2 years	<i>Clinical:</i> Normal presentation <i>Radiographic:</i> Normal presentation
Failure		Diseased
<i>Clinical:</i> Presence of symptoms <i>Radiographic:</i> A decrease in the periradicular rarefaction OR Unchanged periradicular rarefaction OR An appearance of new rarefaction or an increase in the initial rarefaction		Radiolucency has emerged or persisted without change, even when the clinical presentation is normal OR Clinical signs or symptoms are present, even if the radiographic presentation is normal
Uncertain		Healing
<i>Radiographic:</i> There were ambiguous or technically unsatisfactory control radiographs that could not, for some reason, be repeated OR The tooth was extracted prior to the 3-year follow-up owing to the unsuccessful treatment of another root of the tooth		<i>Clinical:</i> Normal presentation <i>Radiographic:</i> Reduced radiolucency

to be compared directly with the more traditionally used binary outcomes of success or failure. In this system of designation, given that the PAI score 2 represents periodontal ligament widening, it effectively signals adoption of the “loose” threshold. The longitudinal follow-up⁸⁶ of 14 cases presenting with widened apical periodontal ligament (PAI score at 2) for 10 years revealed unfavorable future healing in only a small proportion of the cases (28%, 4/14).

OUTCOME MEASURES FOR PERIAPICAL SURGERY

Nonsurgical root canal treatment alone may fail to resolve apical periodontitis in a small proportion of cases because it may not allow access to the infection or resolution of the cause of the infection (i.e., root fracture or crestal bone communication to the periapical infection). Where the cause is microbial persistence, the infection may be localized intraradicularly (in the apical canal anatomy [Fig. 17.6] or in the apical dentinal tubules [Fig. 17.7] or extraradicularly [Fig. 17.8]). In these instances, a surgical approach to the periapex may be required in addition to the nonsurgical approach to removing such microbes (see Fig. 17.7).

The success of periapical surgery has been assessed with the same clinical and radiographic criteria as for nonsurgical root canal treatment. However, the radiographic criteria for successful periapical healing are different from those for nonsurgical root canal treatment (Table 17.3; Figs. 17.9–17.11).^{140,186} Moreover, periodontal attachment loss in the form of marginal gingival recession is an additional criterion for measuring the outcome of periapical surgery.

Now that the parameters have been discussed as to the variables that need to be considered when assessing healing

or nonhealing outcomes, the following discussion will describe the expected outcomes of the various endodontic procedures.

Outcomes of Vital Pulp Therapy Procedures

The vital pulp therapy procedures discussed in this chapter include those for managing extensive caries with a high risk of pulpal exposure or pulps exposed through caries or traumatic injuries.

INDIRECT PULP CAPPING (ONE-STEP VERSUS STEPWISE EXCAVATION)

The most conservative management for extensive caries is indirect pulp capping using either the one-step or the stepwise excavation approach. The stepwise excavation approach with initial partial caries removal was advocated to reduce the risk of pulpal exposure and is generally considered to be associated with poorer outcomes. The clinical outcomes of the one-step or stepwise approaches have been compared in three randomized controlled trials on permanent teeth with deep caries.^{19,113,121,122} Stepwise excavation was found to be associated with a lower rate of pulp exposure and higher chance of long-term clinical success than the one-step excavation approach (Table 17.4).^{19,113} Maltz and colleagues,^{121,122} however, found the contrary, and they attributed the low long-term clinical success rate of the stepwise approach to patients being noncompliant in returning for scheduled completion of treatment. Interestingly, comparison of the cost-effectiveness of the one-step versus the stepwise excavation techniques using health-economic

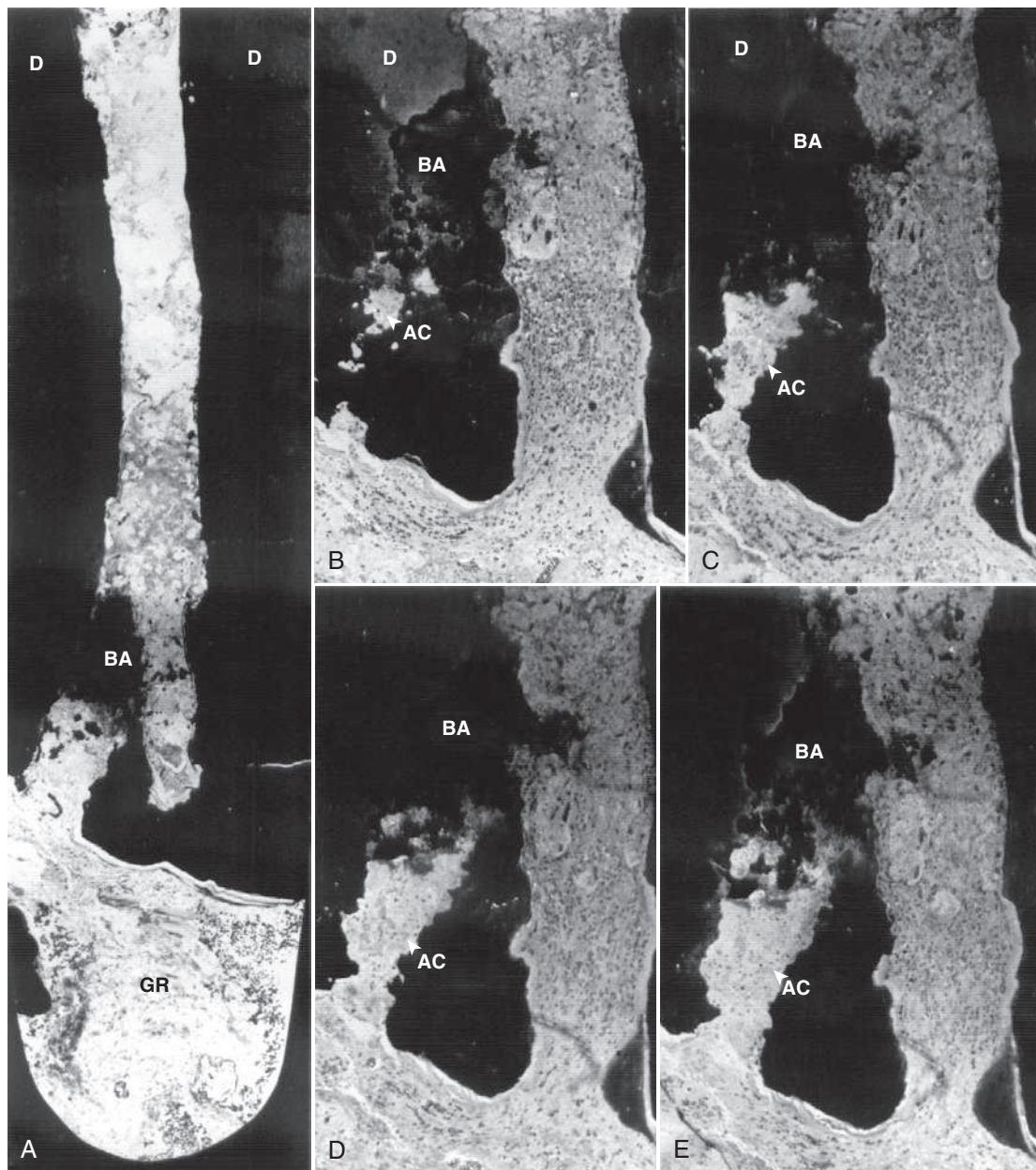


Fig. 17.6 Axial sections through the surgically removed apical portion of the root with a therapy-resistant periapical lesion (GR). Note the cluster of bacteria (BA) visible in the root canal. Parts (B)–(E) show serial semithin sections taken at varying distances from the section plane of (A) to reveal the emerging (B) and gradually widening (C–E) profiles of an accessory root canal (AC). Note that the accessory canal is clogged with bacteria (BA). Original magnification: A, $\times 52$; B–E, $\times 62$. (From Nair PN, Sjogren U, Krey G, Sundqvist G: Therapy-resistant foreign body giant cell granuloma at the periapex of a root-filled human tooth, *J Endod* 16:589, 1990.)

modeling involving simulated treatment of a molar tooth with deep caries in a 15-year-old patient revealed that the one-step procedure accrues lower long-term costs as well as longer tooth retention and pulp vitality.²⁰⁰

Meta-analyses of data from studies^{19,20,65,81,89,113,120-122,141} listed in Table 17.4 revealed that the weighted pooled success rate for indirect pulp capping using the one-step approach (81.7%; 95% confidence interval [CI]: 72.7%, 90.6%) (Fig. 17.12) was similar to that for the stepwise approach (81.9%; 95% CI: 72.1%, 91.7%) (Fig. 17.13).

Calcium hydroxide cement was the preferred lining material for the pulpal surface, whereas zinc oxide–eugenol–based cement was the preferred base material in most

studies. Recently, resin-modified-glass-ionomer liner has also been used, but the type of lining material did not influence the outcome (see Table 17.4). The age of patients, the presence of preoperative pain, and pulpal exposure during excavation were significant negative prognostic factors.¹⁹

DIRECT PULP CAPPING

Direct pulp capping is performed on teeth with pulp exposures caused by caries, caries excavation, or traumatic injuries. Most studies investigating the clinical outcome of direct pulp capping on permanent teeth had excluded teeth with signs and symptoms of irreversible pulpitis and apical



Fig. 17.7 **A**, Adequate root filling demonstrated on radiograph. **B**, View of the periapical surgery and root resection of the tooth shown in **(A)** shows stained root dentine. **C**, Resected root showing stained/infected dentine. **D**, Histologic view of the root end shown in **(C)** showing infected dentinal tubules (S).

pathosis (Table 17.5). Saline, sodium hypochlorite, and chlorhexidine have been reportedly used to irrigate the exposed pulp and to achieve hemostasis. Calcium hydroxide paste and mineral trioxide aggregate (MTA) were the commonly used capping materials (see Table 17.5).

Table 17.3 Classification of Periapical Healing Following Apical Surgery

COMPLETE HEALING

Reformation of a Periodontal Space With

Normal width and lamina dura to be followed around the apex
Slight increase in width of apical periodontal space but less than twice the width of noninvolved parts of the root
Tiny defect in the lamina dura (maximum 1 mm) adjacent to the root filling

Complete Bone Repair With

Bone bordering the apical area does not have the same density as surrounding noninvolved bone
No apical periodontal space can be discerned

Incomplete Healing (Scar Tissue)

The Rarefaction Has Decreased in Size or Remained Stationary With

Bone structure recognized within the rarefaction
The irregular periphery of the rarefaction and demarcation by a compact bone border
The rarefaction located asymmetrically around the apex
Angular connection between the rarefaction and the periodontal space

An Isolated Scar Tissue in the Bone With Findings Above

Uncertain Healing

The Rarefaction Has Decreased in Size With

The size larger than twice the width of the periodontal space
Lamina-dura like bone structures around the border
A circular or semicircular periphery
Symmetric location around the apex as a funnel-shaped extension of the periodontal space
Bone structure discernible within the bony cavity
A collar-shaped increase in width of lamina dura coronal to the radiolucency

Unsatisfactory Healing (Failures)

The rarefaction has enlarged or is unchanged
If a case still demonstrated "uncertain healing" 4 years postoperatively, the treatment should be considered a failure¹⁸⁶

From Rud J, Andreasen JO: A study of failures after endodontic surgery by radiographic, histologic and stereomicroscopic methods, *Int J Oral Surg* 1:311-328, 1972; Molven O, Halse A, Grung B: Observer strategy and the radiographic classification of healing after endodontic surgery, *Int J Oral Maxillofac Surg* 16:432-439, 1987.

A meta-analysis of data from studies^{12,19,21,64,65,72,88,90,92,128,132,133,193,207,254} listed in Table 17.5 revealed the weighted pooled success rate to be 70.1% (95% CI: 59.9%, 80.2%) (Fig. 17.14). The patient's age and sex; tooth location and type; pulp exposure type, size, and its location; and the restoration type, size, and quality did not have a significant influence on success. Although the outcome of direct pulp capping of teeth with immature versus mature apices has not been systematically compared in individual studies, indirect comparison of pooled data from different studies¹ revealed that teeth with immature roots were associated with significantly more successful outcomes.

The type of capping material was another significant prognostic factor,^{90,197,207} with MTA performing superiorly to calcium hydroxide in a randomized controlled trial⁹⁰ and in a systematic review.¹

PULPOTOMY

Earlier studies on the outcome of *partial pulpotomy* (specifying varying degrees of coronal pulp removal) had only

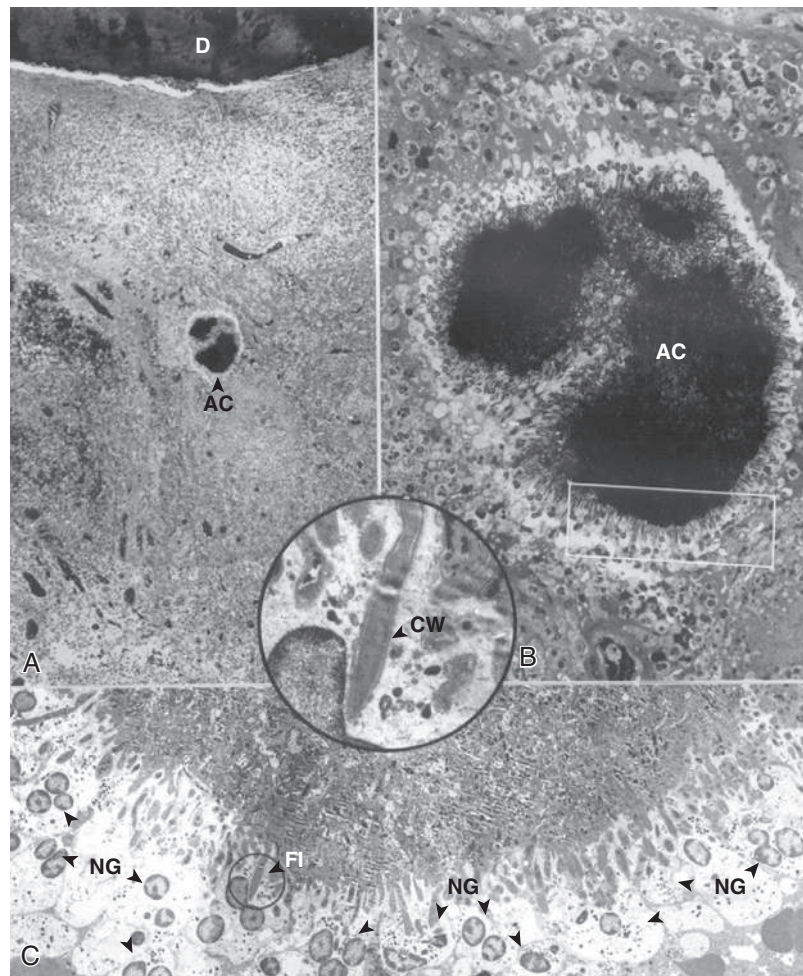


Fig. 17.8 Actinomyces in the body of a human periapical granuloma. The actinomycotic colony (AC) in (A) is magnified in (B). The rectangular area demarcated in (B) is magnified in (C). Note the starburst appearance of the colony with needle-like peripheral filaments surrounded by few layers of neutrophilic granulocytes (NG), some of which contain phagocytosed bacteria. A dividing peripheral filament (FI) is magnified in the inset. Note the typical gram-positive wall cell wall (CW). D, Dentine. Original magnification: A, $\times 60$; B, $\times 430$; C, $\times 1680$; inset, $\times 6700$. (From Nair PR, Schroeder H: Periapical actinomycosis, *J Endod* 10:567, 1984.)

included teeth with vital traumatic pulp exposures; however, more recent studies have also included teeth with carious pulp exposures. Teeth with spontaneous or severe pain and signs and symptoms of apical pathosis were excluded (Table 17.6). On the other hand, the outcomes of *full pulpotomies* (completely removing the coronal pulp and retaining the radicular pulp) had only been investigated in teeth with carious exposures (Table 17.7). Saline irrigation has been preferred for achieving hemostasis, whereas calcium hydroxide, and more recently, MTA, were the preferred pulp capping materials (see Table 17.6).

A meta-analysis of data from studies listed in Tables 17.6^{10,11,19,40,71,125,131,177} and 17.7^{6,30,31,52,58,126,187,192,235,251,255} revealed the weighted pooled success rate to be 79.3% (95% CI: 66.7%, 91.8%) for partial pulpotomies (Fig. 17.15) and 82.4% (95% CI: 69.3%, 95.4%) for full pulpotomies (Fig. 17.16).

The effects of potential prognostic factors for pulpotomy have not been explored systematically except for the pulp capping material. Randomized controlled trials revealed that MTA achieved similar outcomes in partial¹⁷⁷ or full pulpotomies⁵⁸ when compared with calcium hydroxide.

SUMMARY OF PROGNOSTIC FACTORS FOR VITAL PULP THERAPY

In summary, vital pulp therapy performed under guideline standards with optimal coronal seal achieved promising long-term success in teeth with carious, mechanical, or traumatic exposures of healthy pulps.

The most important factors affecting the outcome of vital pulp therapy are preexisting health of the pulp, adequate removal of infected hard or soft tissues, careful operative technique to avoid damage to residual tissues, and elimination of microbial leakage around the final restoration. It can be difficult to gauge the health of the residual pulp as it is a matter of subjective assessment and relies on experience in pulp diagnosis. The degree of pulp bleeding upon exposure is a more reliable tool to judge the status of the pulp than the preoperative clinical signs and symptoms. Continued bleeding after 10 minutes, even after rinsing with sodium hypochlorite solution, may suggest that the residual pulp was still heavily inflamed and a complete pulpectomy may be a more effective treatment modality. Removal of infected tissue is a matter of subjective experience but may be aided by various

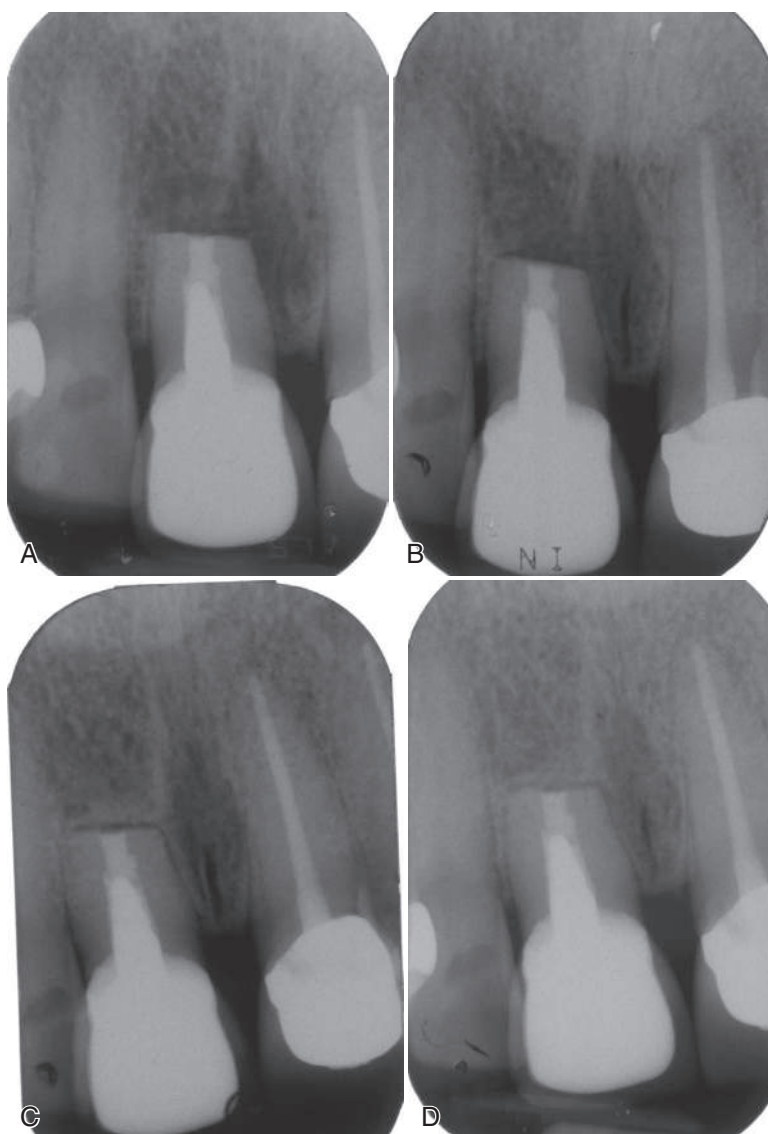


Fig. 17.9 **A**, A maxillary right central incisor having undergone periapical surgery. **B**, Incomplete periapical healing at 1 year postoperatively. Complete healing at 3 (**C**) and 4 years (**D**) postoperatively.



Fig. 17.10 Example of incomplete healing with isolated scar tissue.

dyes. The final factor relies on the correct choice of restorative material and its adequate manipulation to prevent leakage.

Factors such as the age and health of the patient, the size and nature (cariou or traumatic) of pulp exposure and its duration of exposure to the oral environment (up to 48 hours) do not in themselves compromise outcomes of vital pulp therapy.

Outcomes of Nonsurgical Root Canal Treatment

In contrast to other areas of endodontics, the number of studies and extent of investigation of nonsurgical root canal treatment is more comprehensive, yielding a much greater insight even though the quality and scope of the research does not always reach the highest levels.

A systematic review and meta-analysis of the factors affecting primary root canal treatment outcome conducted

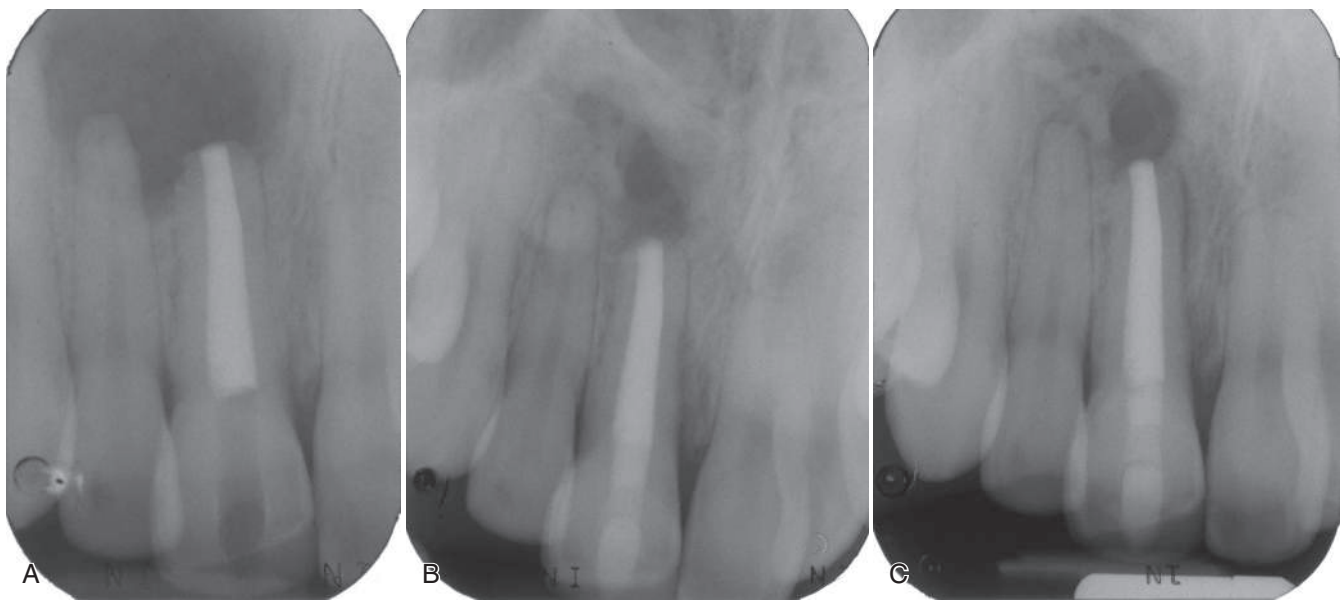


Fig. 17.11 Example of uncertain healing. Periapical radiographs of a maxillary left central incisor taken immediately after periapical surgery (A), at 2 years postoperatively (B), and at 3 years postoperatively (C).

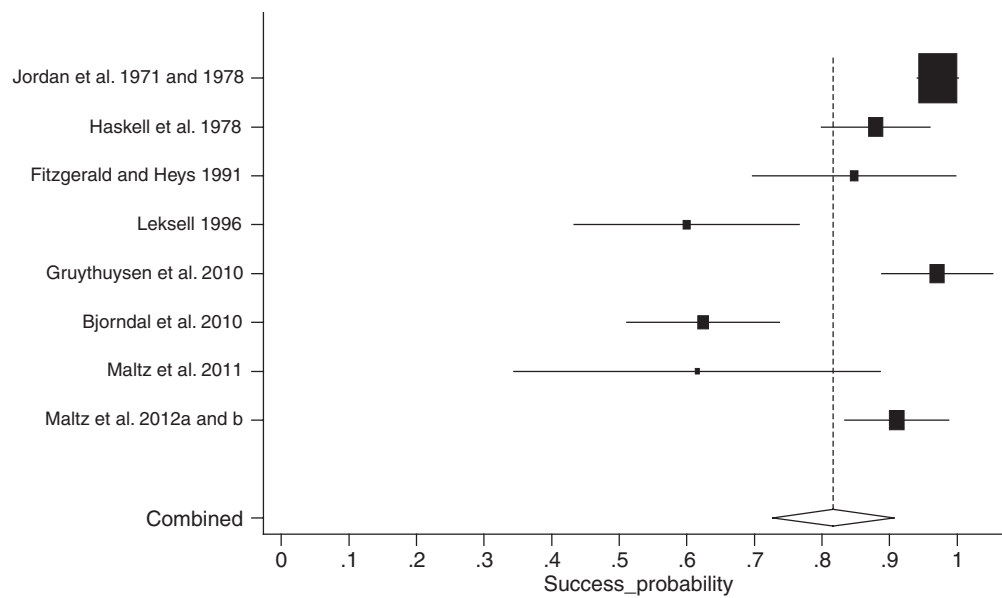


Fig. 17.12 Forest plot showing individual study and weighted pooled probabilities of success for one-step indirect pulp capping.

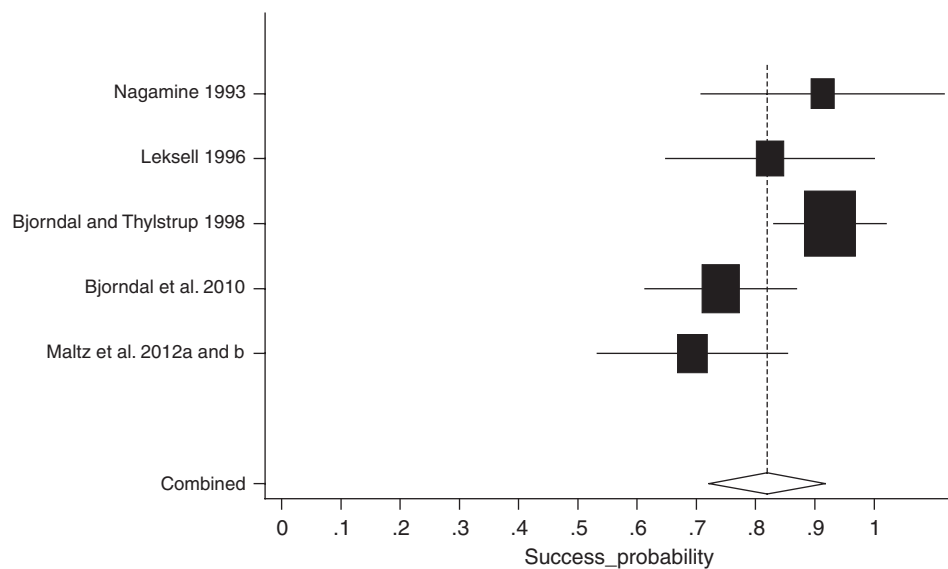


Fig. 17.13 Forest plot showing individual study and weighted pooled probabilities of success for stepwise indirect pulp capping.

Table 17.4 Studies Investigating the Clinical/Radiographic Outcome of Indirect Pulp Capping of Permanent Teeth—cont'd

Study	Study Design	Type of Indirect Pulp Capping	No. of Teeth	Success (%)	Age	Pretreatment Status	Capping Material	Criteria for Success	Duration After Treatment	Notes
Bjorndal and Thylstrup, 1998	Case series	Stepwise	94	87 (92.6%)	Not given	Deep caries	CH/temporary filling	Absence of perforation, asymptomatic	1 year	88/94 without exposure
Bjorndal et al., 2010	RCT	Stepwise	143	106 (74.1%)	29 (25–38)	Deep caries Only provoked pain Vital	CH/GIC	Vital/intact PDL	1 year	Stepwise better than complete; preoperative pain, older patients reduce success
		1-step	149	93 (62.4%)						
Gruythuysen et al., 2010	Case series	1-step	34	33 (97.1%)	Up to 18	Cariou (> 2/3) No spontaneous, persistent pain No apical pathosis	GIC	Survive/ asymptomatic/ intact PDL/no resorption	3 years	
Maltz et al., 2011	Case series	1-step	26	16 (61.5%)		Vital Deep caries	CH/ZOE then composite	Vital	10 years	
Maltz et al., 2012 (a and b)	RCT (multicenter)	1-step	112	111 (18 months) 102 (91.7%) (3 years)	≥ 6	Vital Cariou (> 1/2) No spontaneous, persistent pain No apical pathosis	GIC/amalgam or composite	Vital/intact PDL Vital/intact PDL	18 months, 3 years (clinical and radiographic outcome)	Indirect pulp capping was associated with significantly higher success rate than stepwise, regardless of duration after treatment; the low success rate of stepwise was attributed to patients not returning for completion of treatment
		Stepwise	101	87 (18 months) 70 (69.3%) (3 years)			CH/ZOE then GIC/ amalgam or composite			

CH, Calcium hydroxide; GIC, glass ionomer; PDL, periodontal ligament; RCT, randomized controlled trial; ZOE, zinc oxide eugenol.

Table 17.5 Studies Investigating the Clinical/Radiographic Outcome of Direct Pulp Capping of Permanent Teeth

Study	Study Design	No. of Teeth	Success (%)	Age (Years)	Pretreatment Status	Hemostasis	Capping Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Weiss, 1966	Case series	160	141 (88.0%)	16–67	Not given	Not given	CH + cresatin	ZOE	Vital/asymptomatic/intact PDL	3 years	
Shovelton et al., 1971	RCT	154 (1-step)	115 (74.7%)	15–44	Cariou or traumatic exposure Vital Asymptomatic Molar/premolar	Saline	Corticosteroid + antibiotic, glycerrhetic acid + antibiotics, ZOE, or CH	ZOE/amalgam	Vital/asymptomatic/intact PDL	24 months	There was no significant difference in success rates among the various capping materials
		53 (2-step)	33 (64.7%)								
Haskell et al., 1978	Case series	133	117 (88.0%)	8–74	Cariou exposure Asymptomatic	Not given	CH or penicillin	ZOE	Vital/asymptomatic/intact PDL	>5 years	5- to 22-year survival Success rate was not affected by age and tooth type
Gillien and Schuman, 1985	Case series	17	13 (76.4%)	6–9	Cariou exposure	Not given	CH	Base material not given/amalgam or crown	Asymptomatic/intact PDL	6–12 months	
Horsted et al., 1985	Case series	510	485 (95.1%)	Not given	Exposure during cavity preparation or caries removal No periapical pathosis No pain	2% chloramine or 0.2% CHX	CH	ZOE	Vital/asymptomatic/intact PDL	5 years	Type of exposure and tooth type had no significant effects Older age had lower survival
Fitzgerald and Heys, 1991	Case series	8	6 (75.0%)	20–60	Vital Exposure during caries removal	Sterile cotton pellets	CH	ZnPO ₄ /amalgam or composite	Asymptomatic	12 months	The brand of setting CH paste did not have significant effect
Matsuo et al., 1996	Case series	44	36 (81.2%)	20–69	Cariou exposure No intense pain	10% NaOCl and 3% H ₂ O ₂	CH	ZOE/GIC	Vital/asymptomatic/intact PDL	3 years	

Table 17.5 Studies Investigating the Clinical/Radiographic Outcome of Direct Pulp Capping of Permanent Teeth—Cont'd

Study	Study Design	No. of Teeth	Success (%)	Age (Years)	Pretreatment Status	Hemostasis	Capping Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Santucci, 1999	Case series	29	15 (51.7%)	Not given	Exposure due to caries, or its removal Sensitive to cold or sweat with no other pain No periapical pathosis	Not given	CH	Composite or cast gold restoration	Asymptomatic	4.5 years	
Barthel et al., 2000	Case series	123	29 (23.6%)	10–70	Cariou exposure	3% H ₂ O ₂	CH/ZnPO ₄ or other materials	Not given	Vital/asymptomatic/ intact PDL	5–10 years	Age, tooth type, site of exposure had no significant effects Immediate placement of permanent restoration had significantly higher success rate
Farsi et al., 2006	Case series	30	28 (93.3%)	9–12	Deep caries Potential exposure Reversible pulpitis Intact PDL	Saline	MTA	ZOE/composite	Vital/asymptomatic/ intact PDL/ continuous root development	2 years	
Bogen et al., 2008	Case series	49	48 (98.0%)	7–45	Cariou exposure (0.25–2.5 mm) Reversible pulpitis	5.25/6% NaOCl	MTA	Composite	Dentine Vital/bridge/ asymptomatic/ root development/ intact PDL	1–9 years	
Bjorndal et al., 2010	RCT	22	7 (31.8%)	25–38	Exposure during caries removal Only provoked pain Vital	Saline	CH	GIC	Vital/intact PDL	1 year	

Continued

Table 17.5 Studies Investigating the Clinical/Radiographic Outcome of Direct Pulp Capping of Permanent Teeth—Cont'd

Study	Study Design	No. of Teeth	Success (%)	Age (Years)	Pretreatment Status	Hemostasis	Capping Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Mente et al., 2010	Case series	122	86 (70.5%)	8–78	Carious or mechanical exposure	0.12% CHX	MTA or CH	GIC/composite or crown	Vital/no clinical or radiographic evidence of apical pathosis	12–18 months	Use of MTA for capping and immediate placement of permanent restoration had significantly higher success rate Age; sex; tooth location and type; exposure site and type; restoration type, size, and quality did not have significant effects
Miles et al., 2010	Case series	51	23 (45.1%)	21–85	Carious exposure	2.5% NaOCl	MTA	GIC/composite or amalgam	Vital/asymptomatic/intact PDL	12–27 months	
Hilton et al., 2013	RCT	126	81 (64.3%)	9–90	Carious, traumatic, mechanical exposure	5.25% NaOCl	CH	GIC	Vital/intact PDL/no resorption/not requiring extraction or root canal treatment	2 years	MTA was associated with significantly higher success rate than CH Patient, dentist, tooth, pulp exposure, and pulp capping characteristics did not have significant influence on the results
		144	116 (80.6%)	8–89			MTA				

CH, Calcium hydroxide; GIC, glass ionomer; MTA, mineral trioxide aggregate; NaOCl, sodium hypochlorite; RCT, randomized controlled trial.

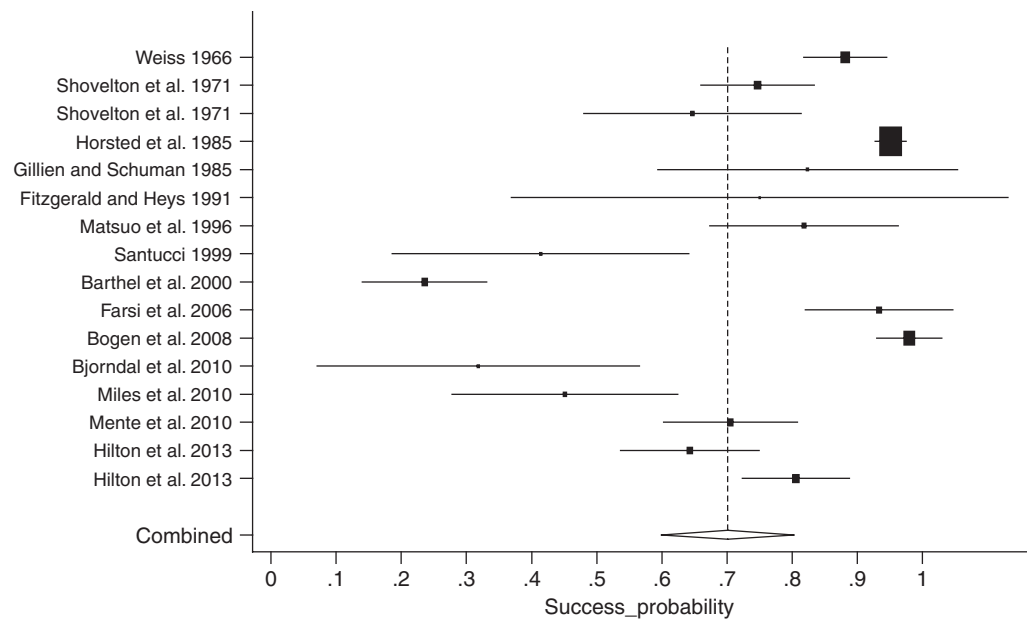


Fig. 17.14 Forest plot showing individual study and weighted pooled probabilities of success for direct pulp capping.

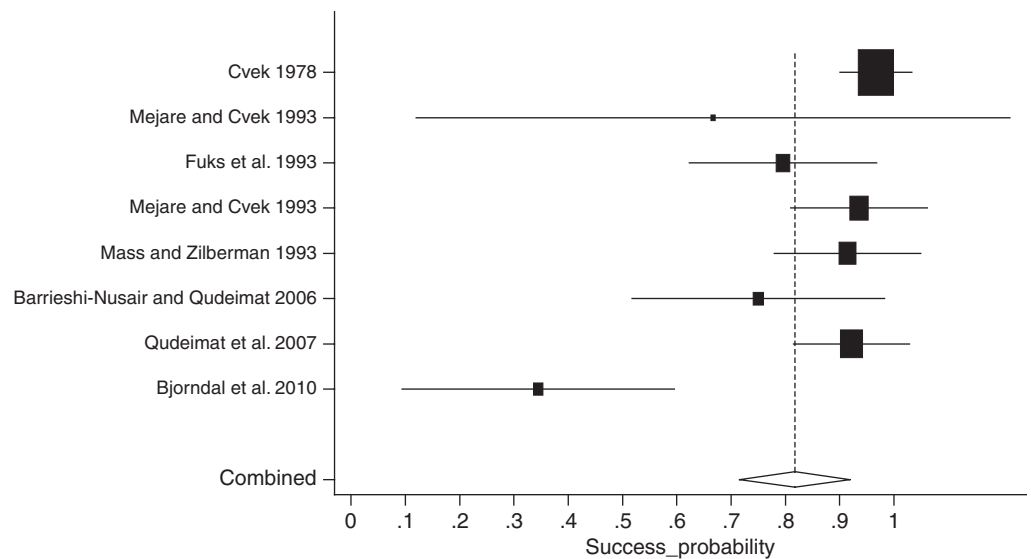


Fig. 17.15 Forest plot showing individual study and weighted pooled probabilities of success for partial pulpotomy.

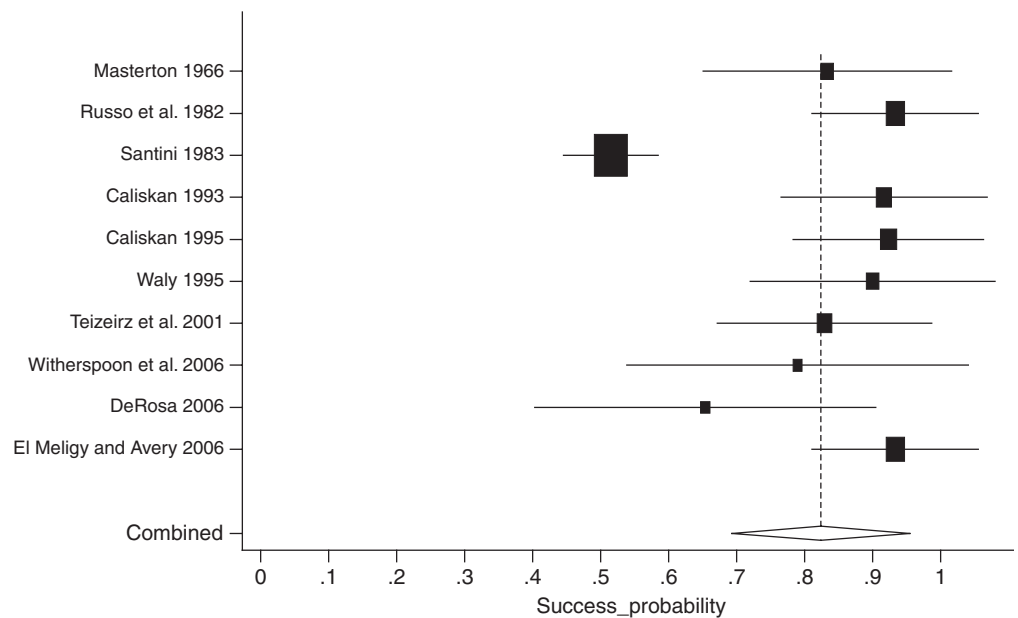


Fig. 17.16 Forest plot showing individual study and weighted pooled probabilities of success for full pulpotomy.

Table 17.6 Studies Investigating the Clinical/Radiographic Outcome of Partial Pulpotomy of Permanent Teeth

Author	No. of Teeth	Success (%)	Age	Pretreatment Status	Exposure Size	Hemostasis	Pulp Dressing Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Cvek, 1978	60	58 (96.7%)	Not given	Traumatic exposure Vital Bleeding wound	0.5–4.0 mm	Saline	1st visit: nonsetting CH 2nd visit: setting CH	1st visit: ZOE 2nd visit: Composite	EPT/asymptomatic/ intact PDL/ continued root development/ hard tissue barrier	14–60 months	Size and duration of exposure, root maturity did not affect outcome
Baratieri et al., 1989	26	26 (100%)	12–44	Exposure due to caries or caries removal Pulp tissue bleeding without signs of degeneration	Not given	Calcium hydroxide solution	CH powder then hard set CH cement	Zinc oxide cement	Asymptomatic/vital	1–2 years	
Fuks et al., 1993	44	35 (79.5%)	Not given	Traumatic exposure Vital Bleeding wound	Not given	Saline	CH	ZOE	Asymptomatic/ bridge/root development/ vital	0.5–4 years	
Mass and Zilberman, 1993	35	32 (91.4%)	7.5–25	Molar Deep caries Asymptomatic No apical pathosis	Less than 1–2 mm in diameter; 2–3 mm deep	Saline	CH	ZOE/amalgam or crown	Asymptomatic/ intact PDL/root development	1–2 years	

Table 17.6 Studies Investigating the Clinical/Radiographic Outcome of Partial Pulpotomy of Permanent Teeth—cont'd

Author	No. of Teeth	Success (%)	Age	Pretreatment Status	Exposure Size	Hemostasis	Pulp Dressing Material	Base/Restorative Material	Criteria for Success	Duration After Treatment	Notes
Mejare and Cvek, 1993	31 (2-step) 6 (1-step)	29 (93.5%) 4 (66.7%)	Not given	Carious exposure Asymptomatic No apical pathosis	N/A	Saline	CH	ZOE	Asymptomatic/ intact PDL/root development	24–140 months	
Barrieshi-Nusair and Qudeimat, 2006	28	21 (75.0%)	7.2–13.1	Carious exposure Molar Reversible pulpitis No apical pathosis	2–4 mm deep	Saline	MTA	GIC/amalgam or crown	Vital/asymptomatic/ intact PCL/root development	1–2 years	
Qudeimat et al., 2007*	23	21 (91.3%)	6.8–13.3	Carious exposure Molar	2–4 mm deep		CH	GIC/amalgam or crown	Absence of signs and symptoms/ intact PDL/root development	24.5–45.6 months	There was no significant difference in outcome between CH and MTA
	28	26 (92.9%)					MTA				
Bjørndal et al., 2010*	29	10 (34.5%)	25–38	Deep caries Only provoked pain Vital Expose during removal	Not given	Saline	CH	GIC	Vital/intact PDL	1 year	

CH, Calcium hydroxide; EPT, electric pulp test; GIC, glass ionomer; MTA, mineral trioxide aggregate; PDL, periodontal ligament; ZOE, zinc oxide eugenol.

NB: All the studies were case series except those labeled with *.

Qudeimat et al. (2007) is a randomized controlled trial comparing calcium hydroxide versus MTA as pulp dressing material.

Bjørndal et al. (2010) is a randomized controlled trial comparing the outcomes of 1-step versus stepwise indirect pulp capping for teeth without exposure, and direct pulp capping versus partial pulpotomy for teeth with exposure during caries removal.

Table 17.7 Studies Investigating the Clinical/Radiographic Outcome of Full Pulpotomy of Permanent Teeth

Author	No. of Teeth	Success (%)	Age	Pretreatment Status	Hemostasis	Capping Material	Base/ Restorative Material	Criteria for Success	Duration After Treatment	Notes
Masterson, 1966	30	25 (83.3%)	6–39	Not given	Not given	CH	Not given	Vital/asymptomatic	1–70 months	
Russo et al., 1982	30	28 (93.3%)	9–28	Carious exposure No apical pathosis	Not given	CH	Not given	Intact PDL	8 weeks	
Santini, 1983	373	192 (51.4%)	Not given	Carious exposure or near exposure Symptomatic	Cotton wool pellet	CH or CH + Ledermix	ZOE	Vital/bridge/ asymptomatic	6 months	Sex and medicament had no significant effect Poor healing was associated with age < 7.5
Caliskan, 1993	24	22 (91.7%)	10–22	Hyperplastic pulpitis	Saline	CH	ZOE/Amalgam or composite	Vital/asymptomatic/ dentine bridge/ intact PDL	1–4 years	
Caliskan, 1995	26	24 (92.3%)	10–24	Carious exposure Asymptomatic Apical pathosis	Saline	CH	ZOE	Asymptomatic/bridge/ root development/ vital	16–72 months	
Waly, 1995	20	18 (90.0%)		Carious exposure Molar		CH-glutaraldehyde CH	Not given	Not given	5 years	
Teixeira et al., 2001	41	34 (82.9%)	6–16	Deep caries or exposed pulp With or without apical pathosis	Not given	CH	GIC	Vital/asymptomatic/ dentine bridge/ intact PDL	24–32 weeks	
DeRosa, 2006	26	17 (65.4%)		Not given		CH	Amalgam	Asymptomatic	14–88 months	
El Meligy and Avery, 2006	15	13 (86.7%)	6–12	Carious or traumatized teeth Immature apex No apical pathosis	Saline	CH	ZOE/Amalgam or composite	Asymptomatic/intact PDL/no resorption/ root development	1 year	
	15	15 (100%)				MTA				
Witherspoon et al., 2006	19	15 (79.0%)	7–16	Carious or traumatic exposure Irreversible pulpitis	6% NaOCl	MTA	Not given	Vital/asymptomatic/ intact PDL/root development	1 year	
Asgary and Ehsani, 2009	12	12 (100%)	14–62	Carious Irreversible pulpitis	Saline	NEC	Permanent rest	Asymptomatic/intact PDL	13–20 months	

CH, Calcium hydroxide; GIC, glass ionomer; MTA, mineral trioxide aggregate; NaOCl, sodium hypochlorite; NEC, new endodontic cement.

NB: All the studies were case series except El Meligy and Avery (2006), which was a randomized controlled trial comparing calcium hydroxide and MTA as pulp dressing material.

by the authors revealed the following: the mean success rate was 83% when vital pulpectomy was performed (Fig. 17.17), which reduced to 72% when the root canal treatment procedure was used to eradicate the established infection associated with a periapical lesion (Fig. 17.18).

FACTORS AFFECTING PERIAPICAL HEALTH OR HEALING FOLLOWING ROOT CANAL TREATMENT

The factors influencing the maintenance of periapical health or periapical healing of preexisting lesions following root canal treatment may be broadly classified into patient factors (age, sex, general health, tooth anatomy, preoperative pulpal and periapical status), treatment factors (operator variables, canal enlargement, irrigation, medication, culture test, and obturation), and restorative factors. Some factors had a profound impact on success rates, whereas others showed a negligible effect. Patient factors characterizing the nature of the disease showed the most significant effect (periapical status), whereas most of the treatment factors were found to exert a less significant effect; exceptions were the apical extent of root canal treatment relative to the root canal terminus. In addition, the quality of the postoperative restorative care also exerted a profound influence on the outcome of treatment.

Patient Factors

Patient's age and sex consistently had no significant effect on outcome, whereas some specific health conditions (diabetes,^{54,67} compromised immune response¹²³) apparently had a significant influence. The evidence for the effect of host immune response characterized by the general health of the patient is, however, weak. Emerging evidence indicates that the host response measured by polymorphisms of various genes involved in periapical healing may have an effect on outcomes (Table 17.8).^{142,183,212,213} The importance of the host response to maintenance of periapical health or periapical healing was also supported by the statistically significant clustering effect of multiple teeth within the same patient in a prospective study.¹⁵⁰

The widespread perception that single-rooted teeth with less complicated anatomy should benefit with more predictable and favorable outcomes proves to be untrue. Having accounted for potential confounding factors such as the presence of periapical disease, tooth type does not seem to exert a strong influence on success rates. This would appear to be counterintuitive but may be explained by the logical inference that canal complexities in the apical anatomy probably play a more dominant role than other complexities, such as the number and curvature of canals.

The presence and size of a periapical lesion seem to have the most negative effect on periapical health/healing; it therefore follows that these factors must be accounted for when analyzing the influence of any other factor. The profound change in success rates once a periapical lesion becomes established is interesting, as it is correlated with the establishment of infection in the apical canal anatomy. This seems to suggest that once the apical canal complexities become infected, it may be much more difficult to eradicate the infection. The negative influence of large periapical lesions has a reasonable biologic explanation: the diversity of

bacteria (by number of species and their relative abundance) is greater in teeth with larger periapical lesions.²³⁰ An endodontic infection is more likely to persist in canals with a higher number of bacteria present preoperatively.²⁷ In addition, larger lesions may represent longer-standing root canal infections that may have penetrated deeper into dentinal tubules and accessory anatomy in the complex canal system²⁰⁶ where mechanical and chemical decontamination procedures may not be so effective. Larger lesions may also represent cystic transformation.¹⁴⁵ Finally, the host response may also play a part, as patients with larger lesions may innately respond less favorably to residual bacteria.¹⁴⁷ This speculation may crystallize into distinct questions for further biologic research into the nature of interactions among host, bacterial infection, and treatment intervention.

Most of the other investigated preoperative factors (pain, tooth tenderness to percussion, soft-tissue tenderness to palpation, soft-tissue swelling and sinus tract, periodontal probing defect of endodontic origin, root resorption) are in fact different clinical manifestations of periapical disease.²⁵⁴ They may therefore act as surrogate measures or complement "presence and size of periapical lesion" in measuring the effect of severity of periapical disease within a broad continuous spectrum. Of these, only presence of preoperative pain,⁶⁸ sinus tract,¹⁵⁰ swelling,¹⁵⁰ and apical resorption²²⁹ have been found to be significant prognostic factors that have been associated with significantly reduced success rates in root canal treatment.

The biologic explanation for the negative impact of sinus tract and swelling, either in the acute or chronic form, on periapical healing is interesting, as both represent suppuration and the proliferation of microbiota into the periapical tissues, with the inference being that the host tissues must have become locally overwhelmed. The precise reasons for reduced success rates under these conditions remain unclear but must somehow be related to the nature of the host-microbial interaction.

Treatment Factors

Operator. Although the impact of operator qualification and skill has not been specifically investigated, systematic review has shown that the involved clinicians may be grouped into undergraduate students, general practitioners, postgraduate students, and specialists. Studies show a clear trend in superior outcomes by greater experience and training. Clearly, technical skills play an important role, but this is difficult and often impossible to quantify. In addition, the technical abilities must be augmented by the overall understanding of the biologic issues and the quest for superior treatment by the operator. If clinicians do not feel that they can provide optimal treatment, it is incumbent upon them to refer the patient to a more qualified practitioner.

Isolation. The use of rubber dams in modern root canal treatment is widely accepted, and the justification seems almost empirical. One study on retreatment²⁴⁴ analyzed the influence of rubber dam use compared to cotton roll isolation and found significantly higher success rates with the former approach. Another study reported a significantly higher success rate of root canal-treated teeth⁷⁶ when a rubber dam was employed during post placement compared

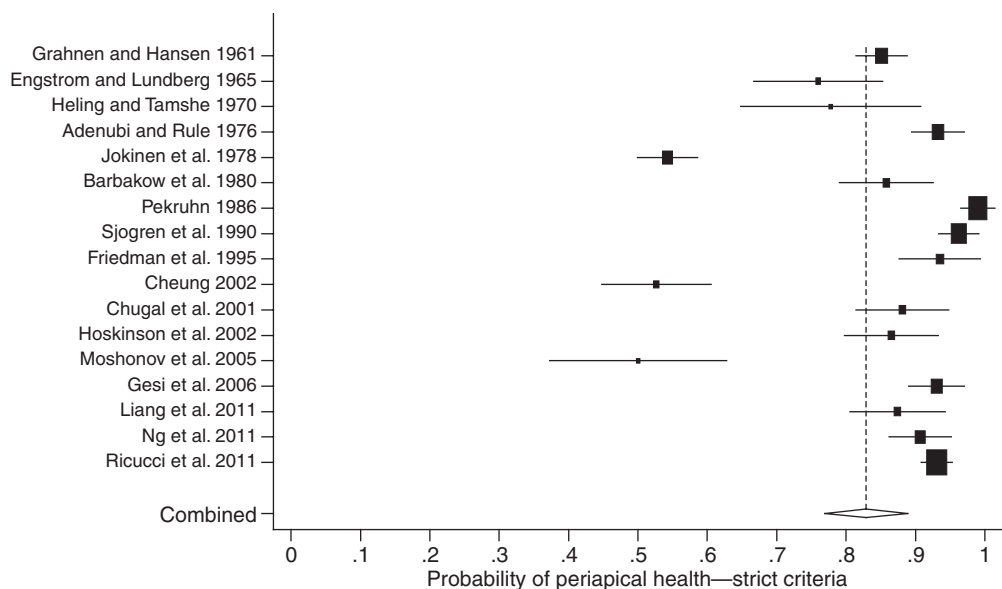


Fig. 17.17 Forest plot showing results of pooled and individual study's probability of maintained periapical health for preoperatively vital teeth undergoing root canal treatment (pooled probability = 0.83; 95% confidence interval: 0.77, 0.89).

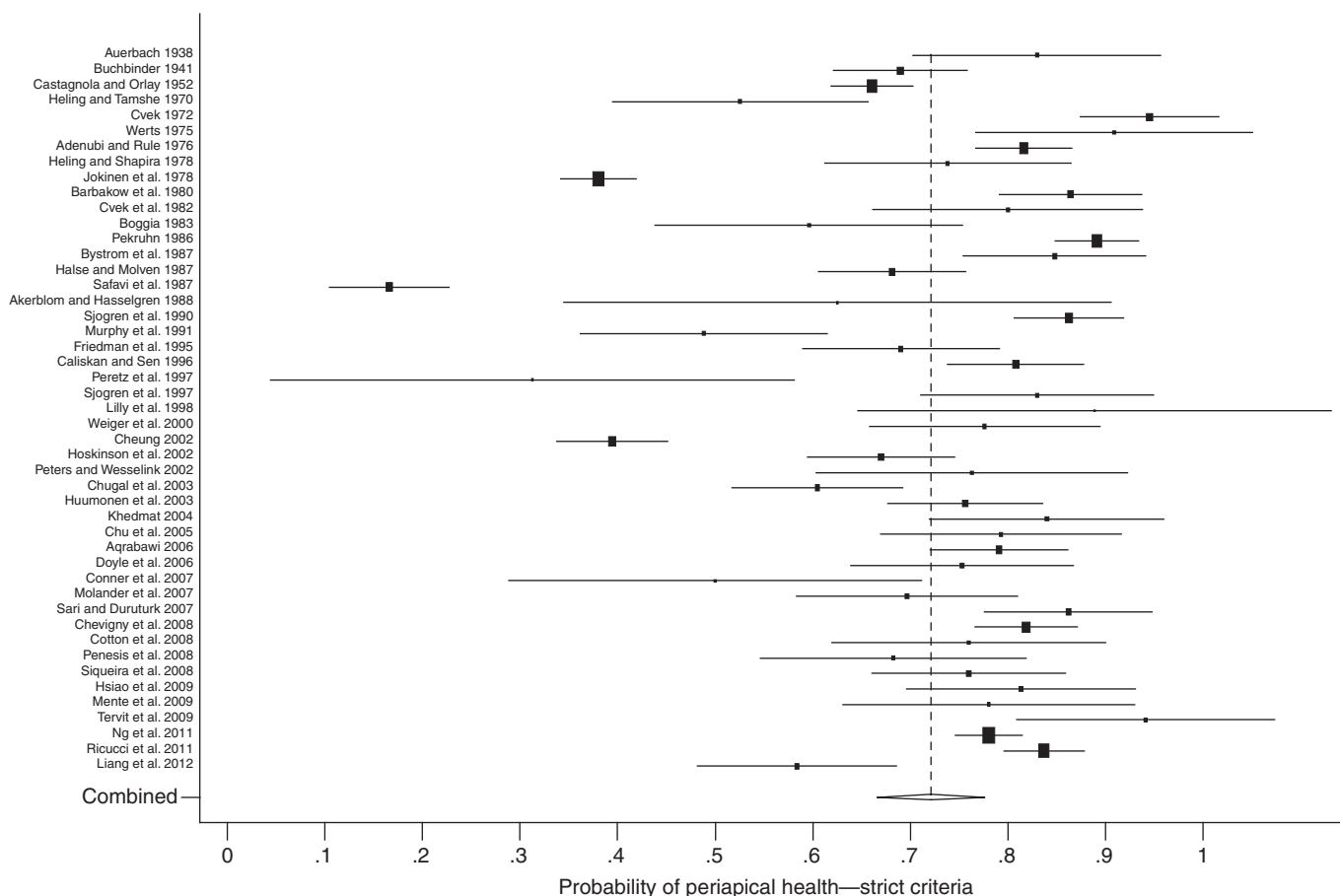


Fig. 17.18 Forest plot showing results of pooled and individual study's probability of periapical health for teeth associated with nonvital pulps and periapical radiolucencies undergoing root canal treatment (pooled probability = 0.72; 95% confidence interval: 0.67, 0.78).

Table 17.8 Type of Genes Investigated in Studies on SNPs and Periapical Healing

Study	Gene Function	Genes	Findings
Siqueira et al., 2009	Interleukin 1 Fcγ receptor	IL-1α, IL-1β FcγRIIa, FcγRIIIb	No significant association Significant association
Morsani et al., 2011	Interleukin	IL-1β	Significant association
Siqueira et al., 2011	Fcγ receptor	FcγRIIIa	No significant association
Rôças et al., 2014	Pattern recognition receptors	CD14, TLR4	No significant association

SNP, Single nucleotide polymorphism.

to when it was not. Perhaps as a consequence, the principal justification for rubber dam use is based on medicolegal implications of root canal instrument ingestion or aspiration by the patient.⁶¹

Magnification and Illumination. Endodontists have repeatedly reinforced the value of magnification and illumination during root canal treatment,¹⁶⁶ but a systematic review failed to draw any objective conclusions on their influence as no article was identified in the current literature that satisfied their inclusion criteria.⁴⁶ A prospective study investigated this factor,¹⁵⁰ but researchers found only an insignificant influence on the final outcome. Use of a microscope may sometimes assist location of the second mesiobuccal canal in maxillary molars, but this only made a small difference to the success rates associated with mesiobuccal roots, when a periapical lesion was present.¹⁵⁰ The true benefit of a microscope can only be verified through a randomized controlled trial. However, canal negotiation with less tooth structure removal and fewer procedural accidents is favorable and seems intuitively more consistent with the use of superior magnification and illumination.

Mechanical Preparation: Size, Taper, Extent, and Procedural Errors. The root canal system may be mechanically prepared to a requisite size and taper¹⁹⁹ using a variety of instruments of different cutting designs, tips, tapers, and materials of construction. Their efficacy is often tested in laboratory studies, and the instruments and their utility may have well-characterized properties.⁹⁵ Investigation of the influence of the type of instrument used for canal enlargement has been undertaken in one nonrandomized prospective study, but the outcome is likely subjective because of many factors, including the protocol adopted for teaching technical skills.¹⁵⁰ In this study, the better success rates for hand or rotary NiTi instruments compared with stainless steel instruments¹⁵⁰ were attributable to the fact that tactile skills training was achieved through a preliminary focus on the use of stainless steel files to develop tactile sensitivity and consistency. Only on demonstration of this competency did the trainees progress to NiTi instruments. More importantly, such senior students may also have had a better understanding of the biologic rationale for root canal treatment. The ability to gain and maintain apical patency as well as to avoid procedural errors was better instilled in the senior students, whereas in selected cases, NiTi instruments appear capable of achieving the same in primary root canal treatment undertaken by undergraduates.¹⁷⁵

A key tenet of the ESE⁶¹ guidelines is that root canal debridement must be extended to the terminus of the canal

system, which is expressed variously as extension to the “apical constriction,” or to “0.5 to 2 mm from the radiographic apex,” or to the “cementodentinal junction.” This guideline is broadly supported by the fact that outcome of treatment is compromised by canal obstruction or failure to achieve patency to the canal terminus.^{150,216,229} Ng and colleagues¹⁵⁰ reported a twofold reduction in the success of treatment when the patency to the canal terminus was not achieved. It could be speculated that the lack of mechanical negotiability of canals may be due to the presence of obstructions caused by “denticles,” tertiary dentine, acute branching or a fine plexus of apical canals, or dentine/organic debris.

The continued debate on the optimal size of apical preparation remains topical in the absence of definitive evidence; the findings from relevant in vitro and clinical studies have been previously reviewed.¹⁵ So far, four clinical outcome studies have considered this issue or have systematically investigated the effect of apical size of canal preparation on treatment outcome.^{93,102,150,188,223,229} One randomized controlled trial revealed that enlargement of the canal to three sizes larger than the first apical binding file was adequate¹⁸⁸ (the mean final size was ISO #30). The observational studies^{95,150,229} had not designed their investigation with apical canal size as their principal focus and neither had they found a statistically significant influence from this factor; nevertheless, they all reported the same inverse trend of decreasing success rates with an increase in size of apical preparation. It was speculated that canal preparation to larger apical sizes may compromise treatment success by generation of more apical dentine debris, which in the absence of an adequate irrigation regimen serves to block apical canal exits that may still be contaminated with bacteria. Continued generation of dentine debris, in the absence of sufficient irrigation, may lead to what is termed *dentine mud*, which ultimately creates a blockage. The impatient or neophyte clinician fails to resist the temptation to force the instrument back to length, resulting in the classically described procedural errors of apical transportation, canal straightening, and perforation. An alternative mechanism is required to explain the higher failures in initially large canals; it is likely that immature roots present a different debridement challenge, where the canal shape is not amenable to planing of the main portions of the canal by conventional instruments. Perhaps an intracanal brush may be a more suitable cleaning device in such teeth. The findings from these studies therefore do not concur with views that more effective bacterial debridement may be achieved with larger apical preparations.^{33,164,184}

The issue of apical preparation size should be considered together with that of the size and taper of the rest of the

canal preparation. Again, there is a paucity of sufficient direct evidence for the influence of degree of canal taper on root canal treatment outcome. The ESE guidelines⁶³ recommend only that canal preparation should be tapered from crown to apex without stipulating any particular degree of taper. Three studies have analyzed the influence of canal preparation taper on primary treatment and retreatment outcome, although again, none had focused their investigation primarily on this factor.^{93,150,220} Smith and coworkers,²²⁰ using loose criteria for determination of success, found that a “flared” preparation (wide taper) resulted in a significantly higher success rate compared with a “conical” preparation (narrow taper); the exact degree of taper was not reported and the effects of other treatment and non-treatment parameters were not controlled. In contrast, Hoskinson and associates⁹³ and Ng and colleagues,¹⁵⁰ using strict criteria, did not find any significant difference in treatment outcome between narrow (0.05) and wide (0.10) canal tapers. The controlled use of stainless steel instruments in a step-back technique may create 0.05 (1 mm step-back) or 0.10 (0.5 mm step-back) tapers, although, of course, uncontrolled use of such instruments may generate a variety of shapes. Ng and colleagues¹⁵⁰ also compared these (0.05 and 0.10) preparation tapers with 0.02, 0.04, 0.06, and 0.08 tapers (generally achieved by using greater taper nickel-titanium instruments) and found no significant effect on treatment outcome. They cautioned that their investigation of the influence of canal preparation taper without randomization could be influenced by the initial size of the canal, the type of instrument used, and operator experience.

Triangulation of the data on the effects of canal preparation size and taper on treatment outcome may intuitively lead to the conclusion that as far as current best evidence indicates, it is not necessary to overenlarge the canal to achieve periapical healing. An apical preparation size of ISO 30 with a 0.05 taper for stainless steel instrumentation or 0.06 taper for NiTi instrumentation is sufficient. Precisely what biologic and hydrodynamic mechanisms underpin such sufficiency is more difficult to define based on available evidence. Although a number of laboratory studies^{4,84,112} have investigated the interaction between canal dimensions and irrigation or obturation dynamics, the precise physical, chemical, or biologic mechanisms that ultimately enable periapical healing remain unknown, although collaborations with fluid dynamics specialists⁸⁴ and (micro)biologists⁸² may ultimately yield a clearer picture.

Procedural errors during root canal preparation include canal blockage, ledge formation, apical zipping and transportation, straightening of canal curvature, tooth or root perforation at the pulp chamber or radicular level, and separation of instruments. Instrument separation during treatment has been found to reduce the success rate significantly^{150,229}; however, the reported prevalence of instrument separation was low (0.5% to 0.9%) in these studies, precluding an analysis of causative factors. A case-control study²²⁵ revealed no significant difference in success rates between periapically involved teeth with or without retained separated instruments. The stage of canal debridement at which instrument separation occurred and the justification for their retention may have implications on the outcome. The coronapical location of a separated instrument and whether the instrument was successfully bypassed were found to have no effect on treatment outcome.

Irrigant. Different chemical agents have been used as irrigants for root canal treatment, singly or in various combinations, both in clinical practice and in the studies reviewed. They have included water, saline and solutions of local anesthetic, sodium hypochlorite, iodine, chloramine, sulfuric acid, EDTA, hydrogen peroxide, organic acid, Savlon, urea peroxide, and Biosept (a quaternary ammonium compound).¹⁵³ Most of the studies had used sodium hypochlorite as an irrigant,¹⁵³ regardless of whether it was primary treatment or retreatment. This is consistent with the ESE guidelines⁶¹ for irrigation, which recommend a solution possessing disinfectant and tissue-dissolving properties.

One prospective study¹⁵⁰ systematically investigated the effect of the irrigant on the success rates of root canal retreatment, which, although not a randomized controlled trial, revealed interesting new findings on the effects of irrigants. Even though a higher concentration of sodium hypochlorite made negligible difference to treatment outcome, the additional use of other specific irrigants had a significant influence on success rates.¹⁵⁰ The finding of a lack of improvement in periapical healing with the use of a higher concentration NaOCl solution is consistent with previous clinical/microbiologic findings.^{29,41} Comparing 0.5% to 5.0% NaOCl solution for irrigation, it was found that the concentration of solution, per se, did not appear to increase the proportion of teeth rendered culture-negative²⁹ or associated with greater periapical healing.⁴¹ As iodine and sodium hypochlorite are both halogen-releasing agents and attack common key protein groups,¹²⁹ the finding that the additional use of 10% povidone-iodine for irrigation had no additional influence on treatment success was as expected. Surprisingly, however, the additional use of 0.2% chlorhexidine solution for irrigation was found to reduce the success of treatment significantly.¹⁵⁰ This finding was in complete contrast to previous reports^{211,252} on its equivalent or superior in vivo antibacterial efficacy when compared with sodium hypochlorite solution. The use of chlorhexidine as a final irrigant following sodium hypochlorite irrigation had been recommended¹⁰⁷ and was justified on several grounds, including its substantivity in root dentin (i.e., prolonged antibacterial effect),¹⁸⁵ relative lack of toxicity,¹¹⁷ and broad-spectrum efficacy.¹³⁰ Not until recently has alternate irrigation with sodium hypochlorite and chlorhexidine solution raised serious concerns because of their interaction product. The interaction product is thought to be an insoluble precipitate containing parachloroaniline, which is cytotoxic and carcinogenic.^{14,24} Apart from mutually depleting the active moiety in the two solutions for bacterial inactivation, the precipitate may cause persistent irritation to the periapical tissue and block dentinal tubules and accessory anatomy, possibly explaining the observed lower success rate when chlorhexidine was used as an additional irrigant.

Ng and associates¹⁵⁰ also found that the additional use of EDTA had a profound effect on improving radiographically observed periapical healing associated with root canal treatment (OR = 1.5 [1.1, 2.0]). In contrast, the observed synergistic effect of sodium hypochlorite and EDTA had been previously demonstrated in terms of bacterial load reduction²⁶ but not periapical healing. The long-term (≥ 2 years) outcome of their cases stratified by canal disinfection protocols²⁵ did not support their microbiologic findings. Their reported success rate for alternate irrigation

with sodium hypochlorite and EDTA solutions (67%) was low when compared to the success rate for irrigation using saline (91%), 0.5% sodium hypochlorite (92%), or 5% sodium hypochlorite (86%) solutions.¹⁶² The reported outcome data were unexpected, as preobturation negative bacterial culture was achieved in all cases. Given the complexity of their study design (clinical and microbiologic), their sample size was restricted to 11 to 15 teeth per group, limiting their outcome data. The synergistic effect of the two disinfectants has been attributed to the chelating properties of the sodium salts of EDTA, and their roles have been reviewed by Zehnder.²⁶¹ EDTA solution assists negotiation of narrow or sclerosed canals by demineralization of root dentine and helps in the removal of compacted debris from noninstrumented canal anatomy. It may also facilitate deeper penetration of sodium hypochlorite solution into dentine by opening dentinal tubules and removing the smear layer from the instrumented surface. Lastly, it may help detach or break up biofilms adhering to root canal walls.⁸⁵

Medicament. Most previous treatment outcome studies have not standardized the type of root canal medicament used in the interappointment period, but the use of several different medicaments has been reported. The list was consistent with that recommended in the ESE guidelines for a medicament with disinfectant properties and included calcium hydroxide, creosote, and iodine solutions.¹⁵³ However, there is an absence of studies investigating the influence of this factor on treatment outcome.

The use of a mixture of calcium hydroxide and chlorhexidine has been tested based on the speculation that the mixture would be more effective against *E. faecalis*.^{13,78,198}

Root Canal Bacterial Culture Results Prior to Obturation. In the past, in various centers of endodontic excellence, completion of root canal treatment by obturation would only be acceptable after a negative culture test was obtained from the canal, confirming the absence of bacteria in the part of the root canal system that could be sampled.^{23,70,143} This practice has fallen out of clinical favor because of the perceived predictability and good prognosis of root canal treatment without microbiologic sampling. Sampling procedures are considered to be lengthy, difficult, and often inaccurate, requiring laboratory support and having low benefit-to-cost ratio.^{138,139} A preobturation negative culture result may increase treatment success twofold (Fig. 17.19). One large study²⁰¹ helped contribute to the demise of the canal-culture test; however, even this study showed a 10% difference in success in favor of the negative culture test when periapical disease was present. The outcome is even worse when a positive culture test result combines with the presence of a periapical lesion.

Numerous studies^{2,7,10,16,26-29,33,37,41,42,60,77,79,99,104,108,110,124,137,154,161,163,168,169,173,179,181,208,209,210,214,215,217,227,248,252,257,258} have evaluated the effect of different stages of root canal treatment on the intraradicular microbiota, both qualitatively and quantitatively (Table 17.9). Some studies merely report positive culture tests, whereas others have

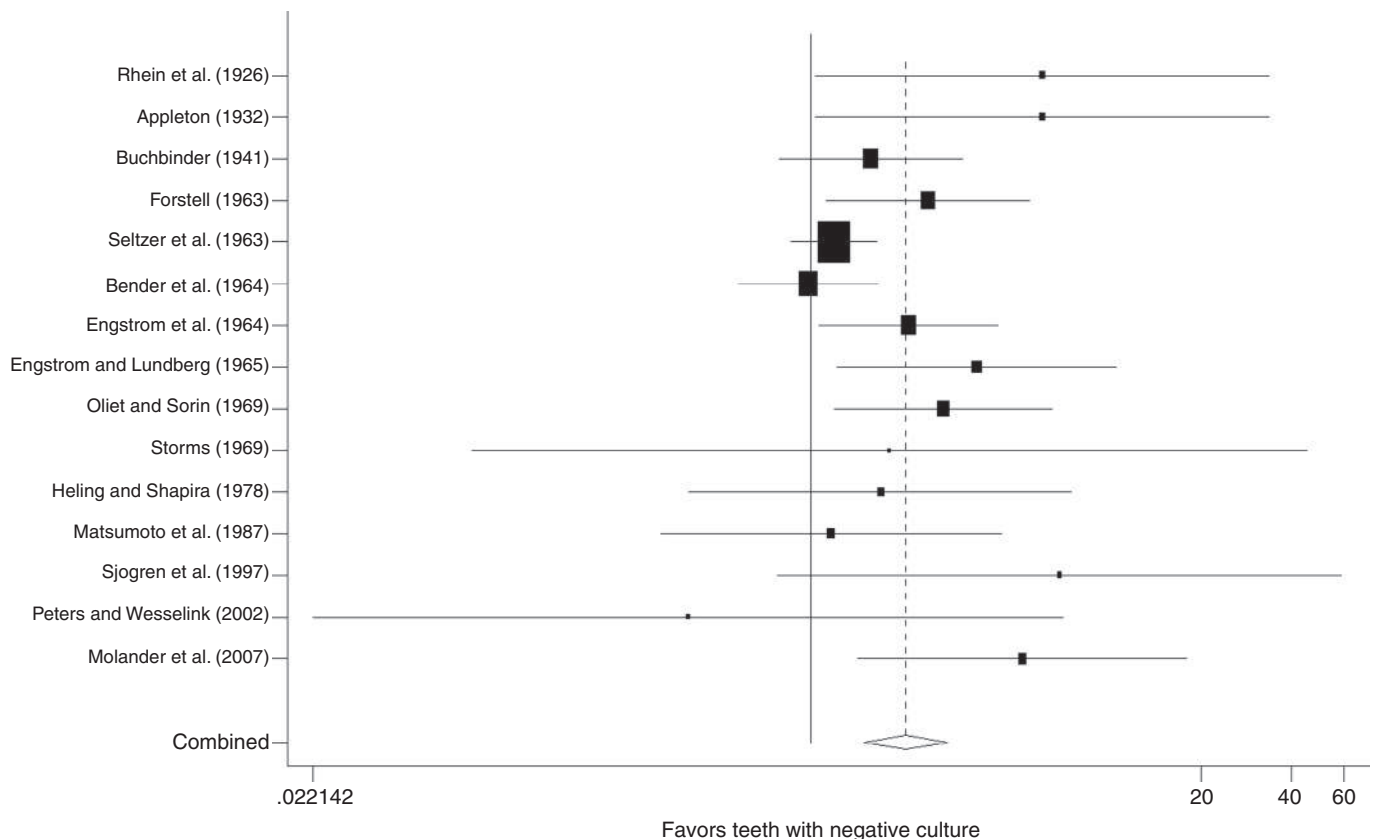


Fig. 17.19 Forest plot showing pooled and individual study's odds ratio (OR) for periapical health of teeth undergoing root canal treatment with preobturation negative versus positive culture test results (pooled OR = 2.1; 95% CI: 1.5, 2.9).

Table 17.9 Summary of Studies Evaluating the Effect of Root Canal Treatment Procedures on Bacterial Presence by Culture

Study	Year	Sample Size	PERCENTAGE OF SAMPLES WITH BACTERIAL PRESENCE		
			At Baseline	After Preparation ± Irrigation	Next Visit (After Dressing±)
Auerbach	1953	60 teeth	93% (56/60)	Chlorinated soda (double strength): 22% (12/56)	—
Ingle and Zeldow	1958	89 teeth	73% (65/89)	H₂O: 70% (62/89) Some initially –ve became +ve after treatment	—
Stewart et al.	1961	77 teeth	100% (77/77)	0.5% NaOCl + Gly-oxide: 2% (1/44) 0.5% NaOCl + 3% H₂O₂: 9% (3/33)	No dressing: 0.5% NaOCl + Gly-oxide: 34% (15/44) 0.5% NaOCl + 3% H ₂ O ₂ : 39% (17/33)
Nicholls	1962	155 teeth	100% (155/155)	Alkaline chloramine: 53% (39/74) H₂O₂ and 2% NaOCl: 50% (30/60) H₂O and 2% NaOCl: 71% (15/21)	—
Grahnén and Krasse	1963	97 teeth	77% (75/97)	NaCl: 72% (23/32) Biocept: 66% (21/32) Nebacin: 36% (12/33) Some initially –ve became +ve after Tx	No dressing: NaCl: 47% (15/32) Biocept: 47% (15/32) Nebacin: 18% (6/33)
Engström	1964	223 teeth (untreated or retreated)	60% (134/223)	Biosept or Iodophor, plus alcohol, chloroform, and 0.5% NaOCl: No data	5% I₂ in 10% IKI: 2nd visit: 43% (58/134); 3rd visit: 22% (29/134); 4th visit: 8% (9/134); 5th visit: 3% (4/134); 6th visit: 2% (3/134); 7th visit: 16% (22/134)
Olgart	1969	207 teeth	72% (149/207)	H₂O₂ and 0.5% NaOCl or H₂O₂ and 1% NaOCl: 43% (88/207)	No dressing: 34% (70/207)
Bence et al.	1973	33 teeth	100% (33/33)	Pre-irritation: 1st file: 93%, enlargement with #3: 14%, #4: 11%, #5: 21% (32% of instruments showed +ve culture, regardless of size) 5.25% NaOCl: 48-h culture: 4% dentine, 10% pp 5-day culture: 8% dentine, 26% pp	No dressing: 8% dentine, 12% pp samples of teeth with negative culture after irrigation
Akpata	1976	20 extracted teeth	100% (20/20)	NaCl: 65% (13/20)	38% CMCP: 20% (2/10) When PP sample –ve, crushed tooth yielded –ve culture; When PP +ve, crushed teeth yielded +ve or –ve cultures
Cvek et al.	1976	108 teeth	NaCl group: 53% (18/34) 0.5% NaOCl group: 63% (29/46) 5% NaOCl group: 79% (22/28)	NaCl: 83% (15/18) 0.5% NaOCl: 59% (17/29) 5% NaOCl: 68% (15/22)	—
Byström and Sundqvist	1981	15 teeth	100% (15/15)	Saline: 100% (15/15)	No dressing: 47% (7/15) (5th visit) Where initial bacteria load high, difficult to eliminate
Byström and Sundqvist	1983	15 teeth	100% (15/15)	0.5% NaOCl: 87% (13/15)	No dressing: 20% (3/15) (5th visit)
Byström and Sundqvist	1985	60 teeth	100% (60/60)	0.5% NaOCl: No data 5% NaOCl: No data 5% NaOCl + 15% EDTA: No data	No dressing: 0.5% NaOCl: 12/20 (2nd visit); 8/20 (3rd visit) 5% NaOCl: 10/20 (2nd visit); 6/20 (3rd visit) 5% NaOCl + 15% EDTA: 11/20 (2nd visit); 3/20 (3rd visit)
Byström et al.	1985	65 teeth	100% (65/65)	0.5% NaOCl: No data 5.0% NaOCl: No data	CH: 0/35 (1 month), 1/35 (2–4 days) CP/CMCP (2 weeks): 10/30

Table 17.9 Summary of Studies Evaluating the Effect of Root Canal Treatment Procedures on Bacterial Presence by Culture—cont'd

Study	Year	Sample Size	PERCENTAGE OF SAMPLES WITH BACTERIAL PRESENCE		
			At Baseline	After Preparation ± Irrigation	Next Visit (After Dressing±)
Sjogren and Sundqvist	1987	31 teeth	100% (31/31)	0.5% NaOCl plus ultrasonic debridement: No data	No dressing: 29% (9/31) at 2nd visit; 23% (7/31) at 3rd visit
Koontongkaew et al.	1988	15 teeth	100% (15/15)	3% H₂O₂/5.25% NaOCl: No data	CMCP: 1-day dressing: 40% (2/5); 3-day dressing: 20% (1/5); 7-day dressing 10% (1/10) No dressing: 60% (3/5) after 1 day, 20% (1/5) after 3 or 7 days
Reit and Dahlen	1988	35 teeth	91% (32/35)	0.5% NaOCl: No data	CH: After 14 days: 23% (8/35); after 21 days: 26% (9/35)
Molander et al.	1990	25 teeth	96% (24/25)	0.04% iodine: No data	Clindamycin: After 14 days: 16% (4/25); after 21 days: 24% (6/25)
Sjogren et al.	1991	30 teeth	100% (30/30)	0.5% NaOCl: 50% (15/30)	CH: 10 min: 50% (6/12) at 1 week later 7 day: 0% (0/18) (none after 1–5 weeks later without dressing)
Orstavik et al.	1991	23 teeth	96% (22/23)	NaCl irrigation and enlarged to: #20–25: 87% (20/23) further to #35–80: No data	CH: 34% (8/23); #35/40: 40% (6/15) #>40: 25% (2/8)
Yared and Bou Dagher	1994	60 teeth	100% (60/60)	1% NaOCl: Enlarged to #25: 73% (22/30) Enlarged to #40: 23% (7/30)	CH: 0% (0/60)
Gomes et al.	1996	42 root canals: Untreated (<i>n</i> = 15) Retreated (<i>n</i> = 27)	95% (40/42)	2.5% NaOCl: No data	Empty canal (7–10 days): 73% (29/40)
Sjogren et al.	1997	55 teeth (single canal)	100% (55/55)	0.5% NaOCl: 40% (22/55)	—
Dalton et al.	1998	46 teeth	100% (46/46)	NaCl + NiTi files: 68% (15/22); NaCl + K-files: 75% (18/24)	—
Reit et al.	1999	50 teeth	84% (42/50)	Enlarged to #35 (curved) or #50 (straight) with 0.5% NaOCl: No data	5% IKI (5–7 days): 44% (22/50) Empty (7 days): 44% (22/50)
Peciulienė et al.	2000	25 teeth	80% (20/25);	2.5% NaOCl and 17% EDTA: No data	Medication unknown: 28% (7/25)
Shuping et al.	2000	42 teeth	98% (41/42)	1.25% NaOCl: 38% (16/42)	CH: 8% (3/40)
Lana et al.	2001	31 teeth	87% (27/31)	2.5% NaOCl: No data	CH: 13% (4/31) Empty for 7 days: 23% (7/31)
Peciulienė et al.	2001	40 teeth	83% (33/40)	2.5% NaOCl and 17% EDTA: 30% (10/33)	CH (10–14 days): 25% (5/20) IKI: 2% I₂ in 4% KI (10 min): 5% (1/20)
Peters et al.	2002	42 teeth	Instrumentation to #20: 100% (42/42)	Enlarged to #35 with 2% NaOCl: 23% (10/42)	CH (4 weeks): 71% (15/21); further irrigation: 43% (9/21)
Card et al.	2002	40 mandibular teeth/canals	95% (38/40)	1% NaOCl Profile instrumentation (0.04 taper): 0/13 of cuspids and bicusps, 5/27 of mesiobuccal canals Further Lightspeed instrumentation to size 57.5–65: 3/27 mesiobuccal canals of molars Only 1/16 of those mesiobuccal canals with detectable communication with the mesiolingual canals had +ve culture after the first preparation using ProFile instruments	No data

Continued

Table 17.9 Summary of Studies Evaluating the Effect of Root Canal Treatment Procedures on Bacterial Presence by Culture—cont'd

Study	Year	Sample Size	PERCENTAGE OF SAMPLES WITH BACTERIAL PRESENCE		
			At Baseline	After Preparation ± Irrigation	Next Visit (After Dressing±)
Kvist et al.	2004	96 teeth	98% (94/96)	0.5% NaOCl: 63% (60/96)	CH (7 days): 36% (16/44) IPI (10 min): 29% (15/52)
Chu et al.	2006	88 canals	99% (87/88)	0.5% NaOCl: No data	CH, Septomixine forte, or Ledermix: 36% (32/88) Exposure of pulp, tooth type, acute versus chronic condition, size of lesion, and type of medication had no significant effect
Paquette et al.	2007	22 teeth (single canal)	100% (22/22)	2.5% NaOCl: 68% (15/22)	2% CHX: 45% (10/22)
Siqueira et al.	2007a	11 teeth (single root)	100% (11/11)	2.5% NaOCl: 55% (6/11)	CH/CPMC: 9% (1/11)
Siqueira et al.	2007b	11 teeth (single root)	100% (11/11)	2.5% NaOCl: 45% (5/11)	CH: 18% (2/11)
Vianna et al.	2007	24 teeth (single root)	100% (24/24)	Saline + 2% CHX gel: 33% (8/24)	2% CHX, CH or mixture: 54% (13/24) Type of medication had no significant effect
Wang et al.	2007	43 canals	91% (39/43)	Saline + 2% CHX gel: 8% (4/39)	2% CHX + CH: 8% (3/36) Size of apical preparation (40 versus 60) had no significant effect.
Markvart et al.	2012	24 teeth	88% (21/24)	2.5% NaOCl: 63% (15/24)	17% EDTA irrigation and 10 min 5% IKI medication: 50% (12/24) Box preparation (#60): 67% (8/12) Cone preparation (#25–30): 33% (4/12)
Xavier et al.	2013	48 teeth (single canal)	100% (40/40)	1% NaOCl: 75% (9/12) 2% CHX: 75% (9/12)	CH: 75% (18/24)

CH, Calcium hydroxide; CP, camphorated phenol; CMCP, camphorated monochlorophenol; NaOCl, sodium hypochlorite; PP, paper point sample; r culture, culture test before obturation.

identified and quantified intraradicular microbiota before and after various stages of treatment.

The effect of the “mechanical preparation” of the canal(s) on the microbiota has been tested using only water or saline as the irrigant. Taken collectively, the studies show that negative cultures were achieved on a weighted pooled average in 31% of the cases (range 0% to 79%). When sodium hypochlorite (concentration range 0.5% to 5%) irrigation supplemented the “mechanical preparation,” the frequency of negative cultures immediately increased to a weighted pooled average of 52% (range 13% to 95%) (see [Table 17.9](#)).

Most studies report culture reversals during the interappointment period when active antibacterial dressing is not used in the root canal system between appointments. The reversals are due to regrowth of residual bacteria or recontamination by bacteria from coronal restoration leakage. When active interappointment antibacterial dressing is used, negative cultures in the subsequent visit were achieved on average in 71% of cases (range 25% to 100%) (see [Table 17.9](#)).

Effect of Persistent Bacteria on Root Canal Treatment Outcome. The bacteria present in preobturation

cultures have included *Enterococcus*, *Streptococcus*, *Staphylococcus*, *Lactobacillus*, *Veillonella*, *Pseudomonas*, *Fusobacterium* species, and yeasts. Studies have been variable as to the relationship between individual species and treatment failure. Although the overall failure rate for cases with positive cultures was 31%, teeth testing positive for *Enterococcus* species had a failure rate of 55%, and teeth with positive cultures for *Streptococcus* species had 90% failures.⁷⁰ In another study, good-quality root canal treatment on 54 teeth with asymptomatic periapical disease gave an overall success rate of 74%, but teeth with positive cultures for *Enterococcus faecalis* achieved a success rate of only 66%.²³¹ The success rate for teeth with no bacteria was 80%, whereas that for teeth with bacteria in the canal before obturation was 33%. These associations cannot be regarded as direct cause-and-effect associations, but they further emphasize the need to determine a relationship between microbial diversity and treatment outcome.

A monkey-model study⁶³ used a four- or five-strain infection to test the effect of debridement and obturation procedures on outcome. When bacteria remained after chemomechanical debridement, 79% of the root canals

were associated with nonhealed periapical lesions, compared with 28% when no bacteria were found to remain. Combinations of several residual bacterial species were more frequently related to nonhealed lesions than were single strains. When no bacteria remained at the end of chemomechanical debridement, healing occurred independently of the quality of the root filling. In contrast, when bacteria remained in the canal system, there was a greater correlation with nonhealing associated with poor-quality root fillings than in technically well-performed fillings. In root canals where bacteria were found after removal of the root filling, 97% had not healed, compared with 18% for those root canal systems with no bacteria detected upon removal of root filling. The study emphasizes the importance of reducing bacteria below detection limits before permanent root filling in order to achieve optimal healing conditions for the periapical tissues. It also reinforces the view that obturation does indeed play a role when residual infection is present.

Regardless of the technique used for obtaining a sample for a canal culture, the presence of a negative culture seems to have a positive impact on treatment outcome. The association of specific species with treatment failure is not well established but the identity of the small group of species isolated from positive cultures is relatively constant and may hold answers to treatment resistance and failure. However, it is important to understand that there are many other factors that can influence root canal treatment outcome.

Root Filling Material and Technique. The interrelationship among the core root filling material, sealer (for filling the gaps between the core material and canal surface), and technique for their placement complicates the investigation of the effect of obturation and technique on treatment outcome. In previous studies on treatment outcome, the most commonly used core root filling material was gutta-percha with various types of sealer or gutta-percha softened in chloroform (chloropercha).¹⁵³ The sealers used may be classified into zinc oxide–eugenol–based, glass ionomer–based, and resin–based types.¹⁵³ Materials such as Resilon, SmartSeal, and MTA have been adopted but have not significantly penetrated the market, except for the use of MTA in surgical repairs or repairs for immature apices. In any case, there is no evidence to show that the nature of root filling material and the technique used for placement has any significant influence on treatment outcome.

Apical Extent of Root Filling. Of the many intraoperative factors associated with success and failure of root canal treatment, the apical extent of the root canal filling material has been the most frequently and thoroughly investigated. In these previous studies, the apical extent of root fillings has been classified into three categories for statistical analyses: more than 2 mm short of radiographic apex (short), 0 to 2 mm within the radiographic apex (flush), and extended beyond the radiographic apex (long).¹⁵³ The apical extent of root filling was found to have a significant influence on the success rates of treatment, regardless of the periapical status.^{150,153} Flush root fillings were associated with the highest success rates, whereas long root fillings were associated with the lowest success rates.

Most previous retrospective studies could not distinguish between the effects of apical extent of instrumentation versus the apical extent of obturation; however, the London Eastman study¹⁵⁰ was able to separate the effect of these two factors and found them both to independently and significantly affect periapical healing. The factors correlated with each other, consistent with the fact that canals are normally filled to the same extent as canal preparation.

Extrusion of cleaning, medication, or filling materials beyond the apical terminus into the surrounding tissues may result in delayed healing or even treatment failure due to a foreign body reaction.^{105,146,218,260} Magnesium and silicon from the talc-contaminated extruded gutta-percha were found to induce a foreign body reaction, resulting in treatment failure.¹⁴⁶ An animal study has shown that large pieces of subcutaneously implanted gutta-percha in guinea pigs were well encapsulated in collagenous capsules, but fine particles of gutta-percha induced an intense, localized tissue response.²¹⁸ The inference that perhaps extrusion of large pieces of gutta-percha may not impact periapical healing was not supported by data from previous studies.^{150,153} The discrepancy may possibly be accounted for by bacterial contamination of the extruded gutta-percha in the clinical data.

The radiographic evidence of “sealer puffs” extruding through the main apical foramina and lateral/accessory canals has been pursued with vigor by some endodontists based on the undaunted belief of its value as “good practice.” Their perception is that this represents a measure of root canal system cleanliness, and they ardently argue that healing would follow, albeit with some delay. The published evidence on the effects of sealer extrusion into the periapical tissues has been contradictory. Friedman and colleagues⁶⁸ found that extrusion of a glass ionomer–based sealer significantly reduced success rates. In contrast, Ng and associates¹⁵⁰ reported that extrusion of a zinc oxide–eugenol–based sealer had no significant effect on periapical healing. The discrepancy may be attributed to the difference in sealer type and the duration of treatment follow-up. The radiographic assessment of the presence or resorption of sealer may be complicated by the radiolucent property of its basic components and the insufficient sensitivity of the radiographic method used to detect small traces of it.¹⁵⁰ It is possible that, in some cases, the radiographic disappearance of extruded sealer may simply be due to resorption of the radio opaque additive, barium sulfate, or its uptake by macrophages, still resident in the vicinity.¹⁴⁶

Extruded glass ionomer–based⁶⁸ zinc oxide–eugenol–based,⁹⁸ silicone–based⁹⁸ sealers or endomethasone²² were not found to be resorbed/absorbed by periapical tissues after 1 year. Traces of calcium hydroxide–based sealer (Sealapex) could still be detected after 3 years.¹⁹⁴ In the latter study, treatments were carried out on primary molar teeth and the canals were obturated with Sealapex without gutta-percha. With longer duration of follow-up, complete resorption of extruded zinc oxide–eugenol–based sealers (Procosol, Roth Elite)⁸ and a resin–based sealer (AH Plus, Dentsply/DeTrey, Konstanz, Germany)¹⁹⁴ was demonstrated in 69% and 45% of the cases after 4 and 5 years, respectively. Ng and associates¹⁵⁰ advanced two explanations for the difference between the effect of extruded core gutta-percha and the zinc oxide–eugenol sealer: the latter is

antibacterial and may kill residual microorganisms, whereas it is also more soluble and readily removed by host cells compared to gutta-percha.

Quality of Root Filling. Another much investigated parameter of obturation in retrospective studies has been the radiographic measure of the “quality of root filling.” The rationale for complete obturation of the root canal system is to prevent recontamination by colonization from the residual infection or newly invading bacteria. Both are supposedly prevented by a “tight” seal with the canal wall and an absence of voids within the body of the material. Quality of root filling may therefore be regarded either as poor root-filling technique or as a surrogate measure of the quality of the entire root canal treatment, because good obturation relies on properly executed preliminary steps in canal preparation. A systematic review¹⁵³ reported that the criteria for judging the quality of root fillings have not been well defined in previous studies.^{43,93,216} An unsatisfactory root filling has been defined as “inadequate seal,” “poor apical seal,” or “radiographic presence of voids.” Nevertheless, satisfactory root fillings were found to be associated with significantly higher success rates than unsatisfactory root fillings.¹⁵³

Acute Exacerbation During Treatment. The causes for interappointment “flare-up” or pain have not been precisely determined, and several hypothetical mechanisms involving chemical, mechanical, or microbial injury to the periradicular tissues, as well as psychological influences, have been suggested as contributory to postpreparation pain or swelling.^{202,203} Although these factors have not been specifically studied in the context of periapical healing, acute “flare-ups” during treatment were not found to be significantly associated with periapical healing in two studies.^{102,216} In contrast, the London Eastman study¹⁵⁰ found that pain or swelling occurred in 15% of cases after chemomechanical debridement and was found to significantly reduce success as measured by periapical healing. This interesting finding may be explained by the hypothesis that “flare-ups” were caused by extrusion of contaminated material during canal preparation. Such material may elicit a foreign body reaction or (transient) extraradicular infection, resulting in treatment failure in a proportion of such cases. Alternatively, acute symptoms may be the result of incomplete chemomechanical debridement at the first appointment, leading to a shift in canal microbial ecology favoring the growth of more virulent microorganisms, leading to further postpreparation pain and treatment failure. The exact biologic mechanisms of failure in these cases remain obscure and warrant further investigation.

Number of Treatment Visits. The number of treatment visits for completing root canal treatment and its effect on periapical healing remains an ongoing controversy. Generally, the argument for single-visit treatment centers around better patient acceptability and cost-effectiveness versus the preference of multiple-visit treatments based on biologic rationale.²²⁴ The premise for multiple-visit treatments has been that primary debridement is not completely effective in eliminating all the adherent bacterial biofilm,¹⁴⁷ and the residual bacteria may multiply and recolonize the canal

system.^{25,29} Therefore the proponents consider it desirable to use the interappointment period to dress the canal with a long-lasting or slow-release antibacterial agent capable of destroying or incapacitating residual bacteria, as well as to take the opportunity to gauge the initial periapical response before root filling. Calcium hydroxide has served in this capacity for many years because of its ability to dissolve organic tissue, kill bacteria, detoxify antigenic material, and act as a slow-release agent because of its low solubility-product in an aqueous environment. However, its antibacterial ability has come under close scrutiny, with advocates suggesting that the material is not suitable for this purpose.¹⁹⁵ A final resolution to this debate is awaited based on robust clinical evidence. Most of the published randomized controlled trials found no significant influence of healing attributable to number of treatment visits, but they all lacked robust statistical power.

The debate about the merits of single- or multiple-visit treatments will continue unabated given the respective strengths of the motivational drivers among the opposing groups. The issue may only be resolved by properly documented, large randomized controlled trials (which are currently unavailable) because undocumented variables (i.e., operator skill, biologic or technical case complexity, and patient compliance) would continue to bias the outcome.

Post Root Canal Treatment Restorative Factors

Effect of Quality and Type of Restoration. The placement of a coronal restoration after root canal obturation is the final step in the management of teeth undergoing root canal treatment. It has been shown to have a major influence on endodontic outcomes. Teeth with satisfactory coronal restorations were found to have significantly better periapical healing compared with those with unsatisfactory restorations (OR = 3.31; 95% CI: 1.07, 10.30).¹⁵³ The term *satisfactory restorations* has been defined as a restoration with no evidence of marginal discrepancy, discoloration, or recurrent caries with absence of a history of decementation.^{93,182}

Given that one of the roles of coronal restorations is to prevent postoperative root canal reinfection via coronal leakage, the criteria for unsatisfactory restoration given by Hoskinson and colleagues⁹³ could not infer coronal leakage when the inner core was still intact. Consequently, the London Eastman study¹⁵⁰ adopted a different classification and definition for unsatisfactory restorations in order to illustrate obvious and potential coronal leakage more accurately. The two groups of unsatisfactory restorations were defined as those with (1) obvious signs of exposed root filling and (2) potential leakage indicated by marginal defects and history of decementation. It is perhaps this strategy that contributed to the finding of a profound effect (OR = 10.7; 95% CI: 3.7, 31.5) of coronal leakage on the endodontic outcome.

A number of investigations have been performed based on comparisons between the types of post root canal treatment restorations, including permanent versus temporary restorations,^{43,68,150} crown versus acrylic restorations,^{43,68,150,216} presence versus absence of posts,^{68,150} and non-abutment versus abutment.^{150,216} Teeth that had been permanently restored were associated with significantly higher success rates than their temporarily restored counterparts

in some studies^{43,68} but not in others.^{38,150} The type of permanent restoration^{43,68,150,216} was found to have no significant influence on the outcome of treatment.

It has often been recommended that it would be wise to provide a subseal over the root filling in case of loss of a permanent or temporary restoration; the subseal would be glass ionomer (GIC) or zinc oxide–eugenol cement.¹⁹⁶ The placement of a GIC or zinc oxide–eugenol (IRM) cement lining coronal to the gutta-percha filling and underneath the permanent core in order to provide an additional antibacterial coronal seal was found in a prospective study to have no beneficial effect on treatment success.¹⁵⁰

In summary, the preceding findings overall support the ESE guidelines⁶¹ that an adequate restoration should be placed after root canal treatment to prevent subsequent bacterial recontamination. Therefore the provision of a good-quality coronal restoration, regardless of type, should be considered the final part of the root canal treatment procedure following obturation.

Use of Root Treated Teeth as Abutments for Prostheses and Occlusal Contacts. Mechanical stress on restorations is a function of the role of individual teeth in the occlusal scheme. The pattern of occlusal loading both in static and dynamic occlusion is dictated by whether the teeth are involved as single units or abutments (bridge/denture) and whether they have holding or guiding contacts. It is reasonable to expect that bridge and denture abutments may be placed under unfavorable loads, as may last-standing teeth in the dental arch.¹²⁷ These teeth may therefore be expected to have lower success rates because of a potential increase in the development of cracks and fractures due to fatigue. This observation has been confirmed for teeth functioning as bridge abutments compared with those restored as individual units following root canal treatment.²¹⁶

Summary of Factors Influencing Periapical Healing Following Nonsurgical Root Canal Treatment

The following factors are considered as having a major impact on periapical health subsequent to root canal treatment:

- Presence (Fig. 17.20) and size (Fig. 17.21) of periapical lesion
- Patency at the canal terminus (achieving patency significantly increased the chance of success twofold)¹⁵⁰
- Apical extent of chemomechanical preparation in relation to the radiographic apex (Fig. 17.22)
- Outcome of intraoperative culture test (see Fig. 17.19)
- Iatrogenic perforation (if present, reduces the odds of success by 30%)¹⁵⁰
- Quality of root canal treatment judged by the radiographic appearance of the root filling (Fig. 17.23)
- Quality of the final coronal restoration (Fig. 17.24)

The following factors are considered as having minimal impact on root canal treatment outcome:

- Age of patient
- Gender of patient
- Tooth morphologic type
- Specific root canal treatment protocol and technique (preparation, irrigation, and obturation material and technique)

Contemporary improvements in mechanical and chemical canal preparation have not resulted in an increase in the success rate for root canal treatment over the past century (Fig. 17.25). This observation may be explained by the currently available techniques not being effective in eliminating the infection in the apical canal anatomy.

It is notable that all factors that have a strong influence on periapical health after root canal treatment are associated in

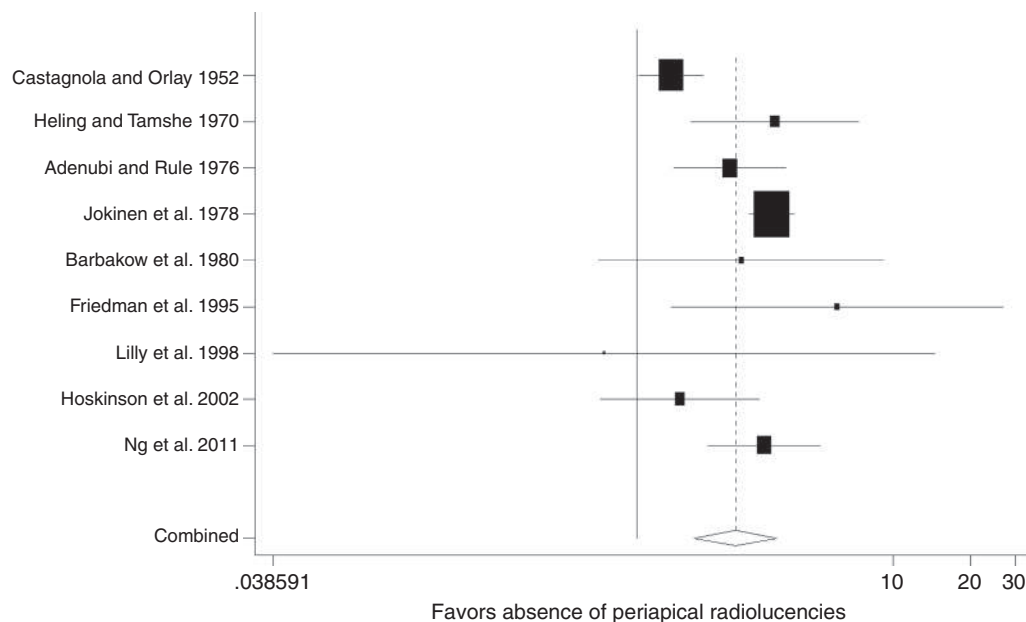


Fig. 17.20 Forest plot showing pooled and individual study's odds ratio (OR) for periapical health of teeth undergoing root canal treatment with pre-operative nonvital pulps and absence of periapical radiolucencies versus teeth with nonvital pulps and presence of periapical radiolucencies (pooled OR = 2.4; 95% CI: 1.7, 3.5).

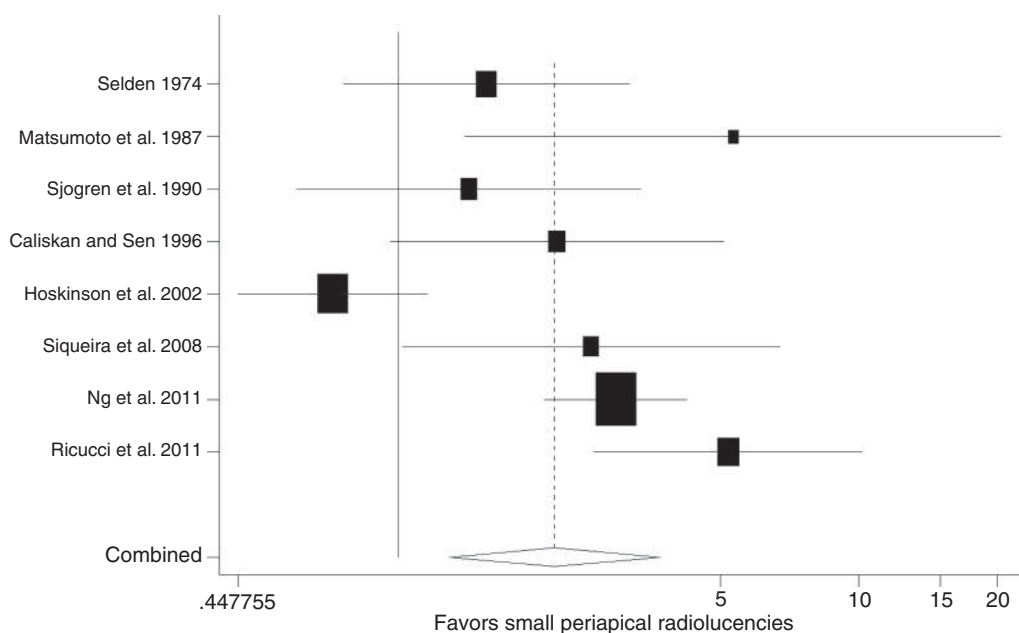


Fig. 17.21 Forest plot showing pooled and individual study's odds ratio (OR) for periapical health of teeth undergoing root canal treatment with pre-operative large (>5 mm) versus small (<5 mm) periapical radiolucencies (pooled OR = 2.2; 95% CI: 1.3, 3.7).

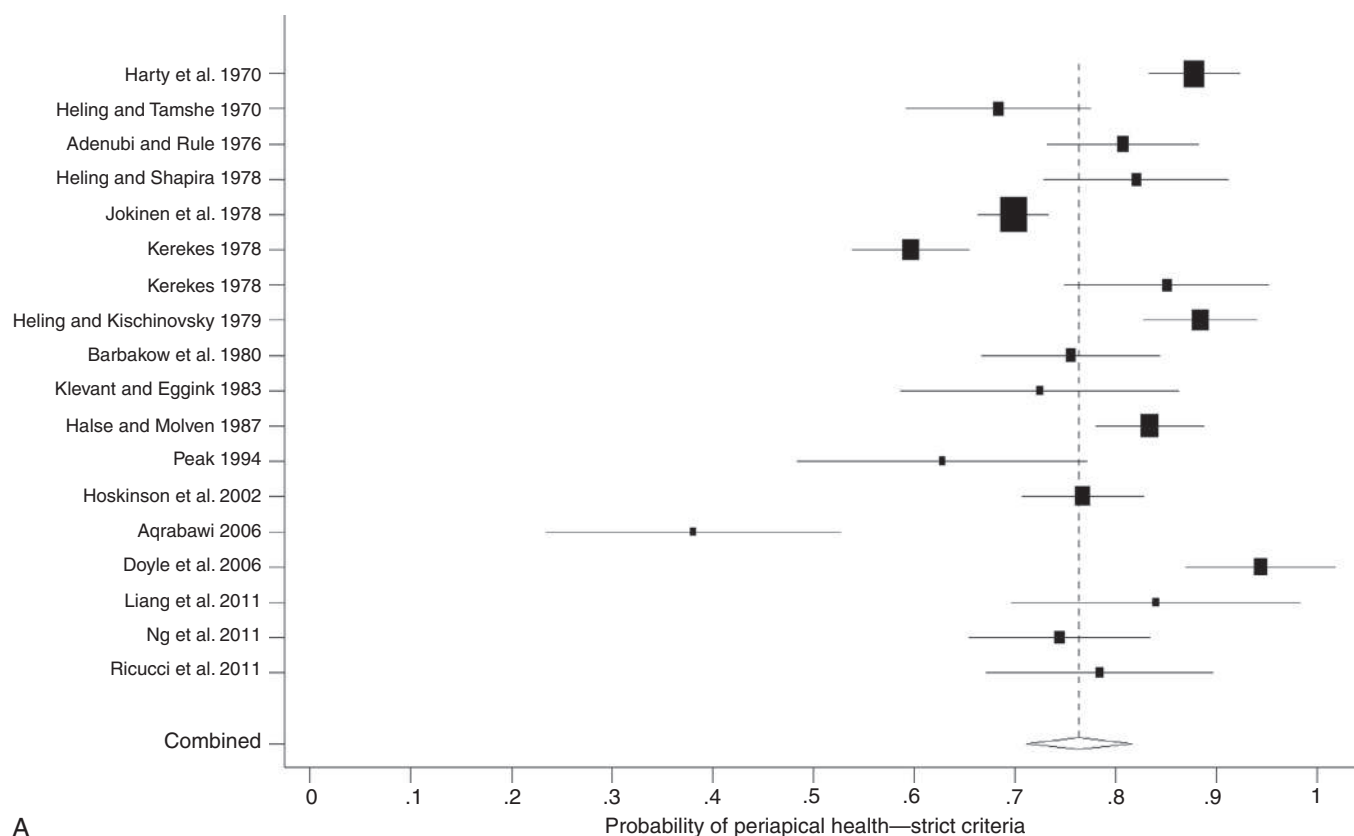
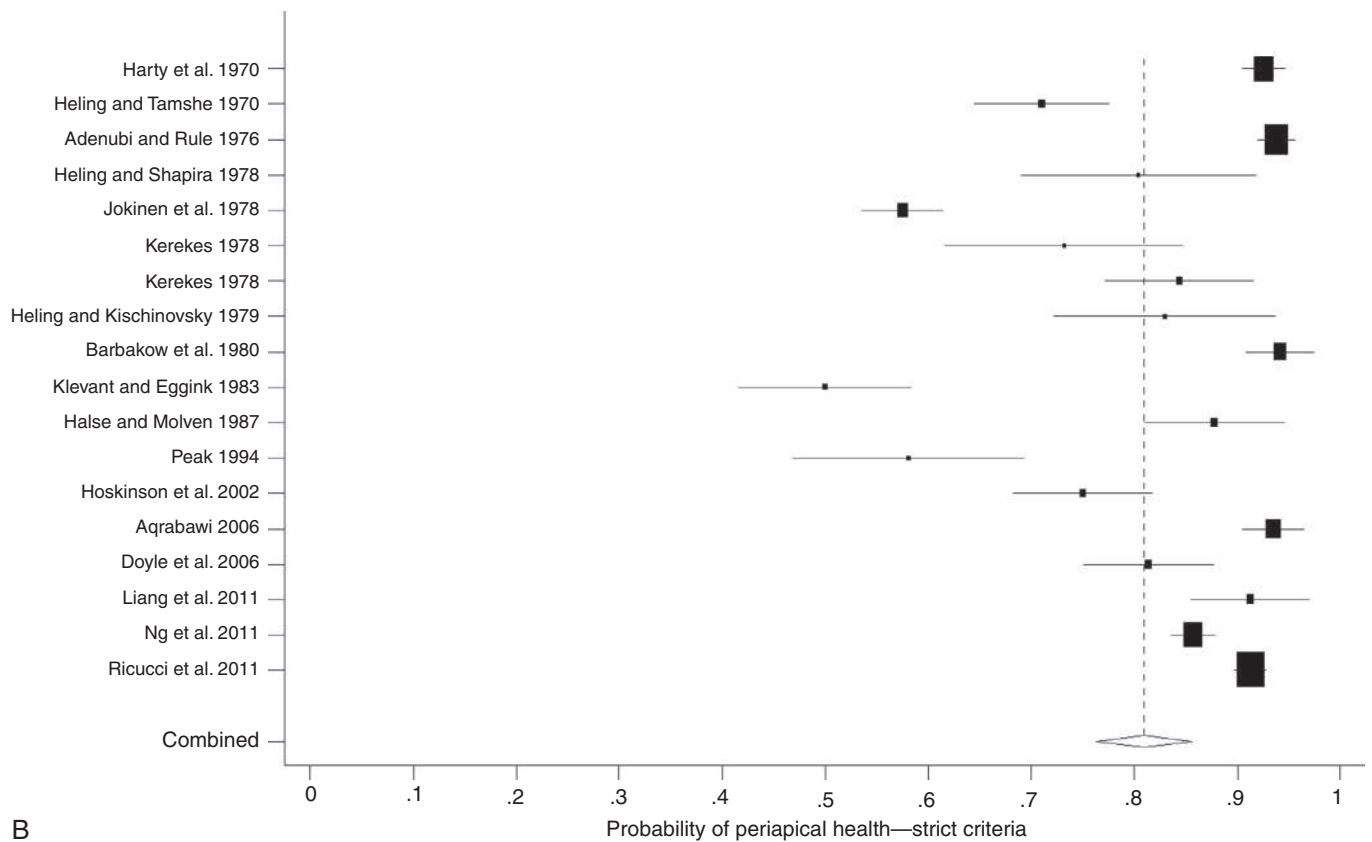
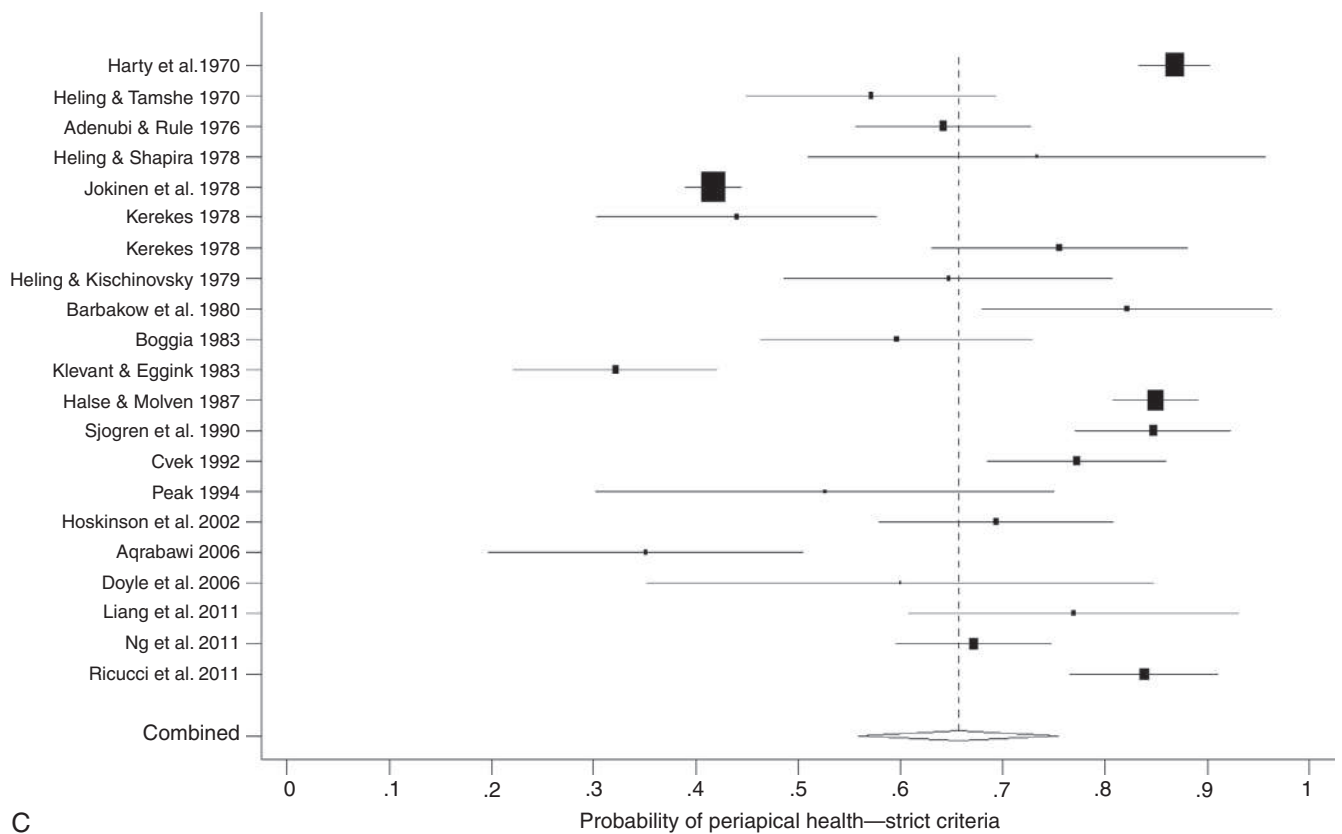


Fig. 17.22 Forest plot showing pooled and individual study's probability of periapical health for teeth undergoing root canal treatment with underextended (0.76 [0.71, 0.82]) (A), flush (0.81 [0.76, 0.86]) (B), or overextended (0.66 [0.56, 0.75]) (C) root fillings.



B



C

Fig. 17.22, cont'd

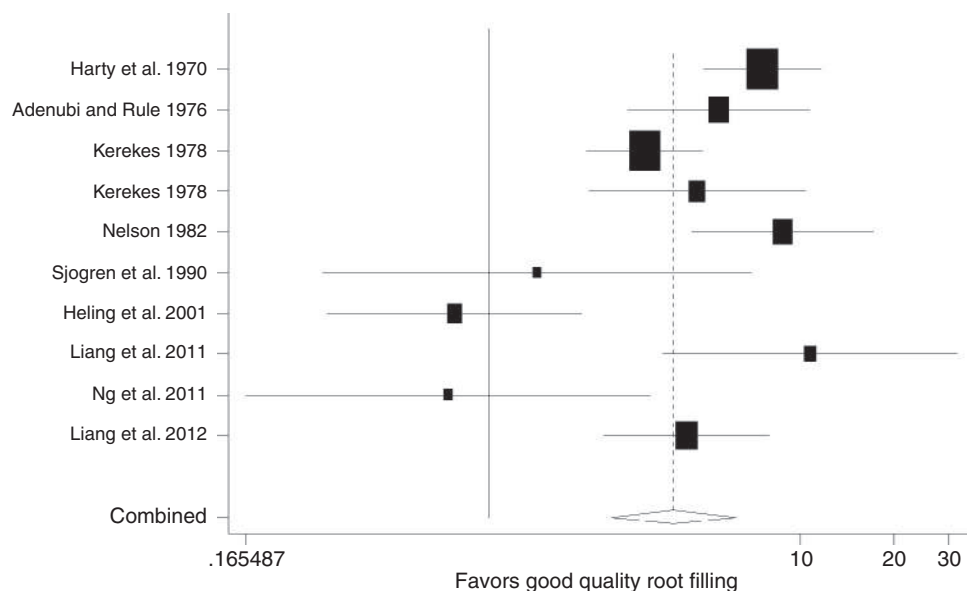


Fig. 17.23 Forest plot showing pooled and individual study's odds ratio (OR) of periapical health for teeth undergoing root canal treatment with good quality versus suboptimal quality root filling (pooled OR = 3.9; 95% CI: 2.5, 6.2).

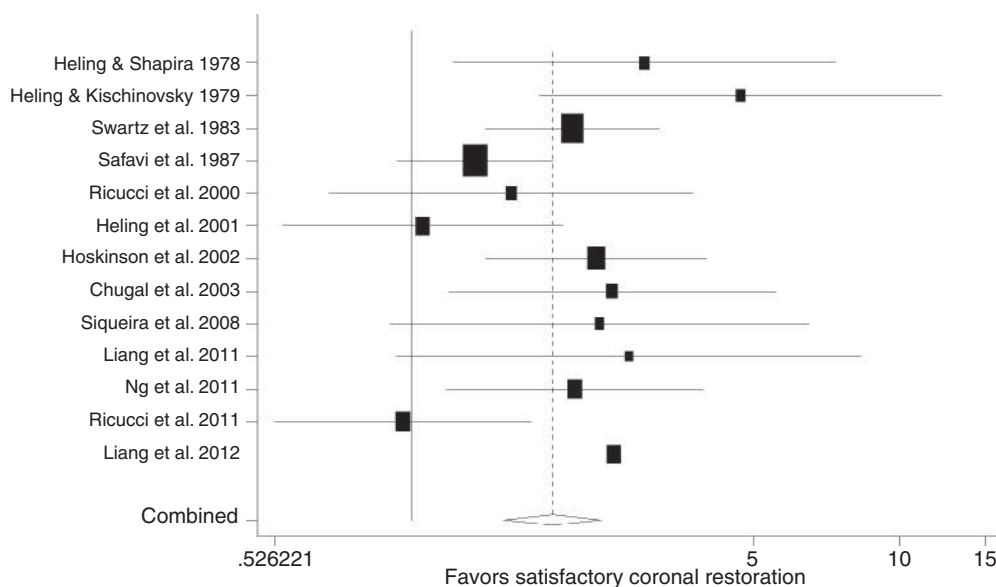


Fig. 17.24 Forest plot showing pooled and individual study's odds ratio (OR) of periapical health for teeth undergoing root canal treatment with satisfactory versus unsatisfactory coronal restoration at follow-up (pooled OR = 1.9; 95% CI: 1.5, 2.5).

some way with a persistent root canal infection. Further improvements in root canal treatment outcomes may therefore be achieved by understanding the nature of the root canal infection (especially apically) and the manner in which the microbiota is altered or eliminated during root canal treatment.

FACTORS AFFECTING TOOTH SURVIVAL FOLLOWING ROOT CANAL TREATMENT

A systematic review and meta-analysis has shown that 93% of endodontically treated teeth survive at 2 years post-operatively; but this survival reduced to 88% at 10 years following treatment (Fig. 17.26). The most common

reasons for such tooth loss were due to problems of endodontic origin, tooth/root fracture, or restoration failure.^{149,151} Consistent with the studies on periapical healing, the considerations affecting the survival of endodontically treated teeth may be divided into patient, intraoperative, and restorative factors.

Patient Factors

Ng and colleagues¹⁵¹ found that teeth in patients suffering from diabetes or receiving systemic steroid therapy had a higher chance of being extracted after root canal treatment. The negative influence of diabetes on tooth survival is consistent with the report by Mindiola and associates,¹³⁵ whereas the influence of steroid therapy had never been

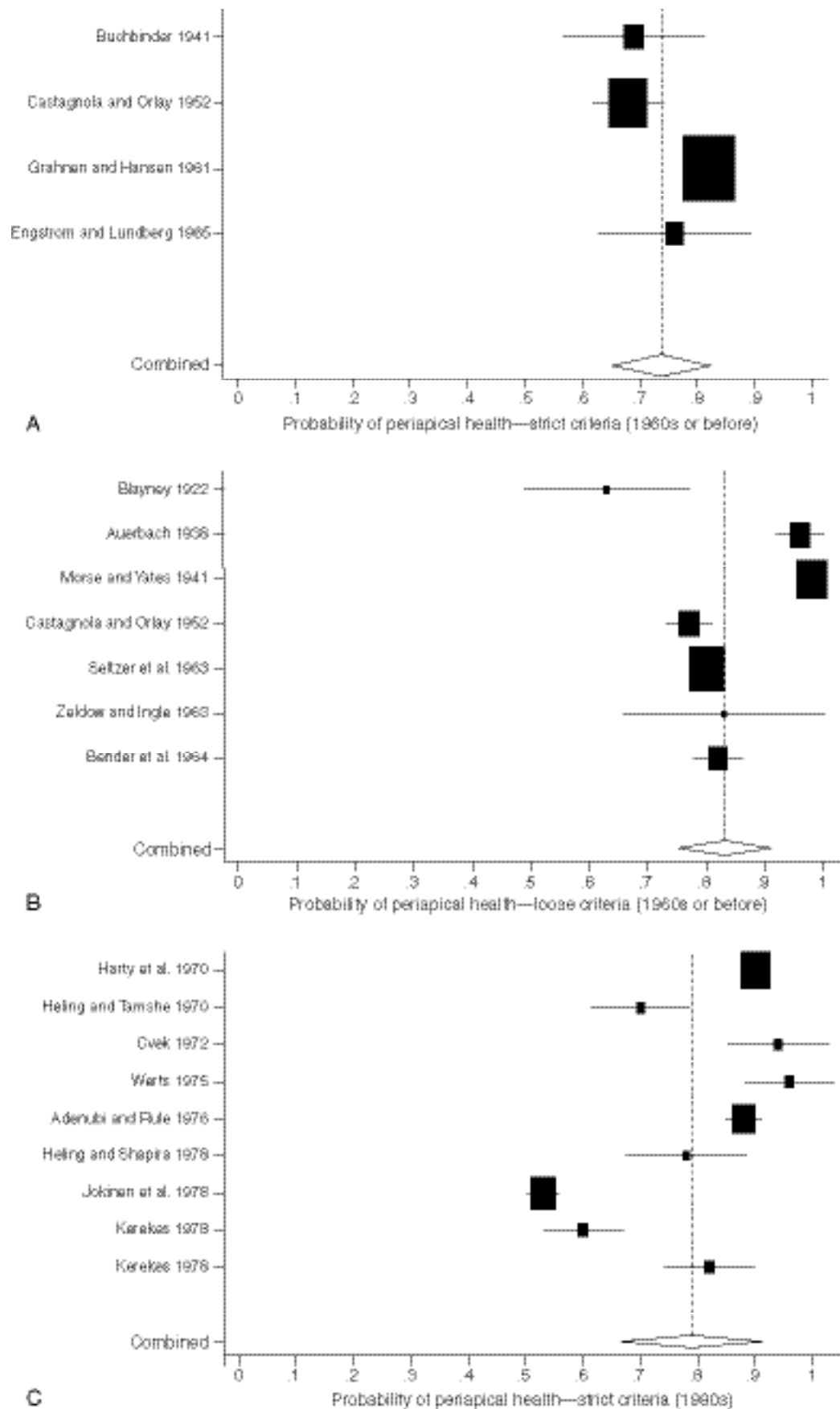


Fig. 17.25 Forest plot showing pooled and individual study's probability of periapical health for teeth undergoing root canal treatment by "decade of publication" and "criteria for success."

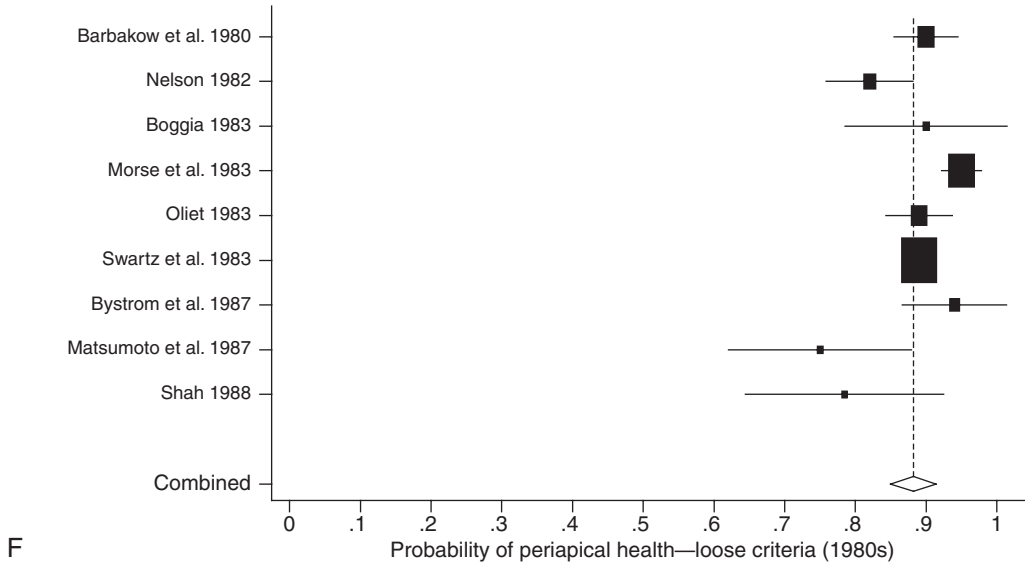
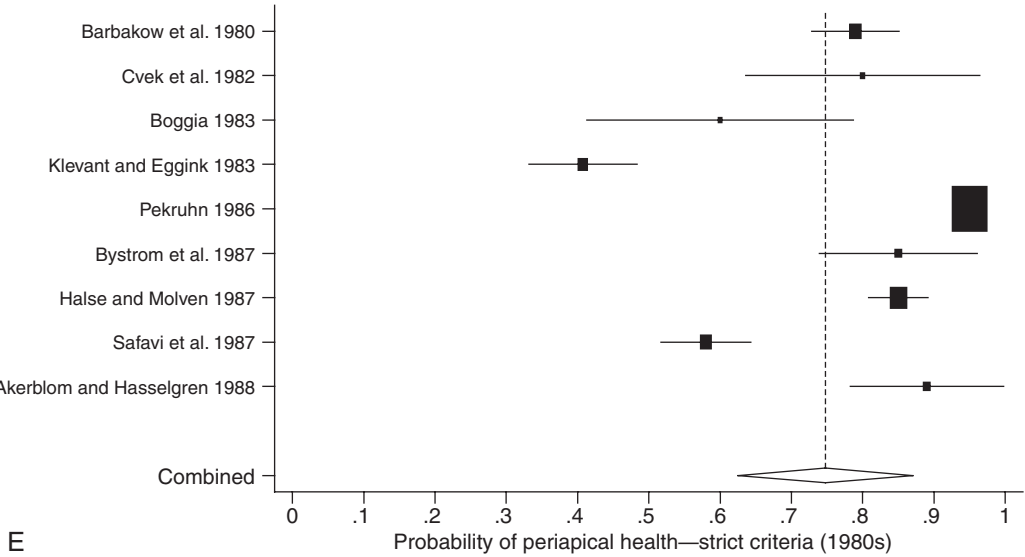
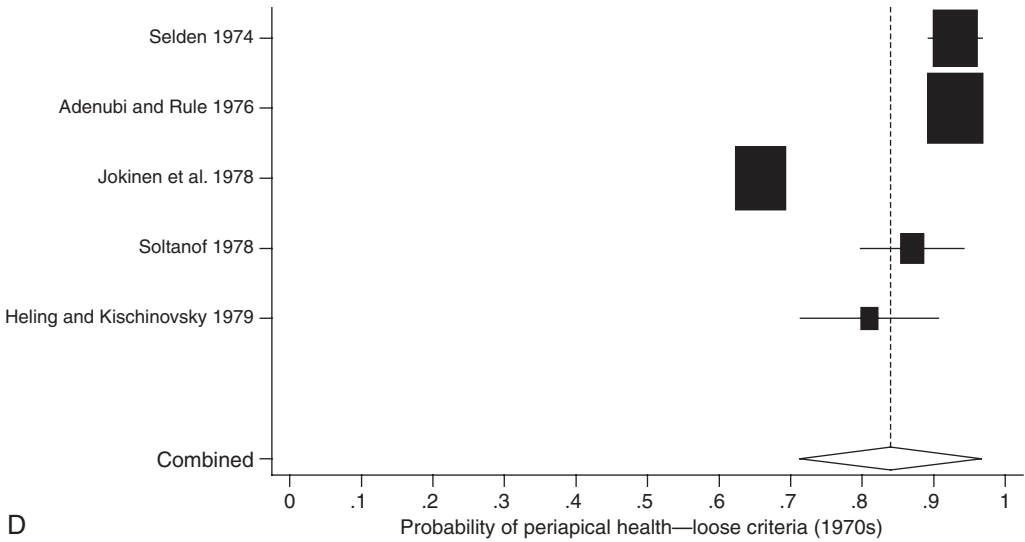


Fig. 17.25, cont'd

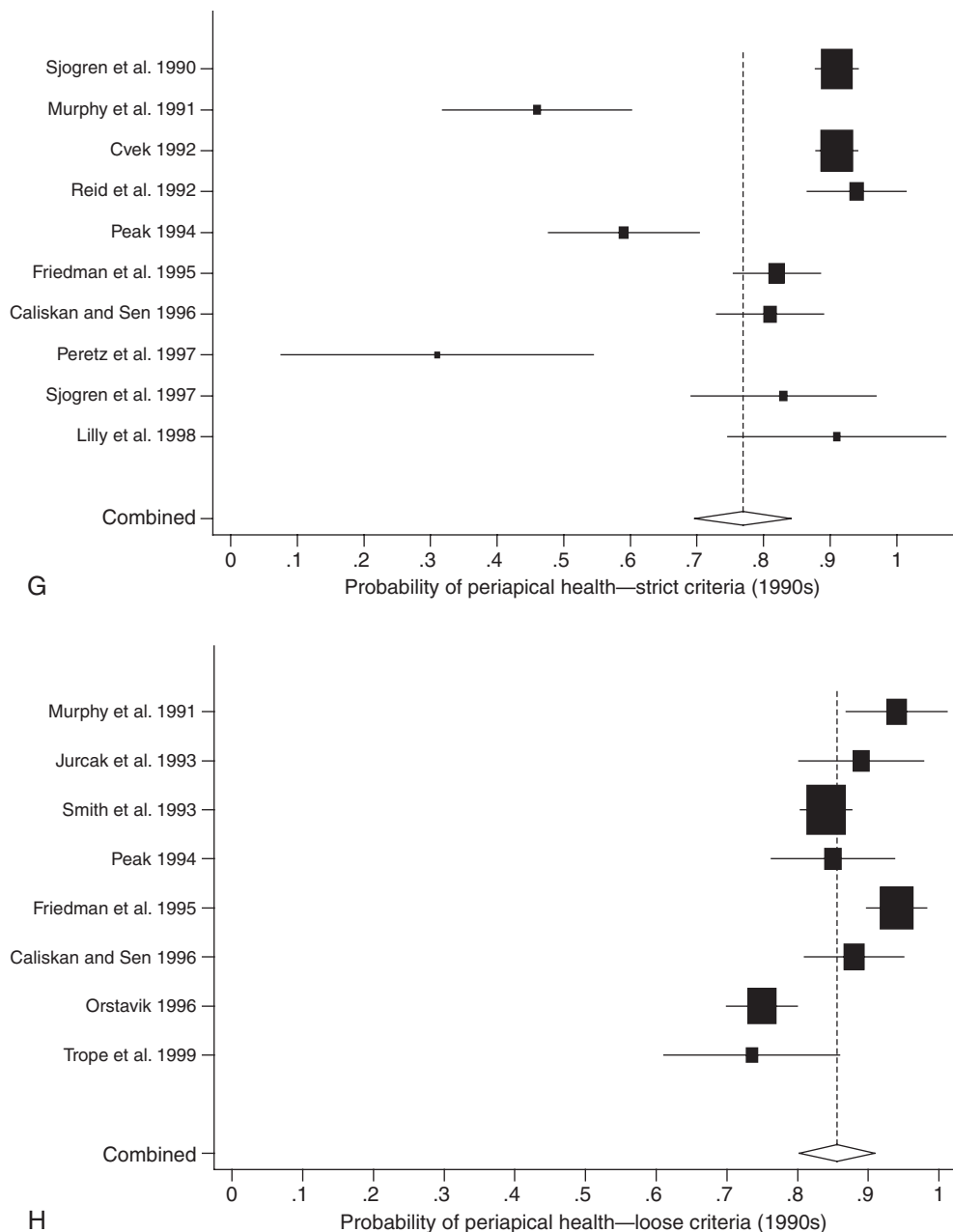


Fig. 17.25, cont'd

reported previously. It may be argued that patients with diabetes are more susceptible to periodontal disease⁷⁴ or have a lower success rate of root canal treatment⁶⁷ because of being immunologically compromised. However, the researchers reported that over 50% of such teeth were extracted due to persistent pain. Some of these observations may be explained by the presence of neuropathy, a debilitating painful complication of diabetes.⁵⁷ It is interesting to note that systemic steroid therapy is often prescribed to control such chronic pain.^{39,51,101}

Tooth Morphologic Type and Location

Tooth types (i.e., location within the arch) may vary with respect to their susceptibility to tooth fracture, a common reason for tooth loss after endodontic treatment. Ng and

coworkers¹⁴⁹ found that tooth type had a significant influence on survival. Maxillary premolars and mandibular molars were found to have the highest frequency of extraction, with tooth fracture being the most common reason. The observation is consistent with previous reports on fracture incidence of maxillary premolars and mandibular molars.^{56,109} The factors “proximal contacts” and “terminal (last standing) teeth” were found to affect tooth survival considerably,^{149,151} but they were significantly correlated with molar teeth. Most of the extractions of terminal teeth or teeth with one or fewer proximal contacts were due to tooth fracture.¹⁵¹ The observation may be explained by the unfavorable distribution of occlusal forces and higher nonaxial stress on terminal teeth and those with fewer than two proximal contacts. Other possible reasons for their

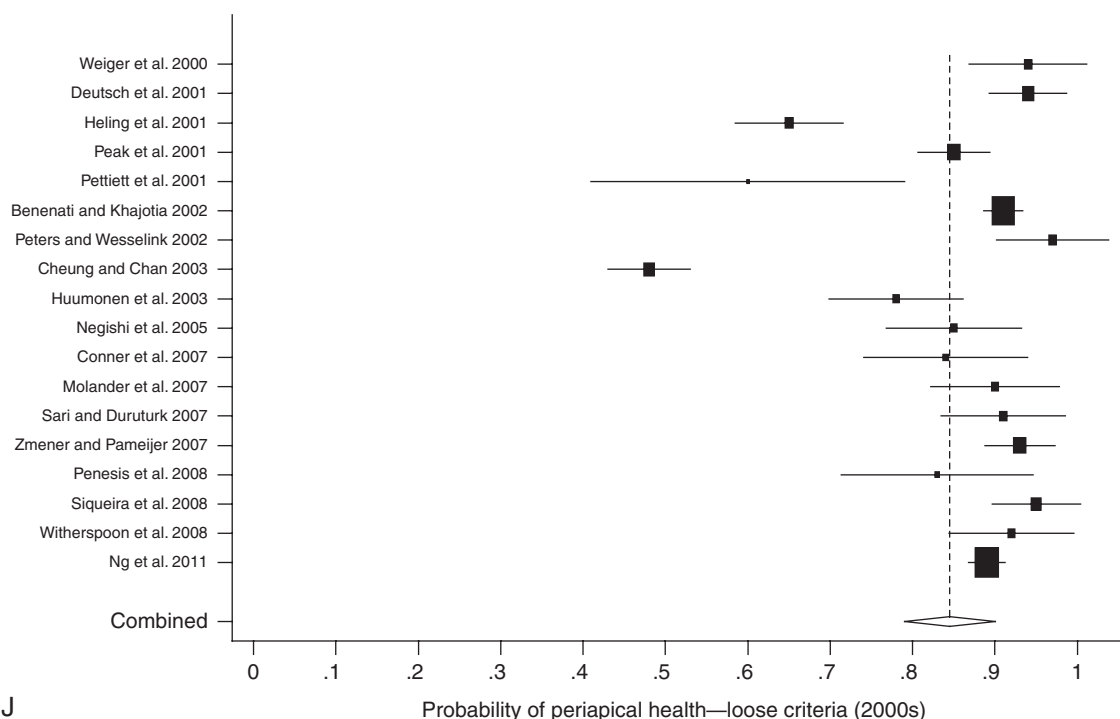
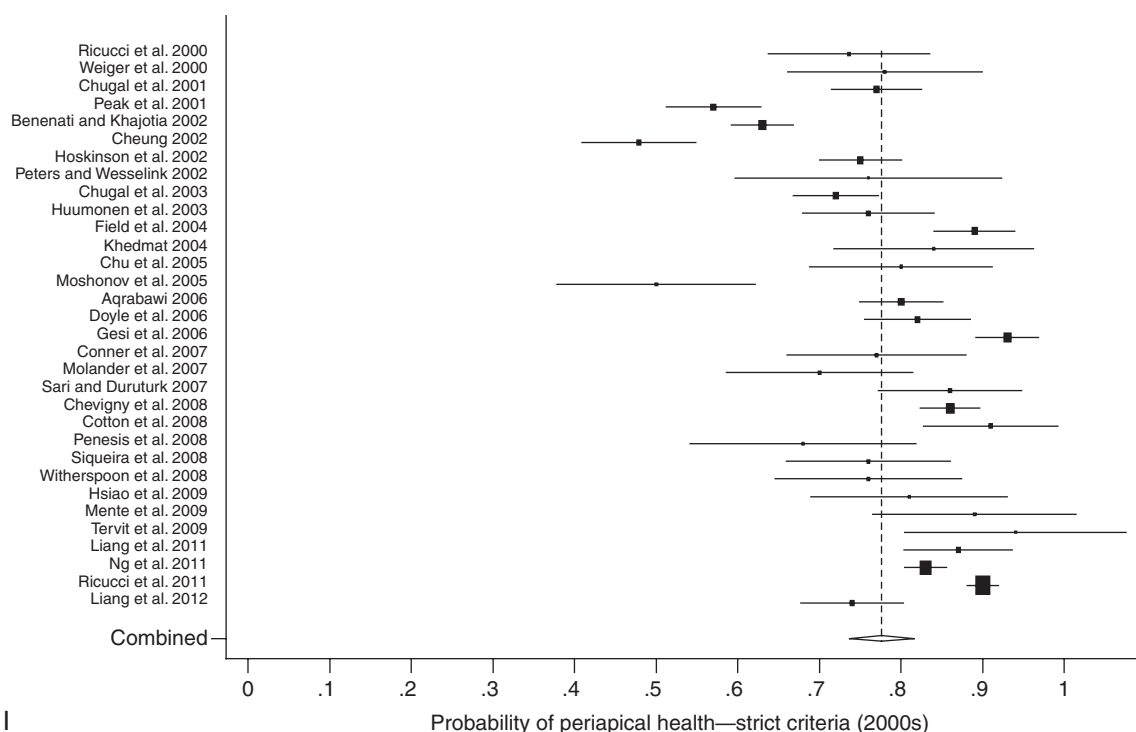


Fig. 17.25, cont'd

higher rate of loss are (1) failure of root canal treatment in a terminal tooth may be accepted more willingly because of little perceived aesthetic value and (2) clinicians may be less likely to offer further treatment on terminal molar teeth due to difficult access. Therefore when restoring molar teeth, the favorable distribution of occlusal forces must be considered, especially on teeth with one or fewer adjacent teeth or on terminal teeth.

Preoperative Conditions of Teeth

The presence of preoperative periapical lesions, which is the most significant prognostic factor for periapical healing, was found to have no significant influence on tooth survival.¹⁵¹ On the other hand, the presence of preoperative periodontal probing defects of endodontic origin, preoperative pain, and preoperative sinus tracts were found to reduce tooth survival.¹⁵¹ These observations are consistent

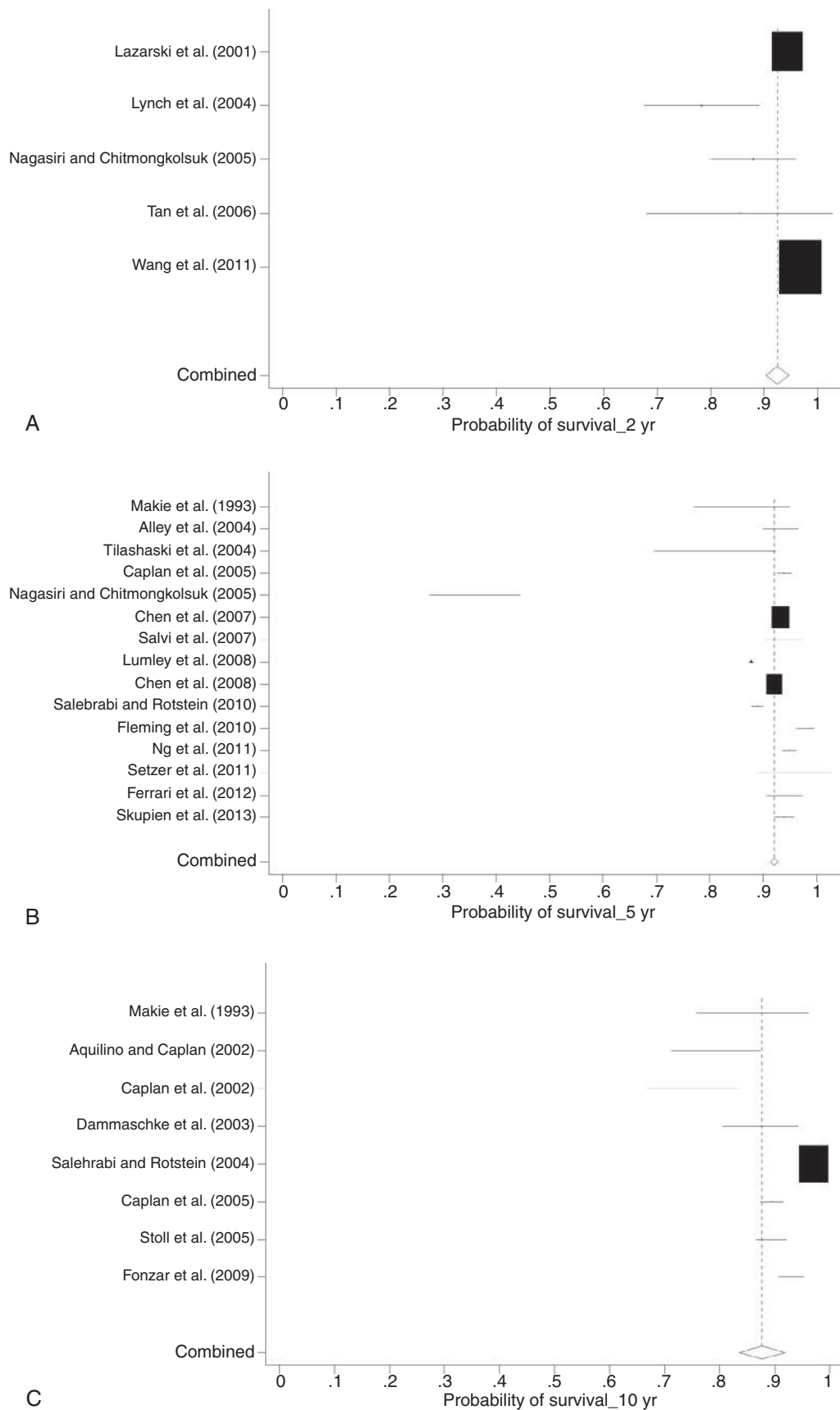


Fig. 17.26 Forest plot showing pooled and individual study's probability of 2-year (pooled probability = 0.93; 95% CI: 0.90, 0.95) (**A**), 5-year (pooled probability = 0.92; 95% CI: 0.91, 0.93) (**B**), and 10-year (pooled probability = 0.88; 95% CI: 0.84, 0.92) (**C**) survival of teeth undergoing root canal treatment.

with a previous report that the mere presence of a persistent periapical lesion was not a sufficient reason for dentists and patients to opt for further treatment, either retreatment or extraction.¹⁸⁰ The negative impact of preoperative pain on survival outcome highlights the importance of accurate pain diagnosis. In some instances, the pain may have been of nonendodontic origin and therefore would persist after treatment.¹⁷⁶ In other instances, preoperative pain of endodontic origin may persist following treatment, as a result of peripheral or central sensitization. Therefore effective pain diagnosis and management for patients presenting with preoperative pain are crucial.

The presence of preoperative cervical resorption and perforation was also found to significantly reduce tooth survival.¹⁵¹ This was as expected because tooth fracture and reinfection due to leakage through the resorption and perforating defects are likely sequelae in such cases. In the presence of reinfection, clinicians are more inclined to suggest extraction due to the intuitive perception of poor long-term prognosis of such teeth.

Treatment Factors

Considering all of the intraoperative factors, the “lack of patency at apical foramen” and the “extrusion of gutta-percha root filling” were found to be the most significant intraoperative factors in reducing tooth survival.¹⁵¹ In the presence of persistent problems and knowing that the treatment objective of cleaning to the canal terminus could not be achieved, patients and dentists may be more likely to opt for extraction sooner than later.

Restorative Factors

Protection of teeth with crowns or cast restorations has not been shown to influence periapical healing; however, the placement of good cores had a positive effect on endodontic outcome. In contrast, placement of crowns or cast restorations was found to improve tooth survival.^{149,151,189} This suggests that crowns and cast restorations help prevent tooth fracture, whereas the mere placement of a satisfactory core is sufficient to prevent reinfection after endodontic treatment. Unfortunately, the study was not able to investigate the interrelationship between tooth morphologic type, the extent of tooth tissue loss after treatment, or the type of final restoration. Although the clinical inference from these findings is that cast restorations should preferably be placed on all teeth after root canal treatment, this is probably a gross exaggeration of the true need. On the basis of laboratory¹⁷⁸ and clinical findings,¹⁴⁴ posterior teeth with compromised marginal ridges (mesial or distal), together with heavy occlusal loading evidenced by faceting, may benefit from full coverage restorations. The restoration design should attempt to preserve as much remaining tooth tissue as possible; the implication is that the so-called nonaesthetic but technically demanding partial veneer onlays and partial coverage crowns would be the restorations of choice for root-treated teeth. In anterior teeth, the missing tooth tissue may often be adequately replaced with composite restorative materials. A crown is only indicated when tooth structure or aesthetics become compromised.

The use of cast post and cores for retention of restorations have also been found to reduce tooth survival.^{149,151} It may be speculated that the presence of a post has different

effects on anterior versus posterior teeth as they are subjected to different directions and amount of occlusal force. It has been reported that only 12% of the extracted teeth with cast post and cores were incisors or canines. Therefore the inference is that the use of such retention should be particularly avoided in premolar and molar teeth. Alternative treatment options should be considered for molar or premolar teeth lacking sufficient tooth structure.

Ng and colleagues¹⁵¹ observed that teeth functioning as prosthetic abutments had poorer survival rates; however, the number of teeth ($n = 94$) in the study that functioned as abutments was too small a sample to be considered statistically significant. As previously described, the explanation may reside in the excessive and unfavorable distribution of occlusal stresses on abutment teeth. If possible, root-treated teeth should be avoided as abutments for prostheses or in provision of occlusal guidance in excursive movements.

Summary of Factors Influencing the Survival of Teeth Following Root Canal Treatment

The following conditions have been found to significantly improve tooth survival following root canal treatment:

- Nonmolar teeth (Fig. 17.27)
- Teeth with both mesial and distal adjacent teeth (Fig. 17.28)
- Teeth not located as the distal-most tooth in the arch¹⁵¹
- Teeth (molar) with cast restorations after treatment (Fig. 17.29)
- Teeth not requiring cast post and core for support and retention of restoration^{111,151}
- Teeth not functioning as abutments for fixed prosthesis^{3,111,191,151}
- Absence of preoperative deep periodontal probing defects, pain, sinus tract, or perforation¹⁵¹
- Achievement of patency at canal terminus and absence of root-filling extrusion during treatment¹⁵¹

In addition, it is important to ensure favorable distribution of occlusal forces when designing restorations for endodontically treated teeth.

IMPACT OF ROOT CANAL TREATMENT ON QUALITY OF LIFE

The impact of root canal treatment on the OHRQoL of patients has been evaluated using the short form (OHIP-14) or the modified version (OHIP-17) of the Oral Health Impact Profile (OHIP-14) (Table 17.10).²¹⁹ The distinctly positive impact of root canal treatment was apparent, regardless of cultural background of the patient group or the measure used.^{55,73,87,115} As expected, physical pain, psychological discomfort (feeling tense), and disability (difficulty in relaxing) were the most improved domains following treatment.

Outcome of Nonsurgical Retreatment

When root canal treatment fails to resolve periapical disease, it is often considered appropriate to retreat the tooth using conventional approaches first, especially when the previous treatment is technically deficient (Fig. 17.30).

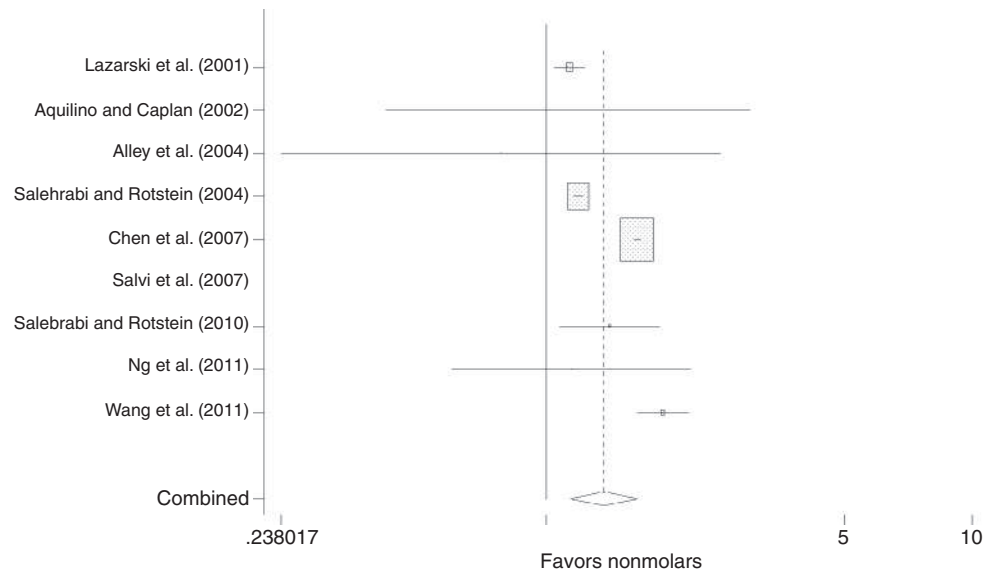


Fig. 17.27 Forest plot showing pooled and individual study's odds ratio (OR) of survival probability for nonmolar versus molar teeth (pooled OR = 1.4; 95% CI: 1.1, 1.6).

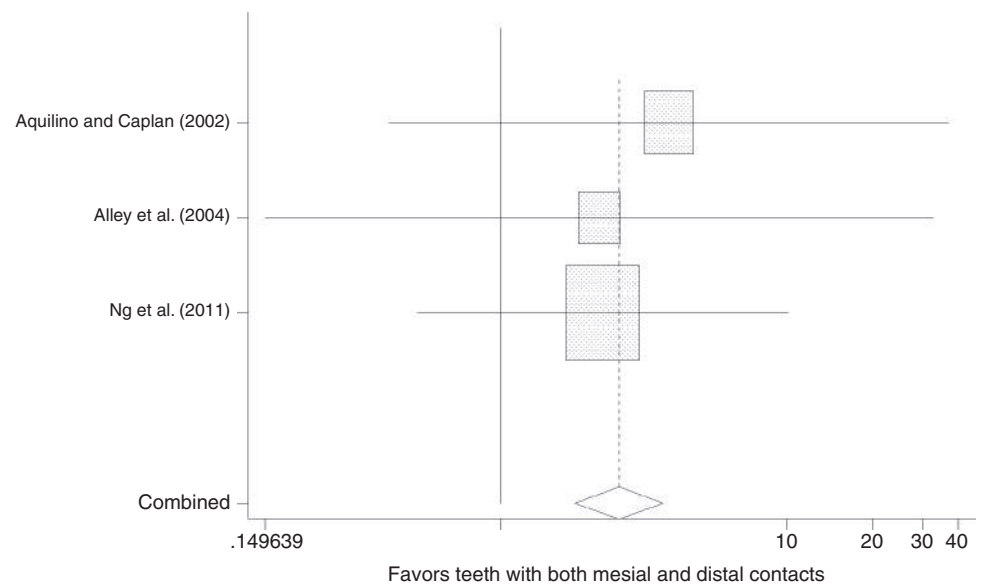


Fig. 17.28 Forest plot showing pooled and individual study's odds ratio (OR) of survival probability for teeth with both mesial and distal contacts present versus absent (pooled OR = 2.6; 95% CI: 1.8, 3.7).

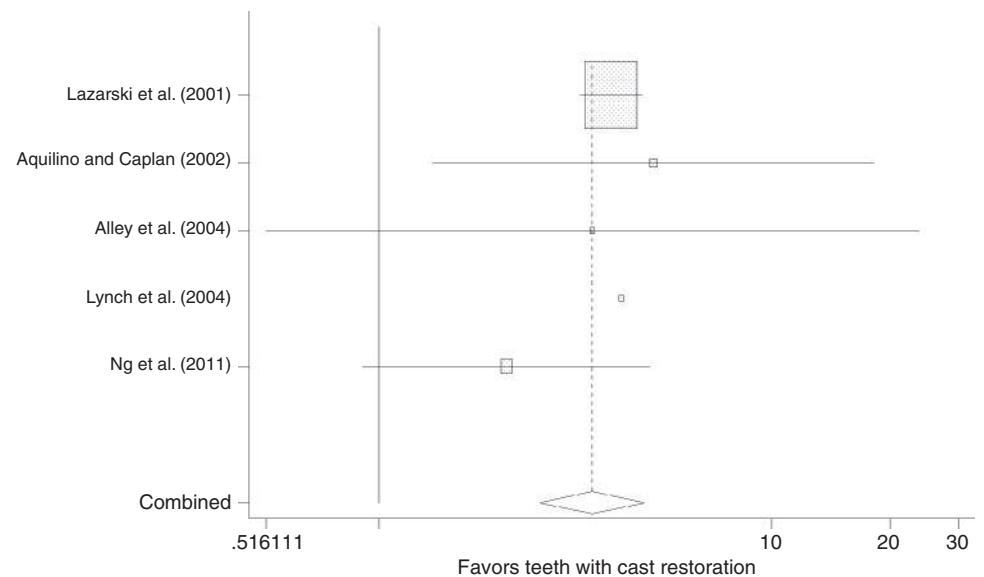


Fig. 17.29 Forest plot showing pooled and individual study's odds ratio (OR) of survival probability for teeth restored with cast versus plastic restoration (pooled OR = 3.5; 95% CI: 2.6, 4.7).

Table 17.10 The OHIP-14 and Modified Oral Health Impact Profile Instrument (OHIP-17)

OHIP-#	Item Question
OHIP-1	Have you had trouble pronouncing words because of your teeth and mouth?
OHIP-2	Have you felt that your sense of taste has worsened because of your teeth or mouth?
OHIP-3	Have you had painful aching in your mouth?
OHIP-4	Have you found it uncomfortable to eat any foods because of your teeth or mouth?
OHIP-5*	Have you had to alter the temperature of the foods that you eat because of your teeth or mouth?
OHIP-6	Have you been self-conscious because of your teeth or mouth?
OHIP-7	Have you felt tense because of your teeth or mouth?
OHIP-8	Has your diet been unsatisfactory because of your teeth or mouth?
OHIP-9	Have you had to interrupt meals because of your teeth or mouth?
OHIP-10	Have you found it difficult to relax because of your teeth or mouth?
OHIP-11*	Have you found it difficult to fall asleep because of your teeth or mouth?
OHIP-12*	Have you ever been awakened by problems with your teeth or mouth?
OHIP-13	Have you been embarrassed because of your teeth or mouth?
OHIP-14	Have you been irritable with other people because of your teeth or mouth?
OHIP-15	Have you had difficulty doing your usual jobs because of problems with your teeth or mouth?
OHIP-16	Have you felt that life in general was less satisfying because of your teeth or mouth?
OHIP-17	Have you been totally unable to function because of your teeth or mouth?

OHIP-14 contains all the items except those with an asterisk (*).

OHIP, Oral Health Impact Profile.

From Dugas NN, Lawrence HP, Teplitsky P, Friedman S: Quality of life and satisfaction outcomes of endodontic treatment, *J Endod* 28:819-827, 2002.

This requires removal of the previous root-filling material and any other material placed for restorative reasons. Correction of any procedural errors may also be required, if possible. All materials must be removed in their entirety to ensure delivery of antibacterial agents to all surfaces of the root canal dentin (see Fig. 17.30). The periapical healing rates of root canal retreatment are generally perceived to be lower compared to primary treatment for the following reasons:

- Obstructed access to the apical infection
- A potentially more resistant microbiota

The outcomes from a range of studies show that the mean weighted success rate of nonsurgical root canal retreatment is 66% (Figs. 17.31 and 17.32), which is about 6% lower than in the case of primary treatment on teeth with apical periodontitis.^{148,150} However, it has also been shown that the survival rate of teeth having undergone

nonsurgical root canal retreatment may be similar to that for primary root canal treatment.¹⁵¹

The factors affecting outcomes of periapical health and tooth survival following root canal retreatment are otherwise identical to those affecting primary root canal treatment. Of the potential prognostic factors unique to retreatment cases, the most significant factor influencing the outcome of treatment is the ability to remove or bypass preexisting root-filling material or separated instruments during retreatment. This is understandable because it would have a direct impact on the ability to achieve canal patency and canal disinfection at the apical terminus.¹⁵⁰

Outcome of Surgical Retreatment

FACTORS AFFECTING PERIAPICAL HEALTH OR HEALING FOLLOWING PERIAPICAL SURGERY AND ROOT-END FILLING

There are a number of published systematic reviews on the prognostic factors for periapical surgery with apical filling.^{45,47,48,155,204,205,239-241,250} The major limitations of this pooled data include the variable length of time for evaluating success following treatment as well as the radiographic criteria used to assess healing. An unpublished meta-analysis¹³⁰ of data from the studies listed in Table 17.11 reveals that the weighted pooled probability of success from periapical surgery with retrograde restorations, based on complete radiographic healing, is 67.5% (95% CI: 62.9%, 72.0%) (Fig. 17.33). The trend shows higher success rates reported in more recent studies. This observation is consistent with the much higher pooled success rate of 92% (95% CI: 86%, 95%) revealed in the meta-analysis of prospective outcome data of surgical endodontic treatments performed by a “modern” technique (using magnification, root-end resection with minimal or no bevel, retrograde cavity preparation with ultrasonic tips, and modern retrograde root canal filling materials).^{240,241} In congruence with these findings, Setzer and colleagues reported that the pooled success rate of treatment using a microsurgical approach (94%; 95% CI: 89%, 98%) was more favorable than traditional root-end surgery (59%; 95% CI: 55%, 63%).²⁰⁵ However, the studies included in the latter meta-analysis differed in design, case selection, duration after treatment, and the provision of preoperative nonsurgical treatment, which may bias the higher success rate of the contemporary treatment.

The unpublished meta-analysis¹³⁰ stratified by duration following treatment reveals that the pooled probability of success based on complete healing plateaued after 2 years postoperatively, being 51% (95% CI: 42%, 60%) after 6 months, 68% (95% CI: 63%, 73%) after 12 months, 76% (95% CI: 67%, 84%) after 24 months, 74% (95% CI: 52%, 95%) after 48 months, and 74% (95% CI: 66%, 82%) after more than 48 months. It is therefore advisable to follow up on cases that have undergone periapical surgery for a minimum period of 2 years and up to 4 years, as suggested in quality guidelines.⁶¹

The factors having a major impact on outcome of periapical surgery with retrograde cavities and fillings in teeth with preoperative periapical radiolucencies revealed in the

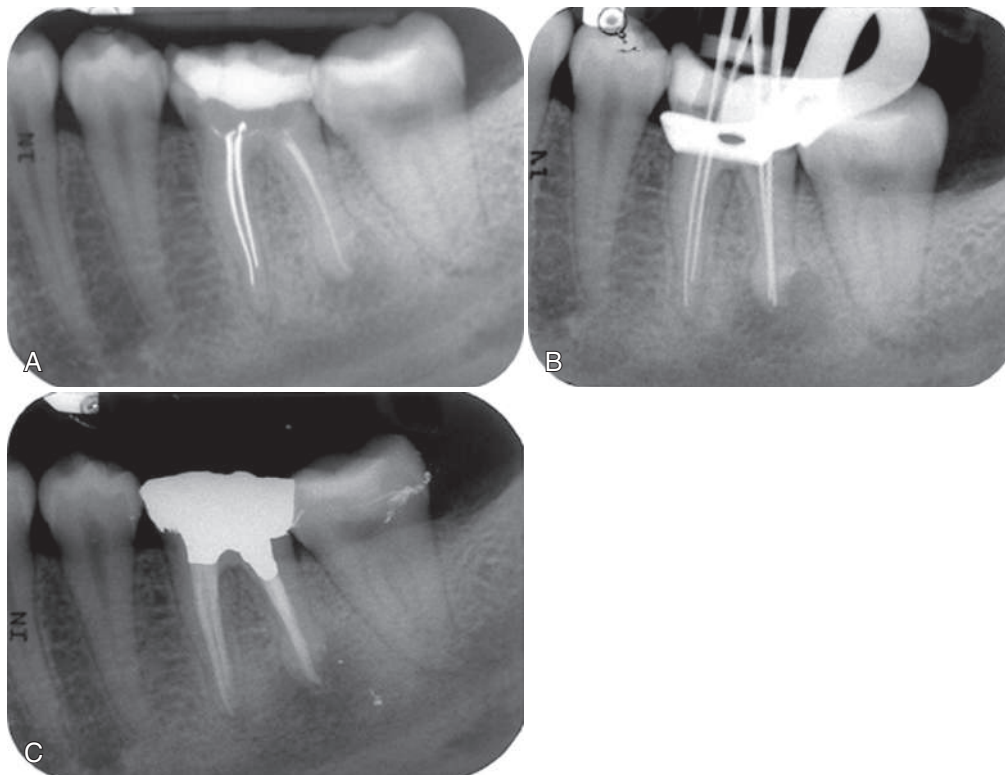


Fig. 17.30 Tooth with technically deficient root canal treatment (A); having undergone (B and C) root canal retreatment.

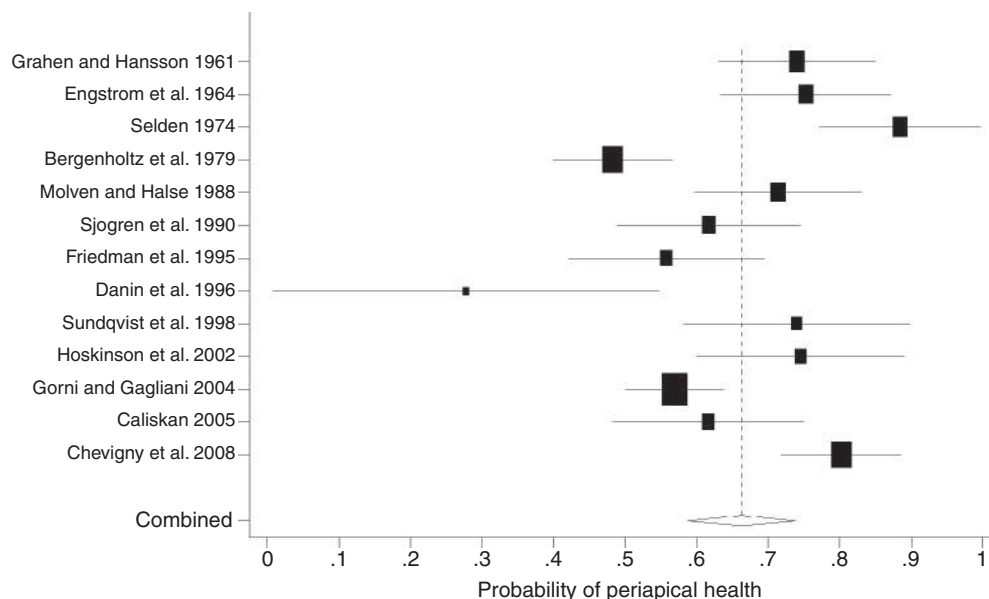


Fig. 17.31 Forest plot showing pooled and individual study's probability of complete periapical healing following root canal retreatment.

(as yet) unpublished meta-analyses¹³⁰ of data from studies listed in Table 17.11 are as follows:

- Small (≤ 5 mm) versus large (> 5 mm) periapical lesion (risk ratio = 1.2; 95% CI: 1.1, 1.3)
- Periapical lesion involving one versus both cortical plates (risk ratio = 1.2; 95% CI: 1.0, 1.5)
- Absence versus presence of previous surgery (risk ratio = 1.2; 95% CI: 1.1, 1.3)
- Using magnification versus without the use of magnification during surgery (risk ratio = 1.5; 95% CI: 1.3, 1.8)
- Root-end resection with minimum versus obvious bevel (risk ratio = 1.3; 95% CI: 1.2, 1.4)
- Use of ultrasonic tip versus bur for retro-cavity preparation (risk ratio = 1.3; 95% CI: 1.2, 1.4)
- Use of retro-filling material with mineral trioxide aggregate (MTA) cement, super ethoxybenzoic acid (EBA)

Table 17.11 Studies Investigating Periapical Healing Following Apical Surgery

Author	Design	Examination	Type of Radiograph	Radiographic Criteria for Success	SAMPLE SIZE			Duration After Treatment (Months)
					Patients	Teeth	Roots	
Harty et al., 1970	Retrospective	C & R	Pa	Other		169		6–60
Nordendram, 1970	Prospective	C & R	Pa	Rud et al., 1972	66	66		6–24
Rud et al., 1972	Prospective	R	Pa	Rud et al., 1972		237		12–180
Finne et al., 1977	Prospective	C & R	Pa	Persson, 1973	156	218		36
Hirsch et al., 1979	Prospective	C & R	Pa	Rud et al., 1972	467	467		6–36
Ioannides and Borstlap, 1983	Retrospective	C & R	Pa	Other	50	50	45	6–60
Allen et al., 1989	Retrospective	C & R	Pa	Rud et al., 1972		175		12–60
Amagasa et al., 1989	Prospective	C & R	Pa	Other	42	64		12–90
Dorn and Gartner, 1990	Retrospective	R	Pa	Other		488		6–120
Grung et al., 1990	Prospective	C & R	Pa	Rud et al., 1972		161		12–96
Lustmann et al., 1991	Retrospective	C & R	Pa	Rud et al., 1972			123	6–96
Rapp et al., 1991	Retrospective	R	Pa	Rud et al., 1972	331	226		6–24
Rud et al., 1991	Retrospective	C & R	Pa	Rud et al., 1972	388	388		12
Waikukul et al., 1991	Prospective	C & R	Pa	Other	34	62		6–24
Pantschev et al., 1994	Prospective	C & R	Pa	Persson, 1973	79	103		36
Jesselen et al., 1995	Prospective	C & R	Pa	Other	67	82		12–60
Rud et al., 1996	Prospective	R	Pa	Rud et al., 1972			347	12–48
Sumi et al., 1996	Retrospective	C & R	Pa	Other	86	157		6–36
Jansson et al., 1997	Retrospective	C & R	Pa	Other	59	59		11–16
Testori et al., 1999	Retrospective	R	Pa	Rud et al., 1972	130	181	302	12–72
Von Arx and Kurt, 1999	Prospective	C & R	Pa	Von Arx and Kurt, 1999	38	43		12
Zuolo et al., 2000	Prospective	C & R	Pa	Molven et al., 1987	106	102		12–48
Pecora et al., 2001	RCT	R	Pa	Rud et al., 1972	20	20		6
Penarrocha et al., 2001	Retrospective	R	Pa	Von Arx and Kurt, 1999	30	31	71	12
Rahbaran et al., 2001	Retrospective	C & R	Pa	Other	154	154		48–108
Rud et al., 2001	Prospective	C & R	Pa	Rud et al., 1972		520	834	6–150
Von Arx et al., 2001	Prospective	C & R	Pa	Other	24	25	39	12
Jensen et al., 2002	RCT	C & R	Pa	Rud et al., 1972		122		
Rubinstein and Kim, 2002	Prospective	C & R	Pa	Rud et al., 1972	52	59	59	68
Tobon et al., 2002	RCT	R	Pa	Rud et al., 1972	25	26		12
Vallecillo et al., 2002	Prospective	R	Pa	Other	29	29		12
Chong et al., 2003	RCT	C & R	Pa	Molven et al., 1987	86	86		12–24
Maddalone and Gagliani, 2003	Prospective	C & R	Pa	Molven et al., 1987	79	120		3–36
Schwartz-Arad et al., 2003	Prospective	R	Pa	Other	101	122		6–45
Platt et al., 2004	RCT	C & R	Pa	Molven et al., 1987	28	34		12
Gagliani et al., 2005	Prospective	C & R	Pa	Rud et al., 1972	164	168	231	60
Lindeboom et al., 2005	RCT	C & R	Pa	Rud et al., 1972	100	100		12
Marti-Bowen et al., 2005	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	52	71	95	6–12
Taschieri et al., 2005	Prospective	C & R	Pa	Rud et al., 1972	32	46		12
Taschieri et al., 2007	Prospective	C & R	Pa	Molven et al., 1987	28	28		12

Table 17.11 Studies Investigating Periapical Healing Following Apical Surgery—cont'd

Author	Design	Examination	Type of Radiograph	Radiographic Criteria for Success	SAMPLE SIZE			Duration After Treatment (Months)
					Patients	Teeth	Roots	
Tsesis et al., 2006	Retrospective	C & R	Pa	Rud et al., 1972	71	88		6–48
Marin-Botero et al., 2006	RCT	C & R	Pa	Rud et al., 1972	30	30		12
Taschieri et al., 2006	RCT	C & R	Pa	Molven et al., 1987	53	71		12
De Lange et al., 2007	RCT	C & R	Pa	Rud et al., 1972	290	290		12
Leco-Berrocá et al., 2007	Prospective	R	Pa and DPT	Other	45	45		6–24
Penarrocha et al., 2007	Prospective	C & R	Pa	Von Arx and Kurt, 1999	235	333	384	6–144
Taschieri et al., 2007a	Prospective	C & R	Pa	Molven et al., 1987	17	27		12
Taschieri et al., 2007b	Prospective	C & R	Pa	Molven et al., 1987	41	59		12
Von Arx et al., 2007	Prospective	C & R	Pa	Rud et al., 1972	200	177		12
Walivaara et al., 2007	Prospective	C & R	Pa	Rud et al., 1972	54	55		12
Garcia et al., 2008	Prospective	C & R	DPT	Von Arx and Kurt, 1999	92	106	129	6–12
Kim et al., 2008	Prospective	C & R	Pa	Molven et al., 1987		148		24
Penarrocha et al., 2008	Prospective	C & R	DPT	Von Arx and Kurt, 1999	278	278		12
Taschieri et al., 2008b	Prospective	C & R	Pa	Molven et al., 1987	27	31		12
Taschieri et al., 2008a	RCT	C & R	Pa	Molven et al., 1987	61	100		25
Christiansen et al., 2009	RCT	C & R	Pa	Molven et al., 1987		25		12
Dominiak et al., 2009	Retrospective	C & R	Pa	Other	106	106		12
Ortega-Sanchez et al., 2009	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	30	30	37	
Pantshev et al., 2009	Retrospective	C & R	Pa	Other		147		12
Waalivaara et al., 2009	RCT	C & R	Pa	Molven et al., 1987	131	147		12
Barone et al., 2010	Prospective	C & R	Pa	PAI		129		48–120
Garcia-Mira et al., 2010	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	75	87		12
Taschieri et al., 2010	Retrospective	C & R	Pa	Molven et al., 1987	76	112		48
Von Arx et al., 2010	Prospective	C & R	Pa	Rud et al., 1972		339		12
Goyal et al., 2011	RCT	C & R	Pa	Rud et al., 1972	25	25		12
Penarrocha et al., 2011	Retrospective	C & R	DPT	Von Arx and Kurt, 1999	150	178	178	12
Song et al., 2011a	Retrospective	C & R	Pa	Molven et al., 1987		441		3–12
Song et al., 2011b	Prospective	C & R	Pa	Molven et al., 1987	42	42		12
Taschieri et al., 2011	Retrospective	C & R	Pa	Molven et al., 1987	33	43		12–48
Waalivaara et al., 2011	RCT	C & R	Pa	Molven et al., 1987	153	194		12–21
Penarrocha et al., 2012	Prospective	C & R	DPT	Von Arx and Kurt, 1999	23	31		12–19
Song and Kim, 2012	RCT	C & R	Pa	Molven et al., 1987	192	192		12
Von Arx et al., 2012	Prospective	C & R	Pa	Rud et al., 1972		170		12–60
Kreisler et al., 2013	Prospective	C & R	Pa	Rud et al., 1972	255	281		6–12
Penarrocha et al., 2013	Retrospective	C & R	Pa and DPT	Von Arx and Kurt	96	139		6–12
Song et al., 2013a	Prospective	C & R	Pa	Molven et al., 1987		344		12–120
Song et al., 2013b	Prospective	C & R	Pa	Molven et al., 1987		135		12
Song et al., 2014	Retrospective	C & R	Pa	Rud et al., 1972		115		12–96
Taschieri et al., 2013	Retrospective	C & R	Pa	Molven et al., 1987		86		6–12
Villa-Machado et al., 2013	Retrospective	C & R	Pa	PAI	154	171		12–192
Li et al., 2014	Retrospective	C & R	Pa	Molven et al., 1987	82	101		48

C, Clinical; DPT, dental pantomogram; Pa, periapical; PAI, periapical index; R, radiographic; RCT, randomized controlled trial.

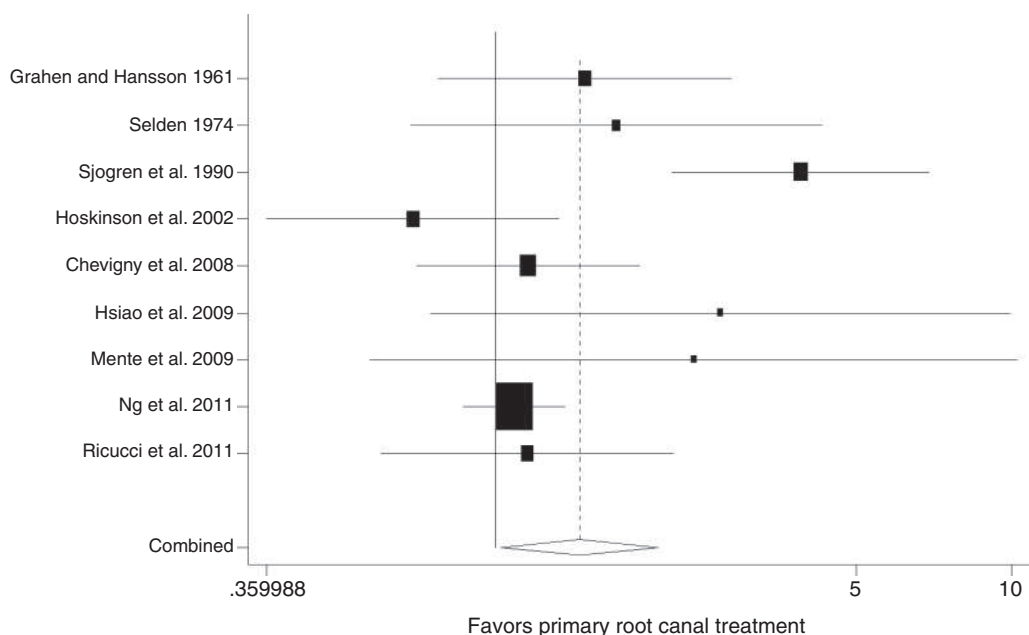


Fig. 17.32 Forest plot showing pooled and individual study's odds ratio of periapical health for teeth with preoperative periapical radiolucency undergoing primary root canal treatment versus root canal retreatment (pooled OR = 1.5; 95% CI: 1.0, 2.1).

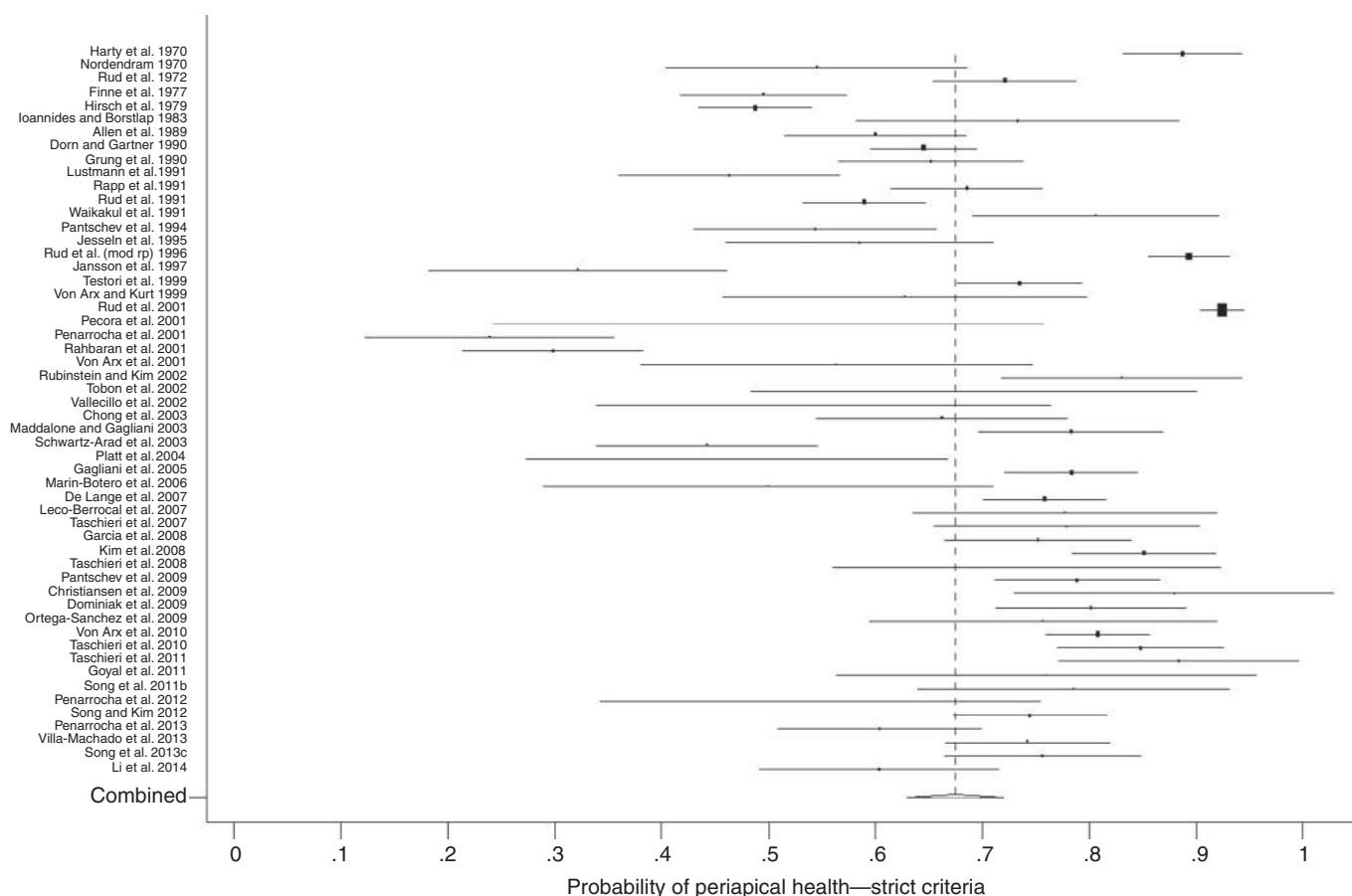


Fig. 17.33 Forest plot showing pooled and individual study's probability of complete periapical healing following apical surgery.

cement, or intermediate restorative material (IRM) versus amalgam; the use of MTA resulted in a similar outcome to SuperEBA (risk ratio = 1.0; 95% CI: 0.99, 1.1) or IRM (risk ratio = 1.1; 95% CI: 0.98, 1.1); SuperEBA and IRM were associated with significantly higher chances of success than amalgam (risk ratio = 1.2; 95% CI: 1.1, 1.3) as the retro-filling material

The preoperative presence of signs or symptoms (risk ratio = 1.2; 95% CI: 1.1, 1.3),²⁴⁹ periodontal status (risk ratio = 2.1; 95% CI: 1.1, 3.8),²²² and quality of the coronal restoration (risk ratio = 1.6; 95% CI: 1.2, 2.1)²⁴⁹ have also been revealed as significant prognostic factors by individual studies.

The following factors have minimal effect on surgical retreatment⁴³:

- Age of patient
- Gender of patient
- General health of patient
- Tooth type
- Quality of the preexisting root canal filling, as judged radiographically
- Histologic diagnosis of the biopsied periapical lesion (cyst or granuloma)

The use of guided periodontal tissue regenerative membrane or grafting material has been advocated for cases with through-and-through defects (missing buccal and palatal cortical plates), but there are conflicting reports as to their benefits.^{53,162,170,232-234,237,249} The unpublished meta-analyses¹³⁰ of data from these studies did not reveal a significant influence of such approaches on the periapical healing outcome.

FACTORS AFFECTING PERIODONTAL INCISIONAL WOUND HEALING

Periodontal attachment level after periapical surgery is an additional physical outcome measure compared to nonsurgical root canal treatment. Studies have compared the effect of different soft-tissue incision techniques (e.g., intra-sulcular with or without involving interproximal papilla, submarginal, papilla-base).^{106,245-247} As expected, all concluded that negligible marginal recession could be achieved by adopting a flap design that avoided reflection of the interproximal papilla.

The improvement in outcomes of periapical surgery has been attributed to modern surgical techniques plus greater biologic awareness of clinicians.^{205,250} In addition, and perhaps even more importantly, case selection may be more critical in excluding potential failures, inferring that prognostication may have improved.

References

1. Aguilar P, Linsuwanont P: Vital pulp therapy in vital permanent teeth with cariously exposed pulp: a systematic review, *J Endod* 37:581, 2011.
2. Akpata ES: Effect of endodontic procedures on the population of viable microorganisms in the infected root canal, *J Endod* 2:369, 1976.
3. Alley BS, Kitchens GG, Alley LW, et al: A comparison of survival of teeth following endodontic treatment performed by general dentists or by specialists, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 98:115, 2004.

FACTORS AFFECTING TOOTH SURVIVAL FOLLOWING PERIAPICAL SURGERY AND ROOT-END FILLING

Unlike nonsurgical root canal treatment, the outcome measure of tooth survival had only been adopted by one study²⁵³ on surgical cases at the time of this writing. The reported median survival time for first-time surgery was 92.1 months (95% CI: 40.9, 143.4) and that for resurgery was 39.1 months (95% CI: 6.1, 72.1).²⁵³ The failure events, however, included tooth extraction and clinical and radiographic signs of periapical disease after treatment.

IMPACT OF PERIAPICAL SURGERY ON QUALITY OF LIFE

The impact of periapical surgery on the patient's quality of life has only been evaluated using a questionnaire including three domains: physical function (chewing, talking, sleeping, daily routine, and work), physical pain, and other physical symptoms (swelling, bleeding, nausea, bad taste/breath).⁴⁹ It was concluded that the papilla-base incision flap design resulted in a lower impact on physical pain and other symptoms within the first week postoperation. The impact on physical pain has also been explored by two other studies^{35,36}; both reported the postoperative pain was of relatively short duration, with the intensity peaking at 3 to 5 hours postoperatively and progressively decreasing with time with no significant influencing factors identified.

Concluding Remarks

The procedures used to maintain pulp vitality and for prevention and treatment of periapical disease are able to achieve excellent outcomes. The outcome data and potential prognostic factors should be considered during treatment options appraisal and planning. Despite the fact that most important prognostic factors are beyond the control of clinicians, optimal outcomes for individual cases may still be achieved by performing the procedure to guideline standards. From a health and economic perspective, conventional root canal treatment is a highly cost-effective treatment as a first-line intervention to extend the life of the tooth with a preoperative periapical lesion.^{103,171} If a root canal treatment subsequently fails, nonsurgical and surgical retreatments are also more cost-effective than replacement with a prosthesis.^{103,171} Ultimately, all sources of evidence must be assessed for biasing influences based on the local cultures, expertise, treatment predilection, and funding sources.

4. Allison DA, Weber CR, Walton RE: The influence of the method of canal preparation on the quality of apical and coronal obturation, *J Endod* 5:298, 1979.
5. Arai Y, Honda K, Iwai K, et al: Practical model "3DX" of limited cone-beam X-ray CT for dental use, *Comput Assist Radiol Surg* 1230:671, 2001.
6. Asgary S, Ehsani S: Permanent molar pulpotomy with a new endodontic cement: a case series, *J Conserv Dent* 12:31, 2009.
7. Auerbach M: Antibiotics vs. instrumentation in endodontics, *N Y State Dent J* 19:225, 1953.

8. Augsburger RA, Peters DD: Radiographic evaluation of extruded obturation materials, *J Endod* 16:492, 1990.
9. Bader JD, Shugars DA: Variation, treatment outcomes, and practice guidelines in dental practice, *J Dent Educ* 59:61, 1995.
10. Baratieri LN, Monteiro S Jr, Caldeira de Andrada MA: Pulp curettage: surgical technique, *Quintessence Int* 20:285, 1989.
11. Barrieshi-Nusair KM, Qudeimat MA: A prospective clinical study of mineral trioxide aggregate for partial pulpotomy in cariously exposed permanent teeth, *J Endod* 32:731, 2006.
12. Barthel CR, Rosenkranz B, Leuenberg A, et al: Pulp capping of carious exposures: treatment outcome after 5 and 10 years: a retrospective study, *J Endod* 26:525, 2000.
13. Basrani B, Tjaderhane L, Santos JM, et al: Efficacy of chlorhexidine- and calcium hydroxide-containing medicaments against *Enterococcus faecalis* in vitro, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 96:618, 2003.
14. Basrani BR, Manek S, Sodhi RN, et al: Interaction between sodium hypochlorite and chlorhexidine gluconate, *J Endod* 33:966, 2007.
15. Baugh D, Wallace J: The role of apical instrumentation in root canal treatment: a review of the literature, *J Endod* 31:333, 2005.
16. Bence R, Madonia JV, Weine FS, et al: A microbiologic evaluation of endodontic instrumentation in pulpless teeth, *Oral Surg Oral Med Oral Pathol* 35:676, 1973.
17. Bender IB, Seltzer S, Soltanof W: Endodontic success: a reappraisal of criteria, 1, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 22:780, 1966.
18. Bender IB, Seltzer S, Soltanof W: Endodontic success: a reappraisal of criteria, 2, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 22:790, 1966.
19. Bjørndal L, Reit C, Bruun G, et al: Treatment of deep caries lesions in adults: randomized clinical trials comparing stepwise vs. direct complete excavation, and direct pulp capping vs. partial pulpotomy, *Eur J Oral Sci* 118:290, 2010.
20. Bjørndal L, Thylstrup A: A practice-based study on stepwise excavation of deep carious lesions in permanent teeth: a 1-year follow-up study, *Community Dent Oral Epidemiol* 26:122, 1998.
21. Bogen G, Kim JS, Bakland LK: Direct pulp capping with mineral trioxide aggregate: an observational study, *J Am Dent Assoc* 139:305; quiz 305, 2008.
22. Boggia R: A single-visit treatment of septic root canals using periapically extruded endomethasone, *Br Dent J* 155:300, 1983.
23. Buchbinder M, Wald A: An improved method of culturing root canals, *J Am Dent Assoc* 26:1697, 1939.
24. Bui TB, Baumgartner JC, Mitchell JC: Evaluation of the interaction between sodium hypochlorite and chlorhexidine gluconate and its effect on root dentin, *J Endod* 34:181, 2008.
25. Byström A: *Evaluation of endodontic treatment of teeth with apical periodontitis*, Sweden, 1986, University of Umeå.
26. Byström A, Claesson R, Sundqvist G: The antibacterial effect of camphorated paramonochlorophenol, camphorated phenol and calcium hydroxide in the treatment of infected root canals, *Endod Dent Traumatol* 1:170, 1985.
27. Byström A, Sundqvist G: Bacteriologic evaluation of the efficacy of mechanical root canal instrumentation in endodontic therapy, *Scand J Dent Res* 89:321, 1981.
28. Byström A, Sundqvist G: Bacteriologic evaluation of the effect of 0.5 percent sodium hypochlorite in endodontic therapy, *Oral Surg Oral Med Oral Pathol* 55:307, 1983.
29. Byström A, Sundqvist G: The antibacterial action of sodium hypochlorite and EDTA in 60 cases of endodontic therapy, *Int Endod J* 18:35, 1985.
30. Caliskan MK: Success of pulpotomy in the management of hyperplastic pulpitis, *Int Endod J* 26:142, 1993.
31. Caliskan MK: Pulpotomy of carious vital teeth with periapical involvement, *Int Endod J* 28:172, 1995.
32. Camps J, Pommel L, Bukiet F: Evaluation of periapical lesion healing by correction of gray values, *J Endod* 30:762, 2004.
33. Card SJ, Sigurdsson A, Orstavik D, et al: The effectiveness of increased apical enlargement in reducing intracanal bacteria, *J Endod* 28:779, 2002.
34. Chambers D: Outcomes-based practice: how outcomes-based practices get better, *Dental Economics* 3:34, 2001.
35. Chong BS, Pitt Ford TR: Postoperative pain after root-end resection and filling, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 100:762, 2005.
36. Christiansen R, Kirkevang LL, Horsted-Bindslev P, et al: Patient discomfort following periapical surgery, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 105:245, 2008.
37. Chu FC, Leung WK, Tsang PC, et al: Identification of cultivable microorganisms from root canals with apical periodontitis following two-visit endodontic treatment with antibiotics/steroid or calcium hydroxide dressings, *J Endod* 32:17, 2006.
38. Chugal NM, Clive JM, Spangberg LS: Endodontic treatment outcome: effect of the permanent restoration, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 104:576, 2007.
39. Colman I, Friedman BW, Brown MD, et al: Parenteral dexamethasone for acute severe migraine headache: meta-analysis of randomised controlled trials for preventing recurrence, *BMJ* 336:13591, 2008.
40. Cvek M: A clinical report on partial pulpotomy and capping with calcium hydroxide in permanent incisors with complicated crown fracture, *J Endod* 4:232, 1978.
41. Cvek M, Hollender L, Nord CE: Treatment of non-vital permanent incisors with calcium hydroxide. VI. A clinical, microbiological and radiological evaluation of treatment in one sitting of teeth with mature or immature root, *Odontol Revy* 27:93, 1976.
42. Dalton BC, Orstavik D, Phillips C, et al: Bacterial reduction with nickel-titanium rotary instrumentation, *J Endod* 24:763, 1998.
43. de Cheigny C, Dao TT, Basrani BR, et al: Treatment outcome in endodontics: the Toronto study—phase 4: initial treatment, *J Endod* 34:258, 2008.
44. de Paula-Silva FW, Wu MK, Leonardo MR, et al: Accuracy of periapical radiography and cone-beam computed tomography scans in diagnosing apical periodontitis using histopathological findings as a gold standard, *J Endod* 35:10092, 2009.
45. Del Fabbro M, Taschieri S: Endodontic therapy using magnification devices: a systematic review, *J Dent* 38:269, 2010.
46. Del Fabbro M, Taschieri S, Lodi G, et al: Magnification devices for endodontic therapy, *Cochrane Database Syst Rev* CD005969, 2009.
47. Del Fabbro M, Taschieri S, Lodi G, et al: Magnification devices for endodontic therapy, *Cochrane Database Syst Rev* CD005969, 2009.
48. Del Fabbro M, Taschieri S, Testori T, et al: Surgical versus non-surgical endodontic retreatment for periradicular lesions, *Cochrane Database Syst Rev* CD005511, 2007.
49. Del Fabbro M, Taschieri S, Weinstein R: Quality of life after microscopic periradicular surgery using two different incision techniques: a randomized clinical study, *Int Endod J* 42:360, 2009.
50. Delano EO, Tyndall D, Ludlow JB, et al: Quantitative radiographic follow-up of apical surgery: a radiometric and histologic correlation, *J Endod* 24:420, 1998.
51. DePalma MJ, Slipman CW: Evidence-informed management of chronic low back pain with epidural steroid injections, *Spine J* 8:45, 2008.
52. DeRosa TA: A retrospective evaluation of pulpotomy as an alternative to extraction, *Gen Dent* 54:37, 2006.
53. Dominiak M, Lysiak-Drwal K, Gedrange T, et al: Efficacy of healing process of bone defects after apicectomy: results after 6 and 12 months, *J Physiol Pharmacol* 60(Suppl 8):51, 2009.
54. Doyle SL, Hodges JS, Pesun JJ, et al: Retrospective cross sectional comparison of initial nonsurgical endodontic treatment and single-tooth implants, *J Endod* 32:822, 2006.
55. Dugas NN, Lawrence HP, Teplitzky P, et al: Quality of life and satisfaction outcomes of endodontic treatment, *J Endod* 28:819, 2002.
56. Eakle WS, Maxwell EH, Braly BV: Fractures of posterior teeth in adults, *J Am Dent Assoc* 112:215, 1986.
57. Edwards JL, Vincent AM, Cheng HT, et al: Diabetic neuropathy: mechanisms to management, *Pharmacol Ther* 120:1, 2008.
58. El-Meligy OA, Avery DR: Comparison of mineral trioxide aggregate and calcium hydroxide as pulpotomy agents in young permanent teeth (apexogenesis), *Pediatr Dent* 28:399, 2006.
59. American Association of Endodontists: *Evolution of the root canal* www.aae.org/uploadedfiles/news_room/press_releases/rcawendodontictimeline.pdf. Accessed June 2015.
60. Engström B: The significance of enterococci in root canal treatment, *Odontol Revy* 15:87, 1964.
61. European Society of Endodontology: Quality guidelines for endodontic treatment: consensus report of the European Society of Endodontology, *Int Endod J* 39:921, 2006.

62. European Society of Endodontology, Patel S, Durack C, et al: European Society of Endodontology position statement: the use of CBCT in endodontics, *Int Endod J* 47:502, 2014.
63. Fabricius L, Dahlen G, Sundqvist G, et al: Influence of residual bacteria on periapical tissue healing after chemomechanical treatment and root filling of experimentally infected monkey teeth, *Eur J Oral Sci* 114:278, 2006.
64. Farsi N, Alamoudi N, Balto K, et al: Clinical assessment of mineral trioxide aggregate (MTA) as direct pulp capping in young permanent teeth, *J Clin Pediatr Dent* 31:72, 2006.
65. Fitzgerald M, Heys RJ: A clinical and histological evaluation of conservative pulpal therapy in human teeth, *Oper Dent* 16:101, 1991.
66. Ford TP: Frequency of radiological review in pulp studies, *Int Endod J* 41:931, 2008.
67. Fouad AF, Burleson J: The effect of diabetes mellitus on endodontic treatment outcome: data from an electronic patient record, *J Am Dent Assoc* 134:43; quiz 117, 2003.
68. Friedman S, Lost C, Zarrabian M, Trope M: Evaluation of success and failure after endodontic therapy using a glass ionomer cement sealer, *J Endod* 21:384, 1995.
69. Friedman S, Mor C: The success of endodontic therapy—healing and functionality, *J Calif Dent Assoc* 32:493, 2004.
70. Frostell G: Clinical significance of the root canal culture. In, *Transactions of Third International Conference of Endodontics*, 1963, pp 112–122.
71. Fuks AB, Gavra S, Chosack A: Long-term followup of traumatized incisors treated by partial pulpotomy, *Pediatr Dent* 15:334, 1993.
72. Gallien GS Jr, Schuman NJ: Local versus general anesthesia: a study of pulpal response in the treatment of cariously exposed teeth, *J Am Dent Assoc* 111:599, 1985.
73. Gatten DL, Riedy CA, Hong SK, et al: Quality of life of endodontically treated versus implant treated patients: a University-based qualitative research study, *J Endod* 37:903, 2011.
74. Genco RJ, Loe H: The role of systemic conditions and disorders in periodontal disease, *Periodontol* 2000 2:98, 1993.
75. Gesi A, Hakeberg M, Warfvinge J, et al: Incidence of periapical lesions and clinical symptoms after pulpectomy—a clinical and radiographic evaluation of 1- versus 2-session treatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 101:379, 2006.
76. Goldfein J, Speirs C, Finkelman M, et al: Rubber dam use during post placement influences the success of root canal-treated teeth, *J Endod* 39:14814, 2013.
77. Gomes BP, Lilley JD, Drucker DB: Variations in the susceptibilities of components of the endodontic microflora to biomechanical procedures, *Int Endod J* 29:235, 1996.
78. Gomes BP, Souza SF, Ferraz CC, et al: Effectiveness of 2% chlorhexidine gel and calcium hydroxide against *Enterococcus faecalis* in bovine root dentine in vitro, *Int Endod J* 36:267, 2003.
79. Grahnen H, Krasse B: The effect of instrumentation and flushing of non-vital teeth in endodontic therapy, *Odontol Rev* 14:167, 1963.
80. Gratt BM, Sickles EA, Gould RG, et al: Xeroradiography of dental structures. IV. Image properties of a dedicated intraoral system, *Oral Surg Oral Med Oral Pathol* 50:572, 1980.
81. Gruythuysen RJ, van Strijp AJ, et al: Long-term survival of indirect pulp treatment performed in primary and permanent teeth with clinically diagnosed deep carious lesions, *J Endod* 36:14903, 2010.
82. Gulabivala K: *Species richness of gram-positive coccoid morphotypes isolated from untreated and treated root canals of teeth associated with periapical disease*, London, 2004, University of London.
83. Gulabivala K, Ng Y: Endodontology. In Wilson N, editor: *Clinical dental medicine 2020*, New Malden, Surrey, 2009, Quintessence, pp 147–182.
84. Gulabivala K, Ng YL, Gilbertson M, et al: The fluid mechanics of root canal irrigation, *Physiol Meas* 31:R49, 2010.
85. Gulabivala K, Patel B, Evans G, et al: Effects of mechanical and chemical procedures on root canal surfaces, *Endodontic Topics* 10:103, 2005.
86. Halse A, Molven O: Increased width of the apical periodontal membrane space in endodontically treated teeth may represent favourable healing, *Int Endod J* 37:552, 2004.
87. Hamasha AA, Hatiwsh A: Quality of life and satisfaction of patients after nonsurgical primary root canal treatment provided by undergraduate students, graduate students and endodontic specialists, *Int Endod J* 46:11319, 2013.
88. Haskell EW, Stanley HR, Chellemi J, et al: Direct pulp capping treatment: a long-term follow-up, *J Am Dent Assoc* 97:607, 1978.
89. Hayashi M, Fujitani M, Yamaki C, et al: Ways of enhancing pulp preservation by stepwise excavation—a systematic review, *J Dent* 39:95, 2011.
90. Hilton TJ, Ferracane JL, Mancl L: Northwest Practice-based Research Collaborative in Evidence-based Dentistry (NWP): Comparison of CaOH with MTA for direct pulp capping: a PBRN randomized clinical trial, *J Dent Res* 92:16SS, 2013.
91. Holroyd J, Gulson A: *The radiation protection implications of the use of cone beam computed tomography (CBCT) in dentistry: what you need to know*, London, 2009, Health Protection Agency.
92. Horsted P, Sandergaard B, Thylstrup A, et al: A retrospective study of direct pulp capping with calcium hydroxide compounds, *Endod Dent Traumatol* 1:29, 1985.
93. Hoskinson SE, Ng YL, Hoskinson AE, et al: A retrospective comparison of outcome of root canal treatment using two different protocols, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:705, 2002.
94. Hughes R: Focal infection revisited, *Rheumatology* 33:370, 1994.
95. Hülsmann M, Peters OA, Dummer PM: Mechanical preparation of root canals: shaping goals, techniques and means, *Endod Topics* 10:30, 2005.
96. Hunter W: The role of sepsis and antiseptics in medicine, *Lancet* 1:79, 1911.
97. Hunter W: Chronic sepsis as a cause of mental disorder, *Br J Psychiatr* 73:549, 1927.
98. Huuomonen S, Lenander-Lumikari M, Sigurdsson A, et al: Healing of apical periodontitis after endodontic treatment: a comparison between a silicone-based and a zinc oxide-eugenol-based sealer, *Int Endod J* 36:296, 2003.
99. Ingle J, Zeldow B: An evaluation of mechanical instrumentation and the negative culture in endodontic therapy, *J Am Dent Assoc* 57:471, 1958.
100. Iqbal MK, Kim S: For teeth requiring endodontic treatment, what are the differences in outcomes of restored endodontically treated teeth compared to implant-supported restorations? *Int J Oral Maxillofac Implants* 22:96, 2006.
101. Kalichman L, Hunter DJ: Diagnosis and conservative management of degenerative lumbar spondylolisthesis, *Eur Spine J* 17:327, 2008.
102. Kerekes K, Tronstad L: Long-term results of endodontic treatment performed with a standardized technique, *J Endod* 5:83, 1979.
103. Kim SG, Solomon C: Cost-effectiveness of endodontic molar retreatment compared with fixed partial dentures and single-tooth implant alternatives, *J Endod* 37:321, 2011.
104. Koontongkaew S, Silapichit R, Thaweboon B: Clinical and laboratory assessments of camphorated monochlorophenol in endodontic therapy, *Oral Surg Oral Med Oral Pathol* 65:757, 1988.
105. Koppang HS, Koppang R, Stolen SO: Identification of common foreign material in postendodontic granulomas and cysts, *J Dent Assoc S Afr* 47:210, 1992.
106. Kreisler M, Gockel R, Schmidt I, et al: Clinical evaluation of a modified marginal sulcular incision technique in endodontic surgery, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 108:e22, 2009.
107. Kuruvilla JR, Kamath MP: Antimicrobial activity of 2.5% sodium hypochlorite and 0.2% chlorhexidine gluconate separately and combined, as endodontic irrigants, *J Endod* 24:472, 1998.
108. Kvist T, Molander A, Dahlen G, et al: Microbiological evaluation of one- and two-visit endodontic treatment of teeth with apical periodontitis: a randomized, clinical trial, *J Endod* 30:572, 2004.
109. Lagouvardos P, Sourai P, Douvitsas G: Coronal fractures in posterior teeth, *Oper Dent* 14:28, 1989.
110. Lana MA, Ribeiro-Sobrinho AP, Stehling R, et al: Microorganisms isolated from root canals presenting necrotic pulp and their drug susceptibility in vitro, *Oral Microbiol Immunol* 16:100, 2001.
111. Lazarski MP, Walker WA III, Flores CM, et al: Epidemiological evaluation of the outcomes of nonsurgical root canal treatment in a large cohort of insured dental patients, *J Endod* 27:791, 2001.
112. Lee SJ, Wu MK, Wesslink PR: The efficacy of ultrasonic irrigation to remove artificially placed dentine debris from different-sized simulated plastic root canals, *Int Endod J* 37:607, 2004.
113. Leksell E, Ridell K, Cvek M, et al: Pulp exposure after stepwise versus direct complete excavation of deep carious lesions in young posterior permanent teeth, *Endod Dent Traumatol* 12:192, 1996.

114. Liang YH, Li G, Wesselink PR, et al: Endodontic outcome predictors identified with periapical radiographs and cone-beam computed tomography scans, *J Endod* 37:326, 2011.
115. Liu P, McGrath C, Cheung GS: Improvement in oral health-related quality of life after endodontic treatment: a prospective longitudinal study, *J Endod* 40:805, 2014.
116. Locker D: Concepts of health, disease and quality of life. In Slade GD, editor: *Measuring oral health and quality of life*, Chapel Hill, North Carolina, 1997, University of North Carolina, pp 11-23.
117. Loe H: Does chlorhexidine have a place in the prophylaxis of dental diseases? *J Periodontol Res Suppl* 12:93, 1973.
118. Lumley PJ, Lucarotti PS, Burke FJ: Ten-year outcome of root fillings in the General Dental Services in England and Wales, *Int Endod J* 41:577, 2008.
119. Mah JK, Danforth RA, Bumann A, et al: Radiation absorbed in maxillofacial imaging with a new dental computed tomography device, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 96:508, 2003.
120. Maltz M: Does incomplete caries removal increase restoration failure? *J Dent Res* 90:541; author reply 542, 2011.
121. Maltz M, Garcia R, Jardim JJ, et al: Randomized trial of partial vs. stepwise caries removal: 3-year follow-up, *J Dent Res* 91:10261, 2012.
122. Maltz M, Henz SL, de Oliveira EF, et al: Conventional caries removal and sealed caries in permanent teeth: a microbiological evaluation, *J Dent* 40:776, 2012.
123. Marending M, Peters OA, Zehnder M: Factors affecting the outcome of orthograde root canal therapy in a general dentistry hospital practice, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 99:119, 2005.
124. Markqvist M, Dahlen G, Reit CE, et al: The antimicrobial effect of apical box versus apical cone preparation using iodine potassium iodide as root canal dressing: a pilot study, *Acta Odontol Scand* 71:786, 2013.
125. Mass E, Zilberman U: Clinical and radiographic evaluation of partial pulpotomy in carious exposure of permanent molars, *Pediatr Dent* 15:257, 1993.
126. Masterton JB: The healing of wounds of the dental pulp: an investigation of the nature of the scar tissue and of the phenomena leading to its formation, *Dent Pract Dent Rec* 16:325, 1966.
127. Matsumoto M, Goto T: Lateral force distribution in partial denture design, *J Dent Res* 49:359, 1970.
128. Matsuo T, Nakanishi T, Shimizu H, et al: A clinical study of direct pulp capping applied to carious-exposed pulps, *J Endod* 22:551, 1996.
129. McDonnell G, Russell AD: Antiseptics and disinfectants: activity, action, and resistance, *Clinical Microbiol Rev* 12:147, 1999.
130. Mehta D, Gulabivala K, Ng Y-L: Personal communication. Outcome of periapical surgery, unpublished manuscript, 2014.
131. Mejare I, Cvek M: Partial pulpotomy in young permanent teeth with deep carious lesions, *Endod Dent Traumatol* 9:238, 1993.
132. Mente J, Geletneký B, Ohle M, et al: Mineral trioxide aggregate or calcium hydroxide direct pulp capping: an analysis of the clinical treatment outcome, *J Endod* 36:806, 2010.
133. Miles JP, Gluskin AH, Chambers D, et al: Pulp capping with mineral trioxide aggregate (MTA): a retrospective analysis of carious pulp exposures treated by undergraduate dental students, *Oper Dent* 35:20, 2010.
134. Miller WD: An introduction to the study of the bacterio-pathology of the dental pulp, *The Dental Cosmos* 36:505-528, 1894.
135. Mindiola MJ, Mickel AK, Sami C, et al: Endodontic treatment in an American Indian population: a 10-year retrospective study, *J Endod* 32:828, 2006.
136. Miyashita H, Worthington HV, Qualtrough A, et al: Pulp management for caries in adults: maintaining pulp vitality (review), *Cochrane Database Syst Rev* 2007.
137. Molander A, Reit C, Dahlen G: Microbiological evaluation of clindamycin as a root canal dressing in teeth with apical periodontitis, *Int Endod J* 23:113, 1990.
138. Molander A, Reit C, Dahlen G: Microbiological root canal sampling: diffusion of a technology, *Int Endod J* 29:163, 1996.
139. Molander A, Reit C, Dahlen G: Reasons for dentists' acceptance or rejection of microbiological root canal sampling, *Int Endod J* 29:168, 1996.
140. Molven O, Halse A, Grung B: Observer strategy and the radiographic classification of healing after endodontic surgery, *Int J Oral Maxillofac Surg* 16:432, 1987.
141. Moore DL: Conservative treatment of teeth with vital pulps and periapical lesions: a preliminary report, *J Prosthet Dent* 18:476, 1967.
142. Morsani JM, Aminoshariae A, Han YW, et al: Genetic predisposition to persistent apical periodontitis, *J Endod* 37:455, 2011.
143. Morse F, Yates M: Follow up studies of root-filled teeth in relation to bacteriologic findings, *J Am Dent Assoc* 28:956, 1941.
144. Nagasiri R, Chitmongkolsuk S: Long-term survival of endodontically treated molars without crown coverage: a retrospective cohort study, *J Prosthet Dent* 93:164, 2005.
145. Nair PN: On the causes of persistent apical periodontitis: a review, *Int Endod J* 39:249, 2006.
146. Nair PN, Sjogren U, Krey G, et al: Therapy-resistant foreign body giant cell granuloma at the periapex of a root-filled human tooth, *J Endod* 16:589, 1990.
147. Nair PNR, Henry S, Cano V, et al: Microbial status of apical root canal system of human mandibular first molars with primary apical periodontitis after "one-visit" endodontic treatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 99:231, 2005.
148. Ng YL, Mann V, Gulabivala K: Outcome of secondary root canal treatment: a systematic review of the literature, *Int Endod J* 41:10266, 2008.
149. Ng YL, Mann V, Gulabivala K: Tooth survival following non-surgical root canal treatment: a systematic review of the literature, *Int Endod J* 43:171, 2010.
150. Ng YL, Mann V, Gulabivala K: A prospective study of the factors affecting outcomes of nonsurgical root canal treatment: part 1: periapical health, *Int Endod J* 44:583, 2011.
151. Ng YL, Mann V, Gulabivala K: A prospective study of the factors affecting outcomes of non-surgical root canal treatment: part 2: tooth survival, *Int Endod J* 44:610, 2011.
152. Ng YL, Mann V, Rahbaran S, et al: Outcome of primary root canal treatment: systematic review of the literature. Part 1. Effects of study characteristics on probability of success, *Int Endod J* 40:921, 2007.
153. Ng YL, Mann V, Rahbaran S, et al: Outcome of primary root canal treatment: systematic review of the literature. Part 2. Influence of clinical factors, *Int Endod J* 41:6, 2008.
154. Nicholls E: The efficacy of cleansing of the root canal, *Br Dent J* 112:167, 1962.
155. Niederman R, Theodosopoulou JN: A systematic review of in vivo retrograde obturation materials, *Int Endod J* 36:577, 2003.
156. Orstavik D: Radiographic evaluation of apical periodontitis and endodontic treatment results: a computer approach, *Int Dent J* 41:89, 1991.
157. Orstavik D: Time-course and risk analyses of the development and healing of chronic apical periodontitis in man, *Int Endod J* 29:150, 1996.
158. Orstavik D, Farrants G, Wahl T, et al: Image analysis of endodontic radiographs: digital subtraction and quantitative densitometry, *Endod Dent Traumatol* 6:6, 1990.
159. Orstavik D, Horsted-Bindslev P: A comparison of endodontic treatment results at two dental schools, *Int Endod J* 26:348, 1993.
160. Orstavik D, Kerekes K, Eriksen HM: Clinical performance of three endodontic sealers, *Endod Dent Traumatol* 3:178, 1987.
161. Orstavik D, Kerekes K, Molven O: Effects of extensive apical reaming and calcium hydroxide dressing on bacterial infection during treatment of apical periodontitis: a pilot study, *Int Endod J* 24:1, 1991.
162. Pantchev A, Nohler E, Tegelberg A: Endodontic surgery with and without inserts of bioactive glass PerioGlas: a clinical and radiographic follow-up, *Oral Maxillofac Surg* 13:21, 2009.
163. Paquette L, Legner M, Fillery ED, et al: Antibacterial efficacy of chlorhexidine gluconate intracanal medication in vivo, *J Endod* 33:788, 2007.
164. Parris J, Wilcox L, Walton R: Effectiveness of apical clearing: histological and radiographical evaluation, *J Endod* 20:219, 1994.
165. Patel S, Dawood A, Mannocci F, et al: Detection of periapical bone defects in human jaws using cone beam computed tomography and intraoral radiography, *Int Endod J* 42:507, 2009.
166. Patel S, Rhodes J: A practical guide to endodontic access cavity preparation in molar teeth, *Br Dent J* 203:133, 2007.

167. Patel S, Wilson R, Dawood A, et al: The detection of periapical pathosis using digital periapical radiography and cone beam computed tomography—part 2: a 1-year post-treatment follow-up, *Int Endod J* 45:711, 2012.
168. Peciuliene V, Balciuniene I, Eriksen HM, et al: Isolation of *Enterococcus faecalis* in previously root-filled canals in a Lithuanian population, *J Endod* 26:593, 2000.
169. Peciuliene V, Reynaud AH, Balciuniene I, et al: Isolation of yeasts and enteric bacteria in root-filled teeth with chronic apical periodontitis, *Int Endod J* 34:429, 2001.
170. Pecora G, De Leonardis D, Ibrahim N, et al: The use of calcium sulphate in the surgical treatment of a 'through and through' periradicular lesion, *Int Endod J* 34:189, 2001.
171. Pennington MW, Vernazza CR, Shackley P, et al: Evaluation of the cost-effectiveness of root canal treatment using conventional approaches versus replacement with an implant, *Int Endod J* 42:874, 2009.
172. Percinoto C, de Castro AM, Pinto LM: Clinical and radiographic evaluation of pulpotomies employing calcium hydroxide and trioxide mineral aggregate, *Gen Dent* 54:258, 2006.
173. Peters LB, van Winkelhoff AJ, Buijs JF, et al: Effects of instrumentation, irrigation and dressing with calcium hydroxide on infection in pulpless teeth with periapical bone lesions, *Int Endod J* 35:13, 2002.
174. Petersson A, Axelsson S, Davidson T, et al: Radiological diagnosis of periapical bone tissue lesions in endodontics: a systematic review, *Int Endod J* 45:783, 2012.
175. Pettiette MT, Delano EO, Trope M: Evaluation of success rate of endodontic treatment performed by students with stainless-steel K-files and nickel-titanium hand files, *J Endod* 27:124, 2001.
176. Polycarpou N, Ng YL, Canavan D, et al: Prevalence of persistent pain after endodontic treatment and factors affecting its occurrence in cases with complete radiographic healing, *Int Endod J* 38:169, 2005.
177. Qudeimat MA, Barrieshi-Nusair KM, Owais AI: Calcium hydroxide vs mineral trioxide aggregates for partial pulpotomy of permanent molars with deep caries, *Eur Arch Paediatr Dent* 8:99, 2007.
178. Reeh ES, Messer HH, Douglas WH: Reduction in tooth stiffness as a result of endodontic and restorative procedures, *J Endod* 15:512, 1989.
179. Reit C, Dahlen G: Decision making analysis of endodontic treatment strategies in teeth with apical periodontitis, *Int Endod J* 21:291, 1988.
180. Reit C, Grondahl HG: Endodontic retreatment decision making among a group of general practitioners, *Scand J Dent Res* 96:112, 1988.
181. Reit C, Molander A, Dahlen G: The diagnostic accuracy of microbiologic root canal sampling and the influence of antimicrobial dressings, *Endod Dent Traumatol* 15:278, 1999.
182. Ricucci D, Russo J, Rutberg M, et al: A prospective cohort study of endodontic treatments of 1,369 root canals: results after 5 years, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 112:825, 2011.
183. Rocas IN, Siqueira JF Jr, Del Aguila CA, et al: Polymorphism of the CD14 and TLR4 genes and post-treatment apical periodontitis, *J Endod* 40:168, 2014.
184. Rollison S, Barnett F, Stevens RH: Efficacy of bacterial removal from instrumented root canals in vitro related to instrumentation technique and size, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 94:366, 2002.
185. Rosenthal S, Spangberg L, Safavi K: Chlorhexidine substantivity in root canal dentin, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 98:488, 2004.
186. Rud J, Andreasen JO: A study of failures after endodontic surgery by radiographic, histologic and stereomicroscopic methods, *Int J Oral Surg* 1:311, 1972.
187. Russo MC, Holland R, de Souza V: Radiographic and histological evaluation of the treatment of inflamed dental pulps, *Int Endod J* 15:137, 1982.
188. Saini HR, Tewari S, Sangwan P, et al: Effect of different apical preparation sizes on outcome of primary endodontic treatment: a randomized controlled trial, *J Endod* 38:1309, 2012.
189. Salehrabi R, Rotstein I: Endodontic treatment outcomes in a large patient population in the USA: an epidemiological study, *J Endod* 30:846, 2004.
190. Salehrabi R, Rotstein I: Epidemiologic evaluation of the outcomes of orthograde endodontic retreatment, *J Endod* 36:790, 2010.
191. Salvi GE, Siegrist Guldener BE, Amstad T, et al: Clinical evaluation of root filled teeth restored with or without post-and-core systems in a specialist practice setting, *Int Endod J* 40:209, 2007.
192. Santini A: Assessment of the pulpotomy technique in human first permanent mandibular molars. Use of two direct inspection criteria, *Br Dent J* 155:151, 1983.
193. Santucci PJ: Dycal versus Nd:YAG laser and Vitrebond for direct pulp capping in permanent teeth, *J Clin Laser Med Surg* 17:69, 1999.
194. Sari S, Okte Z: Success rate of Sealapex in root canal treatment for primary teeth: 3-year follow-up, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 105:e93, 2008.
195. Sathorn C, Parashos P, Messer H: Antibacterial efficacy of calcium hydroxide intracanal dressing: a systematic review and meta-analysis, *Int Endod J* 40:2, 2007.
196. Saunders WP, Saunders EM: Coronal leakage as a cause of failure in root-canal therapy: a review, *Endod Dent Traumatol* 10:105, 1994.
197. Scarfe W: Use of cone-beam computed tomography in endodontics Joint Position Statement of the American Association of Endodontists and the American Academy of Oral and Maxillofacial Radiology, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 111:234, 2011.
198. Schafer E, Bossmann K: Antimicrobial efficacy of chlorhexidine and two calcium hydroxide formulations against *Enterococcus faecalis*, *J Endod* 31:53, 2005.
199. Schilder H: Cleaning and shaping the root canal, *Dent Clin North Am* 18:269, 1974.
200. Schwendicke F, Stolpe M, Meyer-Lueckel H, et al: Cost-effectiveness of one- and two-step incomplete and complete excavations, *J Dent Res* 92:880, 2013.
201. Seltzer S, Bender IB, Turkenkopf S: Factors affecting successful repair after root canal therapy, *J Am Dent Assoc* 67:651, 1963.
202. Seltzer S, Naidorf IJ: Flare-ups in endodontics: I. Etiological factors, *J Endod* 11:472, 1985.
203. Seltzer S, Naidorf IJ: Flare-ups in endodontics: II. Therapeutic measures, *J Endod* 11:559, 1985.
204. Setzer FC, Kohli MR, Shah SB, et al: Outcome of endodontic surgery: a meta-analysis of the literature. Part 2: Comparison of endodontic microsurgical techniques with and without the use of higher magnification, *J Endod* 38:1, 2012.
205. Setzer FC, Shah SB, Kohli MR, et al: Outcome of endodontic surgery: a meta-analysis of the literature. Part 1: Comparison of traditional root-end surgery and endodontic microsurgery, *J Endod* 36:17575, 2010.
206. Shovelton D: The presence and distribution of microorganisms within non-vital teeth, *Br Dent J* 117:101, 1964.
207. Shovelton DS, Friend LA, Kirk EE, et al: The efficacy of pulp capping materials: a comparative trial, *Br Dent J* 130:385, 1971.
208. Shuping GB, Orstavik D, Sigurdsson A, et al: Reduction of intracanal bacteria using nickel-titanium rotary instrumentation and various medications, *J Endod* 26:751, 2000.
209. Siqueira JF Jr, Guimaraes-Pinto T, Rocas IN: Effects of chemomechanical preparation with 2.5% sodium hypochlorite and intracanal medication with calcium hydroxide on cultivable bacteria in infected root canals, *J Endod* 33:800, 2007.
210. Siqueira JF Jr, Magalhaes KM, Rocas IN: Bacterial reduction in infected root canals treated with 2.5% NaOCl as an irrigant and calcium hydroxide/camphorated paramonochlorophenol paste as an intracanal dressing, *J Endod* 33:667, 2007.
211. Siqueira JF Jr, Rocas IN, Paiva SS, et al: Bacteriologic investigation of the effects of sodium hypochlorite and chlorhexidine during the endodontic treatment of teeth with apical periodontitis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 104:122, 2007.
212. Siqueira JF Jr, Rocas IN, Provenzano JC, et al: Relationship between Fcγ receptor and interleukin-1 gene polymorphisms and post-treatment apical periodontitis, *J Endod* 35:11862, 2009.
213. Siqueira JF Jr, Rocas IN, Provenzano JC, et al: Polymorphism of the FcγRIIIa gene and post-treatment apical periodontitis, *J Endod* 37:13458, 2011.
214. Sjogren U, Figdor D, Persson S, et al: Influence of infection at the time of root filling on the outcome of endodontic treatment of teeth with apical periodontitis, *Int Endod J* 30:297, 1997.
215. Sjogren U, Figdor D, Spangberg L, et al: The antimicrobial effect of calcium hydroxide as a short-term intracanal dressing, *Int Endod J* 24:119, 1991.

216. Sjogren U, Hagglund B, Sundqvist G, et al: Factors affecting the long-term results of endodontic treatment, *J Endod* 16:498, 1990.
217. Sjogren U, Sundqvist G: Bacteriologic evaluation of ultrasonic root canal instrumentation, *Oral Surg Oral Med Oral Pathol* 63:366, 1987.
218. Sjogren U, Sundqvist G, Nair PN: Tissue reaction to gutta-percha particles of various sizes when implanted subcutaneously in guinea pigs, *Eur J Oral Sci* 103:313, 1995.
219. Slade GD: Derivation and validation of a short-form oral health impact profile, *Community Dent Oral Epidemiol* 25:284, 1997.
220. Smith CS, Setchell DJ, Hartly FJ: Factors influencing the success of conventional root canal therapy—a five-year retrospective study, *Int Endod J* 26:321, 1993.
221. Sogur E, Baksi BG, Grondahl HG, et al: Detectability of chemically induced periapical lesions by limited cone beam computed tomography, intra-oral digital and conventional film radiography, *Dentomaxillofac Radiol* 38:458, 2009.
222. Song M, Kim SG, Lee SJ, et al: Prognostic factors of clinical outcomes in endodontic microsurgery: a prospective study, *J Endod* 39:14917, 2013.
223. Souza RA, Dantas JC, Brandao PM, et al: Apical third enlargement of the root canal and its relationship with the repair of periapical lesions, *Eur J Dent* 6:385, 2012.
224. Spangberg LS: Evidence-based endodontics: the one-visit treatment idea, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 91:617, 2001.
225. Spili P, Parashos P, Messer HH: The impact of instrument fracture on outcome of endodontic treatment, *J Endod* 31:845, 2005.
226. Stavropoulos A, Wenzel A: Accuracy of cone beam dental CT, intraoral digital and conventional film radiography for the detection of periapical lesions: an ex vivo study in pig jaws, *Clin Oral Investig* 11:101, 2007.
227. Stewart GG, Cobe H, Rappaport H: Study of new medicament in chemomechanical preparation of infected root canals, *J Am Dent Assoc* 63:33, 1961.
228. Stoll R, Betke K, Stachniss V: The influence of different factors on the survival of root canal fillings: a 10-year retrospective study, *J Endod* 31:783, 2005.
229. Strindberg LZ: *The dependence of the results of pulp therapy on certain factors: an analytic study based on radiographic and clinical follow-up examinations*, 1956, Mauritzon.
230. Sundqvist G: *Bacteriological studies of necrotic dental pulps*, Sweden, 1976, University of Umeå.
231. Sundqvist G, Figdor D, Persson S, et al: Microbiologic analysis of teeth with failed endodontic treatment and the outcome of conservative re-treatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 85:86, 1998.
232. Taschieri S, Corbella S, Tsesis I, et al: Effect of guided tissue regeneration on the outcome of surgical endodontic treatment of through-and-through lesions: a retrospective study at 4-year follow-up, *Oral Maxillofac Surg* 15:153, 2011.
233. Taschieri S, Del Fabbro M, Testori T, et al: Efficacy of guided tissue regeneration in the management of through-and-through lesions following surgical endodontics: a preliminary study, *Int J Periodontics Restorative Dent* 28:265, 2008.
234. Taschieri S, Del Fabbro M, Testori T, Weinstein R: Efficacy of xenogeneic bone grafting with guided tissue regeneration in the management of bone defects after surgical endodontics, *J Oral Maxillofac Surg* 65:11217, 2007.
235. Teixeira LS, Demarco FF, Coppola MC, et al: Clinical and radiographic evaluation of pulpotomies performed under intrapulpal injection of anaesthetic solution, *Int Endod J* 34:440, 2001.
236. Tickle M, Milsom K, Qualtrough A, et al: The failure rate of NHS funded molar endodontic treatment delivered in general dental practice, *Br Dent J* 204:E8; discussion 254, 2008.
237. Tobón SI, Arismendi JA, Marín ML, et al: Comparison between a conventional technique and two bone regeneration techniques in periradicular surgery, *Int Endod J* 35:635, 2002.
238. Torabinejad M, Goodacre CJ: Endodontic or dental implant therapy, *J Amer Dent Assoc* 137:973, 2006.
239. Tsesis I, Rosen E, Tamse A, et al: Effect of guided tissue regeneration on the outcome of surgical endodontic treatment: a systematic review and meta-analysis, *J Endod* 37:10395, 2011.
240. Tsesis I, Rosen E, Taschieri S, et al: Outcomes of surgical endodontic treatment performed by a modern technique: an updated meta-analysis of the literature, *J Endod* 39:332, 2013.
241. Tsesis I, Taivishvsky V, Kfir A, et al: Outcome of surgical endodontic treatment performed by a modern technique: a meta-analysis of literature, *J Endod* 35:1505, 2009.
242. Tuna D, Olmez A: Clinical long-term evaluation of MTA as a direct pulp capping material in primary teeth, *Int Endod J* 41:273, 2008.
243. Tyndall DA, Kapa SF, Bagnell CP: Digital subtraction radiography for detecting cortical and cancellous bone changes in the periapical region, *J Endod* 16:173, 1990.
244. Van Nieuwenhuysen JP, Aouar M, D'Hoore W: Retreatment or endodontic monitoring in endodontics, *Int Endod J* 27:75, 1994.
245. Velvart P: Papilla base incision: a new approach to recession-free healing of the interdental papilla after endodontic surgery, *Int Endod J* 35:453, 2002.
246. Velvart P, Ebner-Zimmermann U, Ebner JP: Comparison of papilla healing following sulcular full-thickness flap and papilla base flap in endodontic surgery, *Int Endod J* 36:653, 2003.
247. Velvart P, Ebner-Zimmermann U, Ebner JP: Comparison of long-term papilla healing following sulcular full thickness flap and papilla base flap in endodontic surgery, *Int Endod J* 37:687, 2004.
248. Vianna ME, Horz HP, Conrads G, et al: Effect of root canal procedures on endotoxins and endodontic pathogens, *Oral Microbiol Immunol* 22:411, 2007.
249. Villa-Machado PA, Botero-Ramírez X, Tobón-Arroyave SI: Retrospective follow-up assessment of prognostic variables associated with the outcome of periradicular surgery, *Int Endod J* 46:1063, 2013.
250. Von Arx T, Peñarrocha M, Jensen S: Prognostic factors in apical surgery with root-end filling: A meta-analysis, *J Endod* 36:957, 2010.
251. Waly NG: A five-year comparative study of calcium hydroxide-glutaraldehyde pulpotomies versus calcium hydroxide pulpotomies in young permanent molars, *Egypt Dent J* 41:993, 1995.
252. Wang CS, Arnold RR, Trope M, et al: Clinical efficiency of 2% chlorhexidine gel in reducing intracanal bacteria, *J Endod* 33:1283, 2007.
253. Wang Q, Cheung GS, Ng RP: Survival of surgical endodontic treatment performed in a dental teaching hospital: a cohort study, *Int Endod J* 37:764, 2004.
254. Weiss M: Pulp capping in older patients, *N Y State Dent J* 32:451, 1966.
255. Witherspoon DE, Small JC, Harris GZ: Mineral trioxide aggregate pulpotomies: a case series outcomes assessment, *J Am Dent Assoc* 137:610, 2006.
256. Wu MK, Shemesh H, Wesselink PR: Limitations of previously published systematic reviews evaluating the outcome of endodontic treatment, *Int Endod J* 42:656, 2009.
257. Xavier AC, Martinho FC, Chung A, et al: One-visit versus two-visit root canal treatment: effectiveness in the removal of endotoxins and cultivable bacteria, *J Endod* 39:959, 2013.
258. Yared GM, Dagher FE: Influence of apical enlargement on bacterial infection during treatment of apical periodontitis, *J Endod* 20:535, 1994.
259. Yoshioka T, Kobayashi C, Suda H, et al: An observation of the healing process of periapical lesions by digital subtraction radiography, *J Endod* 28:589, 2002.
260. Yusuf H: The significance of the presence of foreign material periapically as a cause of failure of root treatment, *Oral Surg Oral Med Oral Pathol* 54:566, 1982.
261. Zehnder M: Root canal irrigants, *J Endod* 32:389, 2006.

Figure and Table References

- Adenubi JO, Rule DC: Success rate for root fillings in young patients: a retrospective analysis of treated cases, *Br Dent J* 141:237, 1976.
- Akerblom A, Hasselgren G: The prognosis for endodontic treatment of obliterated root canals, *J Endod* 14:565, 1988.

- Akpata ES: Effect of endodontic procedures on the population of viable microorganisms in the infected root canal, *J Endod* 2:369, 1976.
- Allen RK, Newton CW, Brown CE Jr: A statistical analysis of surgical and nonsurgical endodontic retreatment cases, *J Endod* 15:261, 1989.
- Alley BS, Kitchens GG, Alley LW, Eleazer PD: A comparison of survival of teeth following endodontic treatment performed by general dentists or

- by specialists, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 98:115, 2004.
- Amagasa T, Nagase M, Sato T, Shioda S: Apicoectomy with retrograde gutta-percha root filling, *Oral Surg Oral Med Oral Pathol* 68:339, 1989.
- American Academy of Pediatric Dentistry 2014 Pulp Therapy Subcommittee, Clinical Affairs Committee: *American Academy of Pediatric Dentistry (AAPD) guideline on pulp therapy for primary and immature permanent teeth reference manual*, vol 36, no 6, Chicago, 2014, AAPD, p 242, www.aapd.org/media/Policies_Guidelines/G_Pulp.pdf. Accessed April 19, 2015.
- Appleton JLT: A note on the clinical value of bacteriologically controlling the treatment of periapical infection, *Dental Cosmos* 74:798, 1932.
- Aqrabawi J: Management of endodontic failures: case selection and treatment modalities, *Gen Dent* 53:63, 2005.
- Aqrabawi JA: Outcome of endodontic treatment of teeth filled using lateral condensation versus vertical compaction (Schilder's technique), *J Contemp Dent Pract* 7:17, 2006.
- Aquilino SA, Caplan DJ: Relationship between crown placement and the survival of endodontically treated teeth, *J Prosthet Dent* 87:256, 2002.
- Asgary S, Ehsani S: Permanent molar pulpotomy with a new endodontic cement: a case series, *J Conserv Dent* 12:31, 2009.
- Auerbach M: Antibiotics vs. instrumentation in endodontics, *NY State Dent J* 19:225, 1953.
- Auerbach MB: Clinical approach to the problem of pulp canal therapy, *J Am Dent Assoc* 25:939, 1938.
- Baratieri LN, Monteiro S Jr, Caldeira de Andrada MA: Pulp curettage: surgical technique, *Quintessence Int* 20:285, 1989.
- Barbakow FH, Cleaton-Jones P, Friedman D: An evaluation of 566 cases of root canal therapy in general dental practice. 2. Postoperative observations, *J Endod* 6:485, 1980.
- Barone C, Dao TT, Basrani BB, et al: Treatment outcome in endodontics: the Toronto study—phases 3, 4, and 5: apical surgery, *J Endod* 36:28, 2010.
- Barrieshi-Nusair KM, Qudeimat MA: A prospective clinical study of mineral trioxide aggregate for partial pulpotomy in cariously exposed permanent teeth, *J Endod* 32:731, 2006.
- Barthel CR, Rosenkranz B, Leuenberg A, et al: Pulp capping of carious exposures: treatment outcome after 5 and 10 years: a retrospective study, *J Endod* 26:525, 2000.
- Bence R, Madonia JV, Weine FS, Smulson MH: A microbiologic evaluation of endodontic instrumentation in pulpless teeth, *Oral Surg Oral Med Oral Pathol* 35:676, 1973.
- Bender IB, Seltzer S, Soltanof W: Endodontic success: a reappraisal of criteria 1, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 22:780, 1966.
- Bender IB, Seltzer S, Soltanof W: Endodontic success: a reappraisal of criteria 2, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 22:790, 1966.
- Bender IB, Seltzer S, Turkenkopf S: To culture or not to culture? *Oral Surg Oral Med Oral Pathol* 18:527, 1964.
- Benenati FW, Khajotia SS: A radiographic recall evaluation of 894 endodontic cases treated in a dental school setting, *J Endod* 28:391, 2002.
- Bergenholtz G, Lekholm U, Milthorpe R, et al: Influence of apical overinstrumentation and overfilling on re-treated root canals, *J Endod* 5:310, 1979.
- Bjorndal L, Reit C, Bruun G, et al: Treatment of deep caries lesions in adults: randomized clinical trials comparing stepwise vs. direct complete excavation, and direct pulp capping vs. partial pulpotomy, *Eur J Oral Sci* 118:290, 2010.
- Bjorndal L, Thylstrup A: A practice-based study on stepwise excavation of deep carious lesions in permanent teeth: a 1-year follow-up study, *Community Dent Oral Epidemiol* 26:122, 1998.
- Blayney JR: The clinical results of pulp treatment, *J Natl Dent Assoc* 9:198, 1922.
- Bogen G, Kim JS, Bakland LK: Direct pulp capping with mineral trioxide aggregate: an observational study, *J Am Dent Assoc* 139:305; quiz 305, 2008.
- Boggia R: A single-visit treatment of septic root canals using periapically extruded endomethasone, *Br Dent J* 155:300, 1983.
- Buchbinder M: A statistical comparison of cultured and non-cultured root canal cases, *J Dent Res* 20:93, 1941.
- Bystrom A, Claesson R, Sundqvist G: The antibacterial effect of camphorated paramonochlorophenol, camphorated phenol and calcium hydroxide in the treatment of infected root canals, *Endod Dent Traumatol* 1:170, 1985.
- Bystrom A, Happonen RP, Sjogren U, et al: Healing of periapical lesions of pulpless teeth after endodontic treatment with controlled asepsis, *Endod Dent Traumatol* 3:58, 1987.
- Bystrom A, Sundqvist G: Bacteriologic evaluation of the efficacy of mechanical root canal instrumentation in endodontic therapy, *Scand J Dent Res* 89:321, 1981.
- Bystrom A, Sundqvist G: Bacteriologic evaluation of the effect of 0.5 percent sodium hypochlorite in endodontic therapy, *Oral Surg Oral Med Oral Pathol* 55:307, 1983.
- Bystrom A, Sundqvist G: The antibacterial action of sodium hypochlorite and EDTA in 60 cases of endodontic therapy, *Int Endod J* 18:35, 1985.
- Caliskan MK: Success of pulpotomy in the management of hyperplastic pulpitis, *Int Endod J* 26:142, 1993.
- Caliskan MK: Pulpotomy of carious vital teeth with periapical involvement, *Int Endod J* 28:172, 1995.
- Caliskan MK: Nonsurgical retreatment of teeth with periapical lesions previously managed by either endodontic or surgical intervention, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 100:242, 2005.
- Caliskan MK, Sen BH: Endodontic treatment of teeth with apical periodontitis using calcium hydroxide: a long-term study, *Endod Dent Traumatol* 12:215, 1996.
- Caplan DJ, Cai J, Yin G, et al: Root canal filled versus non-root canal filled teeth: a retrospective comparison of survival times, *J Public Health Dent* 65:90, 2005.
- Caplan DJ, Kolker J, Rivera EM, et al: Relationship between number of proximal contacts and survival of root canal treated teeth, *Int Endod J* 35:193, 2002.
- Card SJ, Sigurdsson A, Orstavik D, et al: The effectiveness of increased apical enlargement in reducing intracanal bacteria, *J Endod* 28:779, 2002.
- Castagnola L, Orlay HG: Treatment of gangrene of the pulp by the Walkhoff method, *Br Dent J* 19:93, 1952.
- Chen SC, Chueh LH, Hsiao CK, et al: An epidemiologic study of tooth retention after nonsurgical endodontic treatment in a large population in Taiwan, *J Endod* 33:226, 2007.
- Chen S-C, Chueh L-H, Hsiao CK, et al: First untoward events and reasons for tooth extraction after nonsurgical endodontic treatment in Taiwan, *J Endod* 34:671, 2008.
- Cheung GS: Survival of first-time nonsurgical root canal treatment performed in a dental teaching hospital, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:596, 2002.
- Cheung GS, Chan TK: Long-term survival of primary root canal treatment carried out in a dental teaching hospital, *Int Endod J* 36:117, 2003.
- Chong BS, Pitt Ford TR, Hudson MB: A prospective clinical study of Mineral Trioxide Aggregate and IRM when used as root-end filling materials in endodontic surgery, *Int Endod J* 36:520, 2003.
- Christiansen R, Kirkevang LL, Horsted-Bindslev P, et al: Randomized clinical trial of root-end resection followed by root-end filling with mineral trioxide aggregate or smoothing of the orthograde gutta-percha root filling—1-year follow-up, *Int Endod J* 42:105, 2009.
- Chu FC, Leung WK, Tsang PC, et al: Identification of cultivable microorganisms from root canals with apical periodontitis following two-visit endodontic treatment with antibiotics/steroid or calcium hydroxide dressings, *J Endod* 32:17, 2006.
- Chu FC, Tsang CS, Chow TW, et al: Identification of cultivable microorganisms from primary endodontic infections with exposed and unexposed pulp space, *J Endod* 31:424, 2005.
- Chugal NM, Clive JM, Spångberg LSW: A prognostic model for assessment of the outcome of endodontic treatment: effect of biologic and diagnostic variables, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 91:342, 2001.
- Chugal NM, Clive JM, Spångberg LS: Endodontic infection: some biologic and treatment factors associated with outcome, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 96:81, 2003.
- Conner DA, Caplan DJ, Teixeira FB, et al: Clinical outcome of teeth treated endodontically with a nonstandardized protocol and root filled with resilon, *J Endod* 33:1290, 2007.
- Cotton TP, Schindler WG, Schwartz SA, et al: A retrospective study comparing clinical outcomes after obturation with Resilon/Epiphany or Gutta-Percha/Kerr sealer, *J Endod* 34:789, 2008.
- Cvek M: Treatment of non-vital permanent incisors with calcium hydroxide, *Odontol Revy* 23:27, 1972.

- Cvek M: A clinical report on partial pulpotomy and capping with calcium hydroxide in permanent incisors with complicated crown fracture, *J Endod* 4:232, 1978.
- Cvek M: Prognosis of luxated non-vital maxillary incisors treated with calcium hydroxide and filled with gutta-percha: a retrospective clinical study, *Endod Dent Traumatol* 8:45, 1992.
- Cvek M, Granath L, Lundberg M: Failures and healing in endodontically treated non-vital anterior teeth with posttraumatically reduced pulpal lumen, *Acta Odontol Scand* 40:223, 1982.
- Cvek M, Hollender L, Nord CE: Treatment of non-vital permanent incisors with calcium hydroxide. VI: A clinical, microbiological and radiological evaluation of treatment in one sitting of teeth with mature or immature root, *Odontol Revy* 27:93, 1976.
- Dalton BC, Orstavik D, Phillips C, et al: Bacterial reduction with nickel-titanium rotary instrumentation, *J Endod* 24:763, 1998.
- Dammaschke T, Steven D, Kaup M, et al: Long-term survival of root-canal-treated teeth: a retrospective study over 10 years, *J Endod* 29:638, 2003.
- Danin J, Strömberg T, Forsgren H, et al: Clinical management of nonhealing periradicular pathosis: surgery versus endodontic retreatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 82:213, 1996.
- de Chevigny C, Dao TT, Basrani BR, et al: Treatment outcome in endodontics: the Toronto study—phase 4: initial treatment, *J Endod* 34:258, 2008.
- de Lange J, Putterts T, Baas EM, et al: Ultrasonic root-end preparation in apical surgery: a prospective randomized study, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 104:841, 2007.
- DeRosa TA: A retrospective evaluation of pulpotomy as an alternative to extraction, *Gen Dent* 54:37, 2006.
- Deutsch AS, Musikant BL, Cohen BI, et al: A study of one visit treatment using EZ-Fill root canal sealer, *Endod Prac* 4:29, 2001.
- Dominik M, Lysiak-Drwal K, Gedrange T, et al: Efficacy of healing process of bone defects after apicectomy: results after 6 and 12 months, *J Physiol Pharmacol* 60 (suppl 8):51, 2009.
- Dorn SO, Gartner AH: Retrograde filling materials: a retrospective success-failure study of amalgam, EBA, and IRM, *J Endod* 16:391, 1990.
- Doyle SL, Hodges JS, Pesun JJ, et al: Retrospective cross sectional comparison of initial nonsurgical endodontic treatment and single-tooth implants, *J Endod* 32:822, 2006.
- Dugas NN, Lawrence HP, Teplitsky P, et al: Quality of life and satisfaction outcomes of endodontic treatment, *J Endod* 28:819, 2002.
- El-Meligy OA, Avery DR: Comparison of mineral trioxide aggregate and calcium hydroxide as pulpotomy agents in young permanent teeth (apexogenesis), *Pediatr Dent* 28:399, 2006.
- Engström B: The significance of enterococci in root canal treatment, *Odontol Revy* 15:87, 1964.
- Engström B, Lundberg M: The correlation between positive culture and the prognosis of root canal therapy after pulpectomy, *Odontol Revy* 16:193, 1965.
- Engström B, Segerstad LHA, Ramstrom G, Frostell G: Correlation of positive cultures with the prognosis for root canal treatment, *Odontol Revy* 15:257, 1964.
- European Society of Endodontology: Quality guidelines for endodontic treatment: consensus report of the European Society of Endodontology, *Int Endod J* 39:921, 2006.
- Farsi N, Alamoudi N, Balto K, et al: Clinical assessment of mineral trioxide aggregate (MTA) as direct pulp capping in young permanent teeth, *J Clin Pediatr Dent* 31:72, 2006.
- Ferrari M, Vichi A, Fadda GM, et al: A randomized controlled trial of endodontically treated and restored premolars, *J Dent Res* 91:72S, 2012.
- Field JW, Gutmann JL, Solomon ES, et al: A clinical radiographic retrospective assessment of the success rate of single-visit root canal treatment, *Int Endod J* 37:70, 2004.
- Finne K, Nord PG, Persson G, et al: Retrograde root filling with amalgam and cavite, *Oral Surg Oral Med Oral Pathol* 43:621, 1977.
- Fitzgerald M, Heys RJ: A clinical and histological evaluation of conservative pulpal therapy in human teeth, *Oper Dent* 16:101, 1991.
- Fleming CH, Litaker MS, Alley LW, et al: Comparison of classic endodontic techniques versus contemporary techniques on endodontic treatment success, *J Endod* 36:414, 2010.
- Fonzar F, Fonzar A, Buttolo P, et al: The prognosis of root canal therapy: a 10-year retrospective cohort study on 411 patients with 1175 endodontically treated teeth, *Eur J Oral Implantol* 2:201, 2009.
- Friedman S, Lost C, Zarrabian M, et al: Evaluation of success and failure after endodontic therapy using a glass ionomer cement sealer, *J Endod* 21:384, 1995.
- Friedman S, Mor C: The success of endodontic therapy—healing and functionality, *J Calif Dent Assoc* 32:493, 2004.
- Frostell G: Clinical significance of the root canal culture, *Transactions of 3rd Int Conferences of Endodontics* 112, 1963.
- Fuks AB, Gavra S, Chosack A: Long-term followup of traumatized incisors treated by partial pulpotomy, *Pediatr Dent* 15:334, 1993.
- Gagliani MM, Gorni FG, Strohmenger L: Periapical resurgery versus periapical surgery: a 5-year longitudinal comparison, *Int Endod J* 2005 38:320.
- Gallien GS Jr, Schuman NJ: Local versus general anesthesia: a study of pulpal response in the treatment of cariously exposed teeth, *J Am Dent Assoc* 111:599, 1985.
- García B, Penarrocha M, Martí E, et al: Periapical surgery in maxillary premolars and molars: Analysis in terms of the distance between the lesion and the maxillary sinus, *J Oral Maxillofac Surg* 66:1212, 2008.
- García-Mira B, Ortega-Sánchez B, Penarrocha-Diago M, et al: Osteotomy versus osteotomy with repositioning of the vestibular cortical in periapical surgery of mandibular molars: a preliminary study, *Medicina Oral, Patología Oral y Cirugía Bucal* 15:e628, 2010.
- Gesi A, Hakeberg M, Warfvinge J, et al: Incidence of periapical lesions and clinical symptoms after pulpectomy—a clinical and radiographic evaluation of 1- versus 2-session treatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 101:379, 2006.
- Gomes BP, Lilley JD, Drucker DB: Variations in the susceptibilities of components of the endodontic microflora to biomechanical procedures, *Int Endod J* 29:235, 1996.
- Gorni FGM, Gagliani MM: The outcome of endodontic retreatment: a 2-yr follow-up, *J Endod* 30:1, 2004.
- Goyal B, Tewari S, Duhan J, et al: Comparative evaluation of platelet-rich plasma and guided tissue regeneration membrane in the healing of apicomarginal defects: a clinical study, *J Endod* 37:773, 2011.
- Grahnén H, Hansson L: The prognosis of pulp and root canal therapy, *Odontol Revy* 12:146, 1961.
- Grahnén H, Krasse B: The effect of instrumentation and flushing of non-vital teeth in endodontic therapy, *Odontol Rev* 14:167, 1963.
- Grung B, Molven O, Halse A: Periapical surgery in a Norwegian county hospital: Follow-up findings of 477 teeth, *J Endod* 16:411, 1990.
- Gruythuysen RJ, van Strijp AJ, Wu MK: Long-term survival of indirect pulp treatment performed in primary and permanent teeth with clinically diagnosed deep carious lesions, *J Endod* 36:1490, 2010.
- Halse A, Molven O: Overextended gutta-percha and Kloroperka N-Ø root canal fillings: radiographic findings after 10–17 years, *Acta Odontol Scand* 45:171, 1987.
- Harty FJ, Parkins BJ, Wengraf AM: Success rate in root canal therapy: a retrospective study on conventional cases, *Br Dent J* 128:65, 1970.
- Haskell EW, Stanley HR, Chellemi J, et al: Direct pulp capping treatment: a long-term follow-up, *J Am Dent Assoc* 97:607, 1978.
- Hayashi M, Fujitani M, Yamaki C, et al: Ways of enhancing pulp preservation by stepwise excavation—a systematic review, *J Dent* 39:95, 2011.
- Heling B, Kischinsky D: Factors affecting successful endodontic therapy, *J Br Endod Soc* 12:83, 1979.
- Heling B, Shapira J: Roentgenologic and clinical evaluation of endodontically treated teeth, with or without negative culture, *Quintessence Int* 11:79, 1978.
- Heling B, Tamshe A: Evaluation of the success of endodontically treated teeth, *Oral Surg Oral Med Oral Pathol* 30:533, 1970.
- Heling I, Bialla-Shenkman S, Turetzky A, et al: The outcome of teeth with periapical periodontitis treated with nonsurgical endodontic treatment: a computerized morphometric study, *Quintessence Int* 32:397, 2001.
- Hilton TJ, Ferracane JL, Mancl L, Northwest Practice-based Research Collaborative in Evidence-based Dentistry (NWP): Comparison of CaOH with MTA for direct pulp capping: a PBRN randomized clinical trial, *J Dent Res* 92:16S, 2013.
- Hirsch J-M, Ahlström U, Henrikson P-Å, et al: Periapical surgery, *Int J Oral Surg* 8:173, 1979.
- Horsted P, Sandergaard B, Thylstrup A, et al: A retrospective study of direct pulp capping with calcium hydroxide compounds, *Endod Dent Traumatol* 1:29, 1985.

- Hoskinson SE, Ng YL, Hoskinson AE, et al: A retrospective comparison of outcome of root canal treatment using two different protocols, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 93:705, 2002.
- Hsiao A, Glickman G, He J: A retrospective clinical and radiographic study on healing of periradicular lesions in patients taking oral bisphosphonates, *J Endod* 35:1525, 2009.
- Huomonen S, Lenander-Lumikari M, Sigurdsson A, et al: Healing of apical periodontitis after endodontic treatment: a comparison between a silicone-based and a zinc oxide-eugenol-based sealer, *Int Endod J* 36:296, 2003.
- Ingle J, Zeldow B: An evaluation of mechanical instrumentation and the negative culture in endodontic therapy, *J Am Dent Assoc* 57:471, 1958.
- Ioannides C, Borstlap WA: Apicoectomy on molars: a clinical and radiographical study, *Int J Oral Surg* 12:73, 1983.
- Jansson L, Sandstedt P, Låftman AC, et al: Relationship between apical and marginal healing in periradicular surgery, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 83:596, 1997.
- Jensen S, Nattestad A, Egde P, et al: A prospective, randomized, comparative clinical study of resin composite and glass ionomer cement for retrograde root filling, *Clin Oral Investig* 6:236, 2002.
- Jesslén P, Zetterqvist L, Heimdahl A: Long-term results of amalgam versus glass ionomer cement as apical sealant after apicoectomy, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 79:101, 1995.
- Jokinen MA, Kotilainen R, Poikkeus P, et al: Clinical and radiographic study of pulpectomy and root canal therapy, *Scand J Dent Res* 86:366, 1978.
- Jordan RE, Suzuki M: Conservative treatment of deep carious lesions, *J Can Dent Assoc* 37:337, 1971.
- Jordan RE, Suzuki M, Skinner DH: Indirect pulp-capping of carious teeth with periapical lesions, *J Am Dent Assoc* 97:37, 1978.
- Jurcak JJ, Bellizzi R, Loushine RJ: Successful single-visit endodontics during Operation Desert Shield, *J Endod* 19:412, 1993.
- Kerekes K: Radiographic assessment of an endodontic treatment method, *J Endod* 4:210, 1978.
- Khedmat S: Evaluation of endodontic treatment failure of teeth with periapical radiolucent areas and factors affecting it, *Front Dent* 1:34, 2004.
- Kim E, Song JS, Jung IY, et al: Prospective clinical study evaluating endodontic microsurgery outcomes for cases with lesions of endodontic origin compared with cases with lesions of combined periodontal-endodontic origin, *J Endod* 34:546, 2008.
- Klevant FJH, Eggink CO: The effect of canal preparation on periapical disease, *Int Endod J* 16:68, 1983.
- Koontongkaew S, Silapichit R, Thaweboon B: Clinical and laboratory assessments of camphorated monochlorophenol in endodontic therapy, *Oral Surg Oral Med Oral Pathol* 65:757, 1988.
- Kreisler M, Gockel R, Aubell-Falkenberg S, et al: Clinical outcome in periradicular surgery: effect of patient- and tooth-related factors: a multicenter study, *Quintessence Int* 44:53, 2013.
- Kvist T, Molander A, Dahlen G, et al: Microbiological evaluation of one- and two-visit endodontic treatment of teeth with apical periodontitis: a randomized, clinical trial, *J Endod* 30:572, 2004.
- Lana MA, Ribeiro-Sobrinho AP, Stehling R, et al: Microorganisms isolated from root canals presenting necrotic pulp and their drug susceptibility in vitro, *Oral Microbiol Immunol* 16:100, 2001.
- Lazarski MP, Walker WA III, Flores CM, et al: Epidemiological evaluation of the outcomes of nonsurgical root canal treatment in a large cohort of insured dental patients, *J Endod* 27:791, 2001.
- Leco-Berrolcal MI, Martinez Gonzalez JM, Donado Rodriguez M: Clinical and radiological course in apicoectomies with the Erbium:YAG laser, *Medicina Oral, Patología Oral y Cirugía Bucal* 12:E65, 2007.
- Leksell E, Ridell K, Cvek M, et al: Pulp exposure after stepwise versus direct complete excavation of deep carious lesions in young posterior permanent teeth, *Endod Dent Traumatol* 12:192, 1996.
- Li H, Zhai F, Zhang R, et al: Evaluation of microsurgery with SuperEBA as root-end filling material for treating post-treatment endodontic disease: a 2-year retrospective study, *J Endod* 40:345, 2014.
- Liang YH, Li G, Wesselink PR, et al: Endodontic outcome predictors identified with periapical radiographs and cone-beam computed tomography scans, *J Endod* 37:326, 2011.
- Liang YH, Yuan M, Li G, et al: The ability of cone-beam computed tomography to detect simulated buccal and lingual recesses in root canals, *Int Endod J* 45:724, 2012.
- Lilly JP, Cox D, Arcuri M, et al: An evaluation of root canal treatment in patients who have received irradiation to the mandible and maxilla, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 86:224, 1998.
- Lindeboom JAH, Frenken J, Kroon FHM, et al: A comparative prospective randomized clinical study of MTA and IRM as root-end filling materials in single-rooted teeth in endodontic surgery, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 100:495, 2005.
- Lumley PJ, Lucarotti PS, Burke FJ: Ten-year outcome of root fillings in the General Dental Services in England and Wales, *Int Endod J* 41:577, 2008.
- Lustmann J, Friedman S, Shaharabany V: Relation of pre- and intraoperative factors to prognosis of posterior apical surgery, *J Endod* 17:239, 1991.
- Lynch CD, Burke FM, Ní Ríordáin R, et al: The influence of coronal restoration type on the survival of endodontically treated teeth, *Eur J Prosthodont Restor Dent* 12:171, 2004.
- Mackie IC, Worthington HV, Hill FJ: A follow-up study of incisor teeth which had been treated by apical closure and root filling, *Br Dent J* 175:99, 1993.
- Maddaloni M, Gagliani M: Periapical endodontic surgery: a 3-year follow-up study, *Int Endod J* 36:193, 2003.
- Maltz M, Alves LS, Jardim JJ, et al: Incomplete caries removal in deep lesions: a 10-year prospective study, *Am J Dent* 24:211, 2011.
- Maltz M, Garcia R, Jardim JJ, et al: Randomized trial of partial vs. stepwise caries removal: 3-year follow-up, *J Dent Res* 91:1026, 2012.
- Maltz M, Henz SL, de Oliveira EF, et al: Conventional caries removal and sealed caries in permanent teeth: a microbiological evaluation, *J Dent* 40:776, 2012.
- Marín-Botero ML, Domínguez-Mejía JS, Arismendi-Echavarría JA, et al: Healing response of apicomarginal defects to two guided tissue regeneration techniques in periradicular surgery: A double-blind, randomized-clinical trial, *Int Endod J* 39:368, 2006.
- Markvart M, Dahlen G, Reit CE, et al: The antimicrobial effect of apical box versus apical cone preparation using iodine potassium iodide as root canal dressing: a pilot study, *Acta Odontol Scand* 71:786, 2013.
- Marti-Bowen E, Penarrocha-Diago M, Garcia-Mira B: Periapical surgery using the ultrasound technique and silver amalgam retrograde filling. A study of 71 teeth with 100 canals, *Medicina Oral, Patología Oral y Cirugía Bucal* 10:E67, 2005.
- Mass E, Zilberman U: Clinical and radiographic evaluation of partial pulpotomy in carious exposure of permanent molars, *Pediatr Dent* 15:257, 1993.
- Masterton JB: The healing of wounds of the dental pulp: an investigation of the nature of the scar tissue and of the phenomena leading to its formation, *Dent Pract Dent Rec* 16:325, 1966.
- Matsumoto T, Nagai T, Ida K, et al: Factors affecting successful prognosis of root canal treatment, *J Endod* 13:239, 1987.
- Matsuo T, Nakanishi T, Shimizu H, Ebisu S: A clinical study of direct pulp capping applied to carious-exposed pulps, *J Endod* 22:551, 1996.
- Mejare I, Cvek M: Partial pulpotomy in young permanent teeth with deep carious lesions, *Endod Dent Traumatol* 9:238, 1993.
- Mente J, Geletneký B, Ohle M, et al: Mineral trioxide aggregate or calcium hydroxide direct pulp capping: an analysis of the clinical treatment outcome, *J Endod* 36:806, 2010.
- Mente J, Hage N, Pfefferle T, et al: Mineral trioxide aggregate apical plugs in teeth with open apical foramina: a retrospective analysis of treatment outcome, *J Endod* 35:1354, 2009.
- Miles JP, Gluskin AH, Chambers D, et al: Pulp capping with mineral trioxide aggregate (MTA): a retrospective analysis of carious pulp exposures treated by undergraduate dental students, *Oper Dent* 35:20, 2010.
- Molander A, Reit C, Dahlen G: Microbiological evaluation of clindamycin as a root canal dressing in teeth with apical periodontitis, *Int Endod J* 23:113, 1990.
- Molander A, Warfvinge J, Reit C, et al: Clinical and radiographic evaluation of one- and two-visit endodontic treatment of asymptomatic necrotic teeth with apical periodontitis: a randomized clinical trial, *J Endod* 33:1145, 2007.
- Molven O, Halse A: Success rates for gutta-percha and Kloropercha N-Ø root fillings made by undergraduate students: radiographic findings after 10-17 years, *Int Endod J* 21:243, 1998.
- Molven O, Halse A, Grung B: Observer strategy and the radiographic classification of healing after endodontic surgery, *Int J Oral Maxillofac Surg* 16:432, 1987.
- Moore DL: Conservative treatment of teeth with vital pulps and periapical lesions: a preliminary report, *J Prosthet Dent* 18:476, 1967.

- Morsani JM, Aminoshariae A, Han YW, et al: Genetic predisposition to persistent apical periodontitis, *J Endod* 37:455, 2011.
- Morse DR, Esposito JV, Pike C, et al: A radiographic evaluation of the periapical status of teeth treated by the gutta-percha-eucapercha endodontic method: a one-year follow-up study of 458 root canals. Part I, *Oral Surg Oral Med Oral Pathol* 55:607, 1983a.
- Morse DR, Esposito JV, Pike C, et al: radiographic evaluation of the periapical status of teeth treated by the gutta-percha-eucapercha endodontic method: a one-year follow-up study of 458 root canals. Part II, *Oral Surg Oral Med Oral Pathol* 56:89, 1983b.
- Morse DR, Esposito JV, Pike C, et al: A radiographic evaluation of the periapical status of teeth treated by the gutta-percha-eucapercha endodontic method: a one-year follow-up study of 458 root canals. Part III, *Oral Surg Oral Med Oral Pathol* 56:190, 1983c.
- Morse F, Yates M: Follow up studies of root-filled teeth in relation to bacteriologic findings, *J Am Dent Assoc* 28:956, 1941.
- Moshonov J, Slutsky-Goldberg I, Gottlieb A, et al: The effect of the distance between post and residual gutta-percha on the clinical outcome of endodontic treatment, *J Endod* 31:177, 2005.
- Murphy WK, Kaugars GE, Collett WK, et al: Healing of periapical radiolucencies after nonsurgical endodontic therapy, *Oral Surg Oral Med Oral Pathol* 71:620, 1991.
- Nagamine M: Studies on treatment of deep caries lesions utilizing polycarboxylate cement combined with tannin fluoride preparation, *J Okayama Dent Soc* 12:1, 1993 [Japanese].
- Nagasiri R, Chitmongkolsuk S: Long-term survival of endodontically treated molars without crown coverage: a retrospective cohort study, *J Prosthet Dent* 93:164, 2005.
- Negishi J, Kawanami M, Ogami E: Risk analysis of failure of root canal treatment for teeth with inaccessible apical constriction, *J Dent* 33:399, 2005.
- Nelson IA: Endodontics in general practice: a retrospective study, *Int Endod J* 15:168, 1982.
- Ng YL, Mann V, Gulabivala K: A prospective study of the factors affecting outcomes of nonsurgical root canal treatment: part 1: periapical health, *Int Endod J* 44:583, 2011.
- Ng YL, Mann V, Gulabivala K: A prospective study of the factors affecting outcomes of non-surgical root canal treatment: part 2: tooth survival, *Int Endod J* 44:610, 2011.
- Nicholls E: The efficacy of cleansing of the root canal, *Br Dent J* 112:167, 1962.
- Nordenram A: Biobond for retrograde root filling in apicoectomy, *Eur J Oral Sci* 78:251, 1970.
- Olgart LG: Bacteriological sampling from root canals directly after chemo-mechanical treatment: a clinical and bacteriological study, *Acta Odontol Scand* 27:91, 1969.
- Oliet S: Single-visit endodontics: a clinical study, *J Endod* 9:147, 1983.
- Oliet S, Sorin SM: Evaluation of clinical results based upon culturing root canals, *J Br Endod Soc* 3:3, 1969.
- Orstavik D: Time-course and risk analyses of the development and healing of chronic apical periodontitis in man, *Int Endod J* 29:150, 1996.
- Orstavik D, Kerekes K, Molven O: Effects of extensive apical reaming and calcium hydroxide dressing on bacterial infection during treatment of apical periodontitis: a pilot study, *Int Endod J* 24:1, 1991.
- Ortega-Sánchez B, Peñarrocha-Diogo M, Rubio-Martínez LA, Vera-Sempere JF: Radiographic morphometric study of 37 periapical lesions in 30 patients: validation of success criteria, *J Oral Maxillofac Surg* 67:846, 2009.
- Pantchev A, Nohlert E, Tegelberg A: Endodontic surgery with and without inserts of bioactive glass PerioGlas: a clinical and radiographic follow-up, *Oral Maxillofac Surg* 13:21, 2009.
- Pantschev A, Carlsson AP, Andersson L: Retrograde root filling with EBA cement or amalgam: a comparative clinical study, *Oral Surg Oral Med Oral Pathol* 78:101, 1994.
- Paquette L, Legner M, Fillery ED, et al: Antibacterial efficacy of chlorhexidine gluconate intracanal medication in vivo, *J Endod* 33:788, 2007.
- Peak JD: The success of endodontic treatment in general dental practice: a retrospective clinical and radiographic study, *Prim Dent Care* 1:9, 1994.
- Peak JD, Hayes SJ, Bryant ST, et al: The outcome of root canal treatment. A retrospective study within the armed forces (Royal Air Force), *Br Dent J* 190:140, 2001.
- Peciuliene V, Balciuniene I, Eriksen HM, et al: Isolation of *Enterococcus faecalis* in previously root-filled canals in a Lithuanian population, *J Endod* 26:593, 2000.
- Peciuliene V, Reynaud AH, Balciuniene I, et al: Isolation of yeasts and enteric bacteria in root-filled teeth with chronic apical periodontitis, *Int Endod J* 34:429, 2001.
- Pecora G, De Leonardis D, Ibrahim N, et al: The use of calcium sulphate in the surgical treatment of a 'through and through' periradicular lesion, *Int Endod J* 34:189, 2001.
- Pekruhn RB: The incidence of failure following single-visit endodontic therapy, *J Endod* 12:68, 1986.
- Peñarrocha M, Carrillo C, Peñarrocha M, et al: Symptoms before periapical surgery related to histologic diagnosis and postoperative healing at 12 months for 178 periapical lesions, *J Oral Maxillofac Surg* 69:e31, 2011.
- Peñarrocha M, Martí E, García B, et al: Relationship of periapical lesion radiologic size, apical resection, and retrograde filling with the prognosis of periapical surgery, *J Maxillofac Oral Surg* 65:1526, 2007.
- Peñarrocha-Diogo M, Maestre-Ferrín L, Peñarrocha-Oltra D, et al: Influence of hemostatic agents upon the outcome of periapical surgery: dressings with anesthetic and vasoconstrictor or aluminum chloride, *Medicina Oral, Patología Oral y Cirugía Bucal* 18:e272, 2013.
- Peñarrocha Diogo M, Ortega Sánchez B, García Mira B, et al: Evaluation of healing criteria for success after periapical surgery, *Medicina Oral, Patología Oral y Cirugía Bucal* 13:143, 2008.
- Peñarrocha-Diogo MA, Ortega-Sanchez B, Garcia-Mira B, et al: A prospective clinical study of polycarboxylate cement in periapical surgery, *Medicina Oral, Patología Oral y Cirugía Bucal* 17:e276, 2012.
- Peñarrocha Diogo M, Sanchis Bielsa JM, Gay Escoda C: Periapical surgery of 31 lower molars based on the ultrasound technique and retrograde filling with silver amalgam, *Medicina Oral* 6:376, 2001.
- Penesis VA, Fitzgerald PI, Fayad MI, et al: Outcome of one-visit and two-visit endodontic treatment of necrotic teeth with apical periodontitis: a randomized controlled trial with one-year evaluation, *J Endod* 34:251, 2008.
- Peretz B, Yakir O, Fuks AB: Follow up after root canal treatment of young permanent molars, *J Clin Pediatr Dent* 21:237, 1997.
- Persson G: Periapical surgery of molars, *Int J Oral Surg* 11:96, 1982.
- Peters LB, van Winkelhoff AJ, Buijs JF, et al: Effects of instrumentation, irrigation and dressing with calcium hydroxide on infection in pulpless teeth with periapical bone lesions, *Int Endod J* 35:13, 2002.
- Peters LB, Wesseling PR: Periapical healing of endodontically treated teeth in one and two visits obturated in the presence or absence of detectable microorganisms, *Int Endod J* 35:660, 2002.
- Pettiette MT, Delano EO, Trope M: Evaluation of success rate of endodontic treatment performed by students with stainless-steel K-files and nickel-titanium hand files, *J Endod* 27:124, 2001.
- Platt AS, Wannfors K: The effectiveness of compomer as a root-end filling: a clinical investigation, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 97:508, 2004.
- Qudeimat MA, Barrieshi-Nusair KM, Owais AI: Calcium hydroxide vs mineral trioxide aggregates for partial pulpotomy of permanent molars with deep caries, *Eur Arch Paediatr Dent* 8:99, 2007.
- Rahbaran S, Gilthorpe MS, Harrison SD, Gulabivala K: Comparison of clinical outcome of periapical surgery in endodontic and oral surgery units of a teaching dental hospital: a retrospective study, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 91:700, 2001.
- Rapp EL, Brown CE Jr, Newton CW: An analysis of success and failure of apicoectomies, *J Endod* 17:508, 1991.
- Reid RJ, Abbott PV, McNamara JR, et al: A five-year study of hydron root canal fillings, *Int Endod J* 25:213, 1992.
- Reit C, Dahlen G: Decision making analysis of endodontic treatment strategies in teeth with apical periodontitis, *Int Endod J* 21:291, 1988.
- Reit C, Molander A, Dahlen G: The diagnostic accuracy of microbiologic root canal sampling and the influence of antimicrobial dressings, *Endod Dent Traumatol* 15:278, 1999.
- Rhein ML, Krasnow F, Gies WJ: A prolonged study of the electrolytic treatment of dental focal infection: a preliminary report, *Dent Cosmos* 68:971, 1926.
- Ricucci D, Grondahl K, Bergenholtz G: Periapical status of root-filled teeth exposed to the oral environment by loss of restoration or caries, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 90:354, 2000.
- Ricucci D, Russo J, Rutberg M, et al: A prospective cohort study of endodontic treatments of 1,369 root canals: results after 5 years, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 112:825, 2011.

- Rocas IN, Siqueira JF Jr, Del Aguila CA, et al: Polymorphism of the CD14 and TLR4 genes and post-treatment apical periodontitis, *J Endod* 40:168, 2014.
- Rubinstein RA, Kim S: Long-term follow-up of cases considered healed one year after apical microsurgery, *J Endod* 28:378, 2002.
- Rud J, Andreasen JO: A study of failures after endodontic surgery by radiographic, histologic and stereomicroscopic methods, *Int J Oral Surg* 1:311, 1972.
- Rud J, Munksgaard EC, Andreasen JO, et al: Retrograde root filling with composite and a dentin-bonding agent. 2, *Endod Dent Traumatol* 7:126, 1991.
- Rud J, Rud V, Munksgaard EC: Long-term evaluation of retrograde root filling with dentin-bonded resin composite, *J Endod* 22:90, 1996.
- Rud J, Rud V, Munksgaard EC: Periapical healing of mandibular molars after root-end sealing with dentine-bonded composite, *Int Endod J* 34:285, 2001.
- Russo MC, Holland R, de Souza V: Radiographic and histological evaluation of the treatment of inflamed dental pulps, *Int Endod J* 15:137, 1982.
- Safavi KE, Dowden WE, Langeland K: Influence of delayed coronal permanent restoration on endodontic prognosis, *Endod Dent Traumatol* 3:187, 1987.
- Salehrabi R, Rotstein I: Endodontic treatment outcomes in a large patient population in the USA: an epidemiological study, *J Endod* 30:846, 2004.
- Salehrabi R, Rotstein I: Epidemiologic evaluation of the outcomes of orthograde endodontic retreatment, *J Endod* 36:790, 2010.
- Salvi GE, Siegrist Guldener BE, Amstad T, et al: Clinical evaluation of root filled teeth restored with or without post-and-core systems in a specialist practice setting, *Int Endod J* 40:209, 2007.
- Santini A: Assessment of the pulpotomy technique in human first permanent mandibular molars: use of two direct inspection criteria, *Br Dent J* 155:151, 1983.
- Santucci PJ: Dycal versus Nd:YAG laser and Vitrebond for direct pulp capping in permanent teeth, *J Clin Laser Med Surg* 17:69, 1999.
- Sari S, Durutürk L: Radiographic evaluation of periapical healing of permanent teeth with periapical lesions after extrusion of AH Plus sealer, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 104:e54, 2007.
- Sawusch RH: Direct and indirect pulp capping with two new products, *J Am Dent Assoc* 104:459, 1982.
- Schwartz-Arad D, Yarom N, Lustig JP, et al: A retrospective radiographic study of root-end surgery with amalgam and intermediate restorative material, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 96:472, 2003.
- Selden HS: Pulpoperiapical disease: diagnosis and healing: a clinical endodontic study, *Oral Surg* 27:271, 1974.
- Seltzer S, Bender IB, Turkenkopf S: Factors affecting successful repair after root canal therapy, *J Am Dent Assoc* 67:651, 1963.
- Setzer FC, Boyer KR, Jeppson JR, et al: Long-term prognosis of endodontically treated teeth: a retrospective analysis of preoperative factors in molars, *J Endod* 37:21, 2011.
- Shah N: Non-surgical management of periapical lesions: a prospective study, *Oral Surg Oral Med Oral Pathol* 66:365, 1988.
- Shovelton DS, Friend LA, Kirk EE, et al: The efficacy of pulp capping materials: a comparative trial, *Br Dent J* 130:385, 1971.
- Shuping GB, Orstavik D, Sigurdsson A, Trope M: Reduction of intracanal bacteria using nickel-titanium rotary instrumentation and various medications, *J Endod* 26:751, 2000.
- Siqueira JF Jr, Guimaraes-Pinto T, Rocas IN: Effects of chemomechanical preparation with 2.5% sodium hypochlorite and intracanal medication with calcium hydroxide on cultivable bacteria in infected root canals, *J Endod* 33:800, 2007.
- Siqueira JF Jr, Magalhaes KM, Rocas IN: Bacterial reduction in infected root canals treated with 2.5% NaOCl as an irrigant and calcium hydroxide/camphorated paramonochlorophenol paste as an intracanal dressing, *J Endod* 33:667, 2007.
- Siqueira JF Jr, Rocas IN, Paiva SS, et al: Bacteriologic investigation of the effects of sodium hypochlorite and chlorhexidine during the endodontic treatment of teeth with apical periodontitis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 104:122, 2007.
- Siqueira JF Jr, Rocas IN, Provenzano JC, et al: Relationship between Fcgamma receptor and interleukin-1 gene polymorphisms and post-treatment apical periodontitis, *J Endod* 35:1186, 2009.
- Siqueira JF Jr, Rocas IN, Provenzano JC, et al: Polymorphism of the FcgammaRIIIa gene and post-treatment apical periodontitis, *J Endod* 37:1345, 2011.
- Siqueira JF Jr, Rôças IN, Riche FN, et al: Clinical outcome of the endodontic treatment of teeth with apical periodontitis using an antimicrobial protocol, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 106:757, 2008.
- Sjogren U, Figdor D, Persson S, et al: Influence of infection at the time of root filling on the outcome of endodontic treatment of teeth with apical periodontitis, *Int Endod J* 30:297, 1997.
- Sjogren U, Figdor D, Spangberg L, et al: The antimicrobial effect of calcium hydroxide as a short-term intracanal dressing, *Int Endod J* 24:119, 1991.
- Sjogren U, Hagglund B, Sundqvist G, et al: Factors affecting the long-term results of endodontic treatment, *J Endod* 16:498, 1990.
- Sjogren U, Sundqvist G: Bacteriologic evaluation of ultrasonic root canal instrumentation, *Oral Surg Oral Med Oral Pathol* 63:366, 1987.
- Skupien JA, Opdam N, Winnen R, et al: A practice-based study on the survival of restored endodontically treated teeth, *J Endod* 39:1335, 2013.
- Smith CS, Setchell DJ, Harty FJ: Factors influencing the success of conventional root canal therapy—a five-year retrospective study, *Int Endod J* 26:321, 1993.
- Soltanoff W: A comparative study of the single-visit and the multiple-visit endodontic procedure, *J Endod* 4:278, 1978.
- Song M, Jung IY, Lee SJ, et al: Prognostic factors for clinical outcomes in endodontic microsurgery: a retrospective study, *J Endod* 37:927, 2011. Erratum in *J Endod* 37:1595, 2011a.
- Song M, Kim E: A prospective randomized controlled study of mineral trioxide aggregate and super ethoxy-benzoic acid as root-end filling materials in endodontic microsurgery, *J Endod* 38:875, 2012.
- Song M, Kim SG, Lee SJ, et al: Prognostic factors of clinical outcomes in endodontic microsurgery: a prospective study, *J Endod* 39:1491, 2013a.
- Song M, Kim SG, Shin SJ, et al: The influence of bone tissue deficiency on the outcome of endodontic microsurgery: a prospective study, *J Endod* 39:1341, 2013b.
- Song M, Nam T, Shin SJ, et al: Comparison of clinical outcomes of endodontic microsurgery: 1 year versus long-term follow-up, *J Endod* 40:490, 2014.
- Song M, Shin SJ, Kim E: Outcomes of endodontic micro-resurgery: a prospective clinical study, *J Endod* 37:316, 2011b.
- Stewart GG, Cobe H, Rappaport H: Study of new medicament in chemomechanical preparation of infected root canals, *J Am Dent Assoc* 63:33, 1961.
- Stoll R, Betke K, Stachniss V: The influence of different factors on the survival of root canal fillings: a 10-year retrospective study, *J Endod* 31:783, 2005.
- Storms JL: Factors that influence the success of endodontic treatment, *J Can Dent Assoc (Tor)* 35:83, 1969.
- Strindberg LZ: *The dependence of the results of pulp therapy on certain factors: an analytic study based on radiographic and clinical follow-up examinations*, 1956, Mauritzon.
- Sumi Y, Hattori H, Hayashi K, et al: Ultrasonic root-end preparation: clinical and radiographic evaluation of results, *J Oral Maxillofac Surg* 54:590, 1996.
- Sundqvist G, Figdor D, Persson S, et al: Microbiologic analysis of teeth with failed endodontic treatment and the outcome of conservative retreatment, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 85:86, 1998.
- Swartz DB, Skidmore AE, Griffin JA: Twenty years of endodontic success and failure, *J Endod* 9:198, 1983.
- Tan L, Chen NN, Poon CY, et al: Survival of root filled cracked teeth in a tertiary institution, *Int Endod J* 2006 39:886.
- Taschieri S, Corbella S, Tsesis I, et al: Effect of guided tissue regeneration on the outcome of surgical endodontic treatment of through-and-through lesions: a retrospective study at 4-year follow-up, *Oral Maxillofac Surg* 15:153, 2011.
- Taschieri S, Del Fabbro M, Testori T, et al: Endodontic surgery with ultrasonic retrotips: one-year follow-up, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 100:380, 2005.
- Taschieri S, Del Fabbro M, Testori T, et al: Endodontic surgery using 2 different magnification devices: preliminary results of a randomized controlled study, *J Oral Maxillofac Surg* 64:235, 2006.
- Taschieri S, Del Fabbro M, Testori T, et al: Efficacy of guided tissue regeneration in the management of through-and-through lesions following surgical endodontics: a preliminary study, *Int J Periodontics Restorative Dent* 28:265, 2008b.

- Taschieri S, Del Fabbro M, Testori T, et al: Endoscopic periradicular surgery: a prospective clinical study, *Br J Oral Maxillofac Surg* 45:242, 2007.
- Taschieri S, Del Fabbro M, Testori T, et al: Endodontic reoperation using an endoscope and microsurgical instruments: one year follow-up, *Br J Oral Maxillofac Surg* 45:582, 2007a.
- Taschieri S, Del Fabbro M, Testori T, et al: Efficacy of xenogeneic bone grafting with guided tissue regeneration in the management of bone defects after surgical endodontics, *J Oral Maxillofac Surg* 65:1121, 2007b.
- Taschieri S, Del Fabbro M, Testori T, et al: Microscope versus endoscope in root-end management: a randomized controlled study, *Int J Oral Maxillofac Surg* 37:1022, 2008a, doi:10.1016/j.ijom.2008.07.001.
- Taschieri S, Machtou P, Rosano G, et al: The influence of previous non-surgical re-treatment on the outcome of endodontic surgery, *Minerva Stomatol* 59:625, 2010.
- Taschieri S, Weinstein T, Tesis I, et al: Magnifying loupes versus surgical microscope in endodontic surgery: a four-year retrospective study, *Aust Endod J* 39:78, 2013.
- Teixeira LS, Demarco FF, Coppola MC, et al: Clinical and radiographic evaluation of pulpotomies performed under intrapulpal injection of anaesthetic solution, *Int Endod J* 34:440, 2001.
- Tervit C, Paquette L, Torneck CD, et al: Proportion of healed teeth with apical periodontitis medicated with two percent chlorhexidine gluconate liquid: a case-series study, *J Endod* 35:1182, 2009.
- Testori T, Capelli M, Milani S, et al: Success and failure in periradicular surgery: a longitudinal retrospective analysis, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 87:493, 1999.
- Tilashalski KR, Gilbert GH, Boykin MJ, et al: Root canal treatment in a population-based adult sample: status of teeth after endodontic treatment, *J Endod* 30:577, 2004.
- Tobón SI, Arismendi JA, Marín ML, et al: Comparison between a conventional technique and two bone regeneration techniques in periradicular surgery, *Int Endod J* 35:635, 2002.
- Trope M, Delano EO, Orstavik D: Endodontic treatment of teeth with apical periodontitis: single vs. multivisit treatment, *J Endod* 25:345, 1999.
- Tsesis I, Rosen E, Schwartz-Arad D, et al: Retrospective evaluation of surgical endodontic treatment: traditional versus modern technique, *J Endod* 32:412, 2006.
- Vallecillo Capilla M, Muñoz Soto E, Reyes Botella C, et al: Periapical surgery of 29 teeth: a comparison of conventional technique, microsaw and ultrasound, *Med Oral* 7:46, 50, 2002.
- Vianna ME, Horz HP, Conrads G, et al: Effect of root canal procedures on endotoxins and endodontic pathogens, *Oral Microbiol Immunol* 22:411, 2007.
- Villa-Machado PA, Botero-Ramírez X, Tobón-Arroyave SI: Retrospective follow-up assessment of prognostic variables associated with the outcome of periradicular surgery, *Int Endod J* 46:1063, 2013.
- von Arx T, Gerber C, Hardt N: Periradicular surgery of molars: a prospective clinical study with a one-year follow-up, *Int Endod J* 34:520, 2001.
- von Arx T, Jensen SS, Hanni S: Clinical and radiographic assessment of various predictors for healing outcome 1 year after periapical surgery, *J Endod* 33:123, 2007.
- von Arx T, Jensen SS, Hanni S, et al: Five-year longitudinal assessment of the prognosis of apical microsurgery, *J Endod* 38:570, 2012.
- von Arx T, Kurt B: Root-end cavity preparation after apicoectomy using a new type of sonic and diamond-surfaced retrotip: a 1-year follow-up study, *J Oral Maxillofac Surg* 57:656, 1999.
- Von Arx T, Peñarrocha M, Jensen S: Prognostic factors in apical surgery with root-end filling: a meta-analysis, *J Endod* 36:957, 2010.
- Waikakul A, Punwutikorn J: Clinical study of retrograde filling with gold leaf: comparison with amalgam, *Oral Surg Oral Med Oral Pathol* 71:228, 1991.
- Wälivaara DA, Abrahamsson P, Fogelin M, Isaksson S: Super-EBA and IRM as root-end fillings in periapical surgery with ultrasonic preparation: a prospective randomized clinical study of 206 consecutive teeth, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 112:258, 2011.
- Wälivaara DA, Abrahamsson P, Isaksson S, et al: Prospective study of periapically infected teeth treated with periapical surgery including ultrasonic preparation and retrograde intermediate restorative material root-end fillings, *J Oral Maxillofac Surg* 65:931, 2007.
- Wälivaara DA, Abrahamsson P, Sämfors KA, et al: Periapical surgery using ultrasonic preparation and thermoplasticized gutta-percha with AH Plus sealer or IRM as retrograde root-end fillings in 160 consecutive teeth: a prospective randomized clinical study, *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 108:784, 2009.
- Waly NG: A five-year comparative study of calcium hydroxide-glutaraldehyde pulpotomies versus calcium hydroxide pulpotomies in young permanent molars, *Egypt Dent J* 41:993, 1995.
- Wang CH, Chueh LH, Chen SC, et al: Impact of diabetes mellitus, hypertension, and coronary artery disease on tooth extraction after nonsurgical endodontic treatment, *J Endod* 37:1, 2011.
- Wang CS, Arnold RR, Trope M, et al: Clinical efficiency of 2% chlorhexidine gel in reducing intracanal bacteria, *J Endod* 33:1283, 2007.
- Weiger R, Rosendahl R, Löst C: Influence of calcium hydroxide intracanal dressings on the prognosis of teeth with endodontically induced periapical lesions, *Int Endod J* 33:219, 2000.
- Weiss M: Pulp capping in older patients, *NY State Dent J* 32:451, 1966.
- Werts R: Endodontic treatment: a five-year follow-up, *Dent Survey* 51:29, 1975.
- Witherspoon DE, Small JC, Harris GZ: Mineral trioxide aggregate pulpotomies: a case series outcomes assessment, *J Am Dent Assoc* 137:610, 2006.
- Witherspoon DE, Small JC, Regan JD, et al: Retrospective analysis of open apex teeth obturated with mineral trioxide aggregate, *J Endod* 34:1171, 2008.
- Xavier AC, Martinho FC, Chung A, et al: One-visit versus two-visit root canal treatment: effectiveness in the removal of endotoxins and cultivable bacteria, *J Endod* 39:959, 2013.
- Yared GM, Dagher FE: Influence of apical enlargement on bacterial infection during treatment of apical periodontitis, *J Endod* 20:535, 1994.
- Zeldow BI, Ingle JI: Correlation of the positive culture to the prognosis of endodontically treated teeth: a clinical study, *J Am Dent Assoc* 66:9, 1963.
- Zmener O, Pameijer CH: Clinical and radiographical evaluation of a resin-based root canal sealer: a 5-year follow-up, *J Endod* 33:676, 2007.
- Zuolo ML, Ferreira MO, Gutmann JL: Prognosis in periradicular surgery: a clinical prospective study, *Int Endod J* 33:91, 2000.